Bulk, big jumps and boundary layers in the occupation law of a lattice Lévy walk

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Abstract

A lattice Lévy walk moves in runs of alternating direction whose lengths are independent with regularly varying tails of index between one and two. We describe the law of the number of right steps after N steps for every value of that number, except for one window that we describe only in part when the runs of right steps have the smaller fluctuation scale. The law has four pieces. They are a stable bulk at the scale of the larger run fluctuation, a big-jump zone on each side in which one run is macroscopic and the bin is the sum of the contributions of a complete run and of a run cut by the horizon, a renewal boundary layer at each end given by the tail of one run times the renewal mass function of the other, and an atom at each endpoint. The boundary layer joins the big-jump zone continuously, the big-jump zone joins the bulk in a double limit, and neither the bulk profile nor the big-jump kernel nor their sum is accurate across that junction. The first run enters each piece through its own criterion, which is sharp in the bulk and at the ends, and we give its exact contribution, including for the equilibrated start of the physics literature. The global mode is an endpoint or a bulk bin according to a comparison of the endpoint atom with the profile height, which for the equilibrated start with equal indices is the golden ratio threshold. For a fresh start, when the renewal mass function of the opposite runs stays above its limit, the interior minimum near an end sits at the stable scale of those runs, with an exact second-order coefficient. The boundary layers identify both run laws. The bulk theorem uses only regular variation and aperiodicity. The big-jump theorem uses the local form of regular variation, and we show that regular variation of the tail alone does not suffice.

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1 Introduction

A lattice Lévy walk moves on $\mathbb{Z}$ in runs of alternating direction. The run lengths are independent, and their tails are regularly varying with an index α between 1 and 2, so a run has a finite mean and an infinite variance. Let A_N be the number of right steps among the first N. The position of the walk after N steps is $2A_N - N$, so the law $P_N(k) = \mathbb{P}(A_N = k)$ on $\{0, \dots, N\}$ is the law of the walk read on its natural lattice. Lévy walks are a standard model of superdiffusion [44], and the same walk describes heat transport in one-dimensional lattices [13, 40] and the occupation time of an alternating renewal process [21, 25]. Its profile has a central peak and two ballistic side peaks [43, 12], and the physics literature describes the part between them by an infinite density [35, 36]. We describe $P_N(k)$ for every k, with proofs, with one exception that we state below.

1.1 The picture

Figure 1 shows P_N for $N = 5000$ on a logarithmic scale. Four pieces make up the row. We call the right steps plus slots.

- The bulk. Near pN, where p is the long-run fraction of plus slots, the law is a stable density at the scale B_N of the larger of the two run-length fluctuations. Here B_N grows like N^{1/α_*} up to a slowly varying factor, with α_* the smaller index.
- The big-jump zones. A count k with $k - pN$ much larger than B_N needs one plus run of macroscopic length, a giant run, and the law of that run gives the bin its size, which is of order $T_+(N)$ on the ballistic range, with T_+ the tail of a plus run. The giant run is either complete and lies in the interior or the final run, censored at slot N, and the bin is the sum of the two contributions.
- The boundary layers. At $k = N - j$ with j fixed, the giant run is the final one, and the only question is where it started. It starts right after the j-th minus slot exactly when a minus run ends there, so the layer is $T_+(N)$ times the renewal mass function $u_-(j)$ of the minus runs.
- The atoms. The bin $k = N$ is the event that the first run covers the row, and its mass is the tail of the first run.

The first run is the preparation of the walk. It may be a fresh run, a run seen from a typical time, or anything else, and it enters each of the four pieces through its own criterion.

1.2 Results

We state the hypotheses in Section 2.2. They are regular variation of the two tails with indices in $(1, 2)$, written (RV), the local form $f_\sigma(n) \sim \alpha_\sigma T_\sigma(n)/n$ of the mass functions, written (LR), and aperiodicity, written (SPAN). The run laws may differ between the two signs, so the two stable scales $b_{+,N}$ and $b_{-,N}$ may differ, and their ratio may oscillate. Every theorem below holds for every relation between the two scales, but one window is described only in part, as we say below. We write s for the sum of the two mean run lengths and p for the fraction of the plus mean.

The bulk. Theorem A says that for a fresh start, under (RV) and (SPAN),

$$\sup_{k\in\mathbb{Z}} \big| B_N P_N(k) - h_N\big((k - pN)/B_N\big)\big| \to 0, \qquad \sup_k P_N(k) \le \frac{C}{B_N},$$

where h_N is the density of $(1-p)\theta_{+,N} Z_+ - p\,\theta_{-,N} Z_-$, with $Z_\pm$ independent totally skewed stable variables. The supremum runs over the whole row. Outside the bulk the statement says only that every bin is $o(1/B_N)$, and the next theorem says what the bins are.

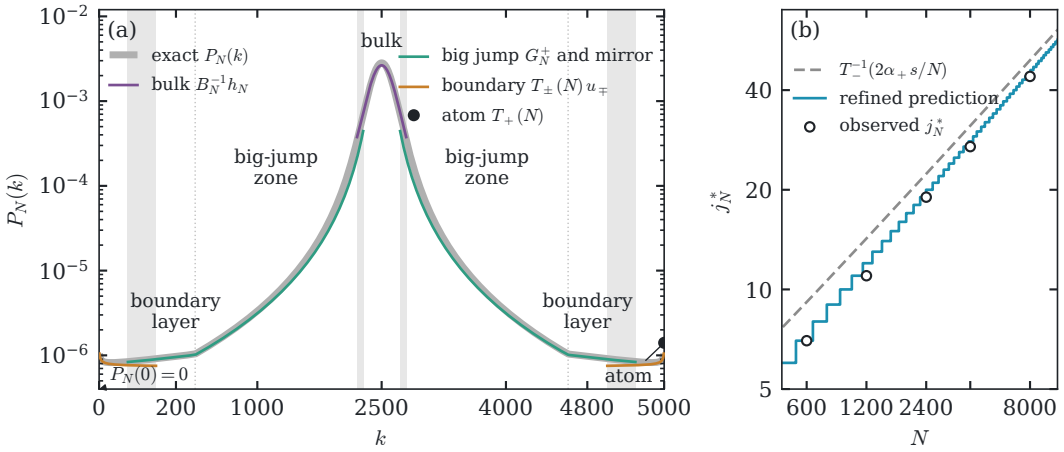


Figure 1: (a) The law $P_N(k)$ for $N = 5000$, run lengths with $f(n) \propto n^{-5/2}$ for both signs and a fresh plus start, on a logarithmic scale, with the four pieces of Theorems A and B. The two ends of the horizontal axis are stretched. (b) The position j_N^* of the interior minimum near the plus end against N, on logarithmic axes, with the law of Theorem E and the refined prediction with $C(j)$ from Theorem 6.1. Section 8 describes the computations.

Big jumps and the boundary layer. Put $L_k = (k - pN)/(1 - p)$, the length of the plus run that carries a typical configuration to the count k. Theorem B says that for a fresh plus start, under (RV) and (SPAN) for both signs and (LR) for the plus sign,

$$P_N(k) = \Big[\gamma(N-k)\, f_+(\lfloor L_k \rfloor) + T_+(\lfloor L_k \rfloor)\, u_-(N-k)\Big](1 + o(1))$$

uniformly on $pN + K\lambda_N B_N \le k \le N$, in the sense that the error tends to zero as $N \to \infty$ and then $K \to \infty$. Here $\gamma(j)$ is an explicit finite sum with $\gamma(j) \sim js/\mu_-^2$, and $\lambda_N = 1 + \log(B_N/b_{+,N})$, which equals 1 when the plus scale is the larger one. The first term is a complete giant run and the second a censored final run. At $k = N - j$ with j fixed the theorem reduces to $P_N(N-j)/T_+(N) \to u_-(j)$, which needs only (RV) for the plus sign. As j grows, $u_-(j) \to 1/\mu_-$ and the layer joins the big-jump zone continuously. The junction with the bulk holds only in the double limit. Remark 4.9 shows that neither the bulk profile nor the big-jump kernel nor their sum approximates P_N with relative error $o(1)$ over the plus side. When the plus scale is the smaller one, on the window $KB_N \le k - pN \le K\lambda_N B_N$ we prove only the bulk estimate and the lower bound $P_N(k) \ge (1 - o(1))G_N(k)$, with G_N the bracket above, and we conjecture that $P_N(k)$ is the sum of the bulk term and $G_N(k)$ there. (LR) suffices for the theorem, and Example 4.11 shows that (RV) alone does not, since the kernel reads f_+ at a single point. The sharp condition is presumably a regularity of f_+ over windows of width B_N, and we leave it open. Read at $k = vN$ the theorem gives the infinite density

$$\frac{P_N(\lfloor vN \rfloor)}{T_+(N)} \to \frac{1}{\mu_-}\Big[\alpha_+(1-\ell)\ell^{-\alpha_+-1} + \ell^{-\alpha_+}\Big], \qquad \ell = \frac{v-p}{1-p},$$

with its censored-run term (Corollary 4.8).

Preparation. Theorem C describes every preparation through the first-run decomposition of Proposition 2.4. Write $\nu_\sigma(n)$ for the probability that the first run has sign σ and length n, and $Q_\sigma(n)$ for the probability that it has sign σ and length at least n. The preparation is invisible in the bulk exactly when $B_n(\nu_+(n) + \nu_-(n)) \to 0$, and in the

boundary layer at the plus end exactly when $\nu_+(n) = o(T_+(n))$. On the big-jump zone up to the end the same condition is sharp, while on its interior only window averages of ν_+ matter, and the window condition is sharp under (LR) for both signs. At the plus end the conditions are nested, since $Q_+(n) = o(T_+(n))$ implies $\nu_+(n) = o(T_+(n))$, which implies $B_n\nu_+(n) \to 0$, the plus half of the bulk criterion. When a condition fails the theorem gives the exact correction, as a directional defect in the bulk, and as an additive term in the big-jump zone and at the end. The stationary preparation, which is the equilibrated start of the physics literature, satisfies the bulk criterion and fails the other two, and Corollary 5.4 gives its corrections in closed form. Corollary 5.5 shows that no trimming of the row by $o(N)$ bins removes the dependence on the preparation for every preparation at once.

The shape. Theorem D compares the endpoint atoms with the bulk for a preparation that satisfies the bulk criterion. An endpoint is the global mode when its atom exceeds H_N/B_N by a fixed factor, with H_N the height of the profile, and the bulk wins when both atoms are smaller by a fixed factor. For the stationary preparation with equal indices this is the golden ratio threshold. The endpoints win for $\alpha < \varphi = (1+\sqrt{5})/2$, the bulk wins for $\alpha > \varphi$, and at $\alpha = \varphi$ with pure power tails the bulk wins by the explicit factor $1/\mathcal{C}(\varphi) = 1.288\ldots$, with $\mathcal{C}$ as in (6.1).

Theorem E locates the interior minimum between the bulk and the plus end for a fresh plus start. Suppose that $u_-(j) > 1/\mu_-$ for every $j \geq 1$. This holds for every strictly log-convex law of the minus runs. Then the minimum sits at $N - j_N^*$ with

$$T_-(j_N^*) \sim \frac{2\alpha_+ s}{N}.$$

So j_N^* grows like the stable scale of the minus runs. At finite N the minimizer of $u_-(j) + C(j)/N$ predicts j_N^* better, and j_N^* is asymptotic to it. Here $C(j)$ is the exact second-order coefficient of Theorem 6.1. When u_- dips below its limit the minimum stays at bounded distance from the end. These results compare bins in different zones. They do not say that the law is monotone between the extrema, and the number of local extrema inside the layers is not determined here.

Identification. Theorem F reads the run laws off the ends. For a fresh plus start, $P_N(N-j)/P_N(N) \to u_-(j)$, and the renewal equation inverts u_- to the law of the minus runs. The minus end then gives the law of the plus runs once it is normalized by the tail of the minus runs. The normalization matters, since the ratios at an end determine a renewal mass function only up to a geometric thinning of the run law (Remark 7.1).

1.3 Related work

Infinite densities and big jumps. Rebenshtok, Denisov, Hänggi and Barkai [35, 36] derived the infinite density of a Lévy walk from the Montroll and Weiss equation, formally and in continuous time. For the two-state velocity model their density [35, Eqs. (8) and (10)] has the shape $|v|^{-1-\alpha}[\alpha - (\alpha-1)|v|]$, and the term $-(\alpha-1)|v|$ is the contribution of the censored final run, so their formula already contains it. What is new in Corollary 4.8 is a proof on the lattice at fixed time, uniform up to the end, together with the boundary layer and the preparation. In the physics model each flight chooses its direction independently, and consecutive flights in the same direction merge into one run. This gives the alternating walk with run laws thinned geometrically with parameter $\frac{1}{2}$, as in Remark 7.1. Vezzani, Barkai and Burioni [38, 39] derived the same tail from a single big jump. Wang, Schulz, Deng and Barkai [42] treat the occupation time of an alternating renewal process with one law for both states and find a boundary layer and an atom at the ends. Theorem C treats the effect of an arbitrary preparation. Wang, Höll and Barkai [41] and Akimoto [1] treat the infinite-mean case and the moments. The

golden ratio appears in Wang, Li and Hänggi [40] as the exponent at which the side peaks and the central peak decay at the same rate, with the edge exponent of Cipriani, Denisov and Politi [9], and Lee-Dadswell, Nickel and Gray [26] found the conduction exponent $(3-\sqrt{5})/2$ by a hydrodynamic argument. The threshold itself is the exponent identity $\alpha-1=1/\alpha$. What Theorem D adds is the comparison of single bins as a theorem and the constant $\mathcal{C}(\varphi)$.

Local limit theorems and renewal theory. Doney [14] proved a bivariate local limit theorem in the stable domain, and Berger [2] proved multivariate local large deviations and strong renewal theorems. His Theorem 3.3 applied to the walk of run lengths, together with the fresh-plus identity of Proposition 2.3, gives the window $|k-pN|\le KB_N$ of Theorem A. What is new there is the uniform bound C/B_N over the whole row and the control of the remote bins under (RV) alone, since Berger's bounds away from the mean direction need a local condition. Melbourne, Pène and Terhesiu [29] prove mixing local limit theorems with multivariate stable limits on Young towers, which give one bin at the stable scale with the first and the last run restricted to fixed lengths. In the i.i.d. renewal setting Theorem A removes these restrictions. We use the local large deviation bound of Caravenna and Doney [6] (see also [30]) for sums of a fixed number of variables. The local one-big-jump theorems for sums of independent variables are due to Denisov, Dieker and Shneer [11] and Berger, Birkner and Yuan [4], and the one under exactly the local condition (LR) is [3, Theorem 2.4]. The proof of Theorem B does not use them. What is new there is the occupation count at a fixed time, where a complete giant run enters with the renewal density of its start and a censored run with the boundary kernel. The second-order coefficient rests on the second-order renewal theorem of Frenk [19], see also Rogozin [37] and Doney [15], and the monotonicity of u_- on de Bruijn and Erdős [10].

The walk. The walk is the directionally reinforced random walk of Mauldin, Monticino and von Weizsäcker [28], with limit laws by Horváth and Shao [22], and the persistent random walk of Cénac, Le Ny, de Loynes and Offret, who prove functional scaling limits, a stable Lévy process for $1<\alpha<2$ [7, 8]. Magdziarz, Scheffler, Straka and Żebrowski [27] prove limit theorems for Lévy walks in continuous space and time. Theorem A is the local form of the marginal of these limits at one time, and the rest of the paper is about the bins that the scaling limits do not see. Godrèche and Luck [21] give the transform of the occupation time, which is the generating function of the fresh-plus identity, and Lamperti [25] proved the arcsine type law for infinite-mean runs.

This series. For runs with geometric lengths the comparison of the ends with the center is Problem 131 of Kagey's Open Problem Collection [24], posed for a Galton board. The walk is then the persistent random walk, whose crossings are treated in [33], and the elephant random walk with rare reversals, which has a Landau profile in place of a stable one, is treated in [34]. The present paper is the heavy-tailed member of the series.

1.4 Method and organization

Everything rests on the exact identities of Section 2, which write $P_N(k)$ through products of renewal kernels of the two signs and reduce every preparation to the two fresh laws. Sections 3 to 7 prove Theorems A to F in order, and Section 8 describes the computations.

2 Setting, hypotheses and exact identities

2.1 Runs, preparation and the occupation count

The walk takes N steps and each step has a sign, $+$ or $-$. We call the positions $1,\dots,N$ slots. The signs come in maximal runs of equal sign, and consecutive runs

have opposite signs. All runs after the first are independent. A run of sign $\sigma \in \{+,-\}$ has a length distributed as X_σ, a random variable on $\{1,2,\dots\}$ with mass function $f_\sigma(n) = \mathbb{P}(X_\sigma = n)$ and survival function

$$T_\sigma(x) = \mathbb{P}(X_\sigma \geq x), \qquad x \in \mathbb{R}.$$

We extend f_σ by zero to nonpositive integers, and we write $\bar{\sigma}$ for the sign opposite to σ.

The first run is the preparation. It has a sign Σ and a length D, and the pair (Σ, D) is independent of all later runs. We put

$$\nu_\sigma(d) = \mathbb{P}(\Sigma = \sigma,\, D = d), \qquad Q_\sigma(N) = \mathbb{P}(\Sigma = \sigma,\, D \geq N),$$

and we assume that D is finite almost surely. A fresh start of sign σ_0 is the preparation $\nu_{\sigma_0} = f_{\sigma_0}$ and $\nu_{\bar{\sigma}_0} = 0$. The stationary preparation has $\nu_\sigma = T_\sigma/s$ with s as in (2.1) below. It is the law of the current run seen from a typical time of a two-sided stationary sequence of runs, and it is the equilibrated start of the physics literature.

Let A_N be the number of plus slots among $1,\dots,N$, and

$$P_N(k) = \mathbb{P}(A_N = k), \qquad 0 \leq k \leq N.$$

The walk that moves right on a plus slot and left on a minus slot is at $2A_N - N$ after N steps. For a fresh start of sign σ we write $F_N^\sigma(k)$ for $P_N(k)$, with $F_N^\sigma(k) = 0$ for $k \notin \{0,\dots,N\}$. We also need the law of the count of the starting sign,

$$\Pi_m^+(b) = F_m^+(b), \qquad \Pi_m^-(b) = F_m^-(m-b),$$

so $\Pi_m^\sigma(b)$ is the probability that a fresh walk of sign σ has exactly b slots of its own sign among its first m slots.

2.2 Hypotheses

The hypotheses carry fixed names, and every statement says which it uses.

(RV) For each sign, T_σ is regularly varying at infinity with index $-\alpha_\sigma$, where $1 < \alpha_\sigma < 2$. We write $\alpha_* = \min(\alpha_+, \alpha_-)$.

(LR) For each sign there is $\alpha_\sigma \in (1,2)$ with $f_\sigma(n) \sim \alpha_\sigma T_\sigma(n)/n$ as $n \to \infty$.

(SPAN) For each sign, the support of X_σ is not contained in any arithmetic progression $b + h\mathbb{Z}$ with $h \geq 2$.

A hypothesis assumed "for the plus sign" is assumed for that sign only. Under (LR) we have $T_\sigma(n+1)/T_\sigma(n) = 1-(\alpha_\sigma+o(1))/n$, so the representation theorem [5, Theorem 1.3.1] shows that (LR) implies (RV) with the same index. (LR) says that the mass function is regularly varying, not only the tail. (SPAN) is the aperiodicity needed for local limit theorems. Under (RV) the runs have finite means, and we put

$$\mu_\sigma = \mathbb{E}X_\sigma, \qquad s = \mu_+ + \mu_-, \qquad p = \mu_+/s, \tag{2.1}$$

so p is the long-run fraction of plus slots and $A_N/N \to p$ in probability.

2.3 Scales and the moving profile

Under (RV) choose for each sign an increasing function $a_\sigma : (0,\infty) \to (0,\infty)$ with $t\,T_\sigma(a_\sigma(t)) \to 1$ as $t \to \infty$. So a_σ is regularly varying with index $1/\alpha_\sigma$, and $a_\sigma(m)$ is the scale of the stable fluctuation of a sum of m runs of sign σ. Put

$$b_{\sigma,N} = a_\sigma(N/s), \qquad B_N = \max(b_{+,N}, b_{-,N}), \qquad \theta_{\sigma,N} = b_{\sigma,N}/B_N \in (0,1], \tag{2.2}$$

and $x_{N,k} = (k - pN)/B_N$. The walk makes about N/s runs of each sign in N slots, so $b_{\sigma,N}$ is the fluctuation scale of the total length of the runs of sign σ, and B_N is the larger of the two. When $\alpha_+ \neq \alpha_-$ one ratio $\theta_{\sigma,N}$ tends to zero, and when the indices are equal the ratios may oscillate, so we keep them in the statements. We collect the facts about regular variation that recur.

Lemma 2.1 (regular variation)**.** *Let f be regularly varying with index ϱ.*

(i) (Uniform convergence.) The supremum of $|f(x)/f(y) - 1|$ over $x, y \geq x_0$ with $|x/y - 1| \leq r$ tends to zero as $x_0 \to \infty$ and $r \to 0$ [5, Theorems 1.2.1 and 1.5.2].

(ii) (Potter's bounds [5, Theorem 1.5.6].) For every $\delta > 0$ there is x_δ with $f(y)/f(x) \leq 2\max\{(y/x)^{\varrho+\delta}, (y/x)^{\varrho-\delta}\}$ for $x, y \geq x_\delta$. Under (RV), for every $c \geq 1$ there is C_c with $T_\sigma(x/c) \leq C_c T_\sigma(x)$ for $x \geq c$.

(iii) [5, Theorem 1.5.3] If $\varrho > 0$ and f is bounded above and away from zero on compact subsets of $[1, \infty)$, then $\sup_{1 \leq y \leq x} f(y) \leq C_f f(x)$ for $x \geq 1$.

(iv) (Karamata [5, Propositions 1.5.8 and 1.5.10].) Under (RV) there is C_{K} such that, for both signs and all $w \geq 1$,

$$\sum_{n<w} n\, T_\sigma(n) \leq C_{\mathrm{K}}\, w^2 T_\sigma(w), \qquad \sum_{n \geq w} T_\sigma(n) \leq C_{\mathrm{K}}\, w T_\sigma(w), \tag{2.3}$$

and $\sum_{n \geq w} T_\sigma(n) \sim w T_\sigma(w)/(\alpha_\sigma - 1)$ as $w \to \infty$.

(v) Under (RV), B_N is nondecreasing and regularly varying with index $1/\alpha_*$. Moreover $B_N T_\sigma(N) \to 0$, $N T_\sigma(b_{\sigma,N}) \to s$ and $B_N \log N = o(N)$.

(vi) Under (RV), for both signs, all large N and all $L \geq b_{\sigma,N}$,

$$\frac{s}{4}\Big(\frac{L}{b_{\sigma,N}}\Big)^{-\alpha_\sigma - 1} \leq N T_\sigma(L) \leq 4s\Big(\frac{L}{b_{\sigma,N}}\Big)^{-(\alpha_\sigma+1)/2}. \tag{2.4}$$

Proof. Parts (i) to (iii) are the cited theorems, and $T_\sigma(x/c) \leq C_c T_\sigma(x)$ follows from (i) for large x and from the positivity of the nonincreasing T_σ for the others. For (iv), $nT_\sigma(n)$ has index $1 - \alpha_\sigma > -1$ and T_σ has index $-\alpha_\sigma < -1$, so the cited propositions give the asymptotics, and positivity gives the bounds for all $w \geq 1$. For (v), each a_σ is increasing, so B_N is nondecreasing. If $\alpha_+ \neq \alpha_-$, the ratio $b_{+,N}/b_{-,N}$ is regularly varying with a nonzero index, so $B_N = b_{\sigma,N}$ for the sign with the smaller index and all large N. If the indices are equal, B_{cN}/B_N lies between the two ratios $b_{\sigma,cN}/b_{\sigma,N}$, which tend to c^{1/α_*}. Then $B_N T_\sigma(N)$ has index $1/\alpha_* - \alpha_\sigma < 0$, and a regularly varying function with a negative index tends to zero [5, Proposition 1.5.1]. The definition of a_σ with $t = N/s$ gives $N T_\sigma(b_{\sigma,N}) \to s$, and $B_N \log N = o(N)$ because $1/\alpha_* < 1$. For (vi), take $\delta = (\alpha_\sigma - 1)/2 \leq \frac{1}{2}$ in (ii). For large N we have $b_{\sigma,N} \geq x_\delta$ and $s/2 \leq N T_\sigma(b_{\sigma,N}) \leq 2s$, and $(L/b_{\sigma,N})^{-\alpha_\sigma - \delta} \geq (L/b_{\sigma,N})^{-\alpha_\sigma - 1}$ for $L \geq b_{\sigma,N}$. □

Let $K_\alpha = -\Gamma(1-\alpha) > 0$ for $1 < \alpha < 2$, and let Z_+ and Z_- be independent with

$$\mathbb{E}e^{itZ_\sigma} = \exp\big(K_{\alpha_\sigma}(-it)^{\alpha_\sigma}\big),$$

with the principal branch. These are the totally skewed stable laws that arise as limits of centered sums of the runs, and we write g_σ for the density of Z_σ. For $\theta = (\theta_+, \theta_-) \in [0,1]^2$ let h_θ be the density of

$$W_\theta = (1-p)\,\theta_+ Z_+ - p\,\theta_- Z_-, \tag{2.5}$$

let Θ be the compact set of $\theta \in [0,1]^2$ with $\max(\theta_+, \theta_-) = 1$, and put $\theta_N = (\theta_{+,N}, \theta_{-,N}) \in \Theta$. The moving profile is $h_N = h_{\theta_N}$. It is a stable density when $\alpha_+ = \alpha_-$, and otherwise a convolution of two stable densities with different indices. We write $H_N = \sup_x h_N(x)$ for its height. For each N let $\tau = \tau_N$ be a sign with $b_{\tau,N} = B_N$, taking $\tau = +$ if the two scales are equal. We call τ the dominant sign and $\bar\tau$ the other sign, and we put $\varsigma_+ = 1$ and $\varsigma_- = -1$.

Lemma 2.2 (the profile)**.** *Assume (RV).*

(a) There are C_1, C_2 with $\sup_N H_N \le C_1$ and $|h_N(x) - h_N(y)| \le C_2|x-y|$ for all N and x, y.

(b) $\varepsilon_h(R) = \sup_N \sup_{|x|\ge R} h_N(x)$ and $\varepsilon'_h(R) = \sup_N \int_{|x|>R} h_N(x)\,dx$ tend to zero as $R \to \infty$.

(c) The map $\theta \mapsto h_\theta$ is continuous from Θ into the bounded continuous functions with the uniform norm. So $h_N \to h_\theta$ uniformly along every subsequence with $\theta_N \to \theta$, and $\|h_{N'} - h_N\|_\infty \to 0$ whenever $N'/N \to 1$.

(d) $\inf_N H_N > 0$, and $\inf_N \min_{|x|\le K} h_N(x) > 0$ for every $K \ge 0$.

(e) For every N and $x \in \mathbb{R}$,

$$h_N(x) = \frac{s}{\mu_{\bar\tau}} \int_{\mathbb{R}} g_{\bar\tau}(y)\, g_\tau\Big(\varsigma_\tau \frac{sx}{\mu_{\bar\tau}} + \frac{\mu_\tau}{\mu_{\bar\tau}} \theta_{\bar\tau,N}\, y\Big) dy. \tag{2.6}$$

Proof. Bounds and continuity. Under (RV) the means are finite, so $0 < p < 1$. For $1 < \alpha < 2$ and the principal branch, $(\mp it)^\alpha = |t|^\alpha e^{\mp i\pi\alpha\,\mathrm{sgn}(t)/2}$, so the real parts of $(-it)^\alpha$ and $(it)^\alpha$ equal $|t|^\alpha \cos\frac{\pi\alpha}{2} < 0$, and $(-ict)^\alpha = c^\alpha(-it)^\alpha$ for $c \ge 0$. Hence W_θ has the characteristic function

$$\widehat{h}_\theta(t) = \exp\Big(K_{\alpha_+}\big((1-p)\theta_+\big)^{\alpha_+}(-it)^{\alpha_+} + K_{\alpha_-}\big(p\,\theta_-\big)^{\alpha_-}(it)^{\alpha_-}\Big),$$

which is continuous in θ for each t. With $c_\sigma = K_{\alpha_\sigma}|\cos\frac{\pi\alpha_\sigma}{2}|$ and $c = \min(c_+(1-p)^{\alpha_+}, c_-p^{\alpha_-}) > 0$, one coordinate of $\theta \in \Theta$ equals 1, so $|\widehat{h}_\theta(t)| \le \Phi(t)$, where $\Phi(t) = 1$ for $|t| < 1$ and $\Phi(t) = e^{-c|t|^{\alpha_*}}$ for $|t| \ge 1$. Both Φ and $|t|\Phi(t)$ are integrable. Fourier inversion gives the continuous density $h_\theta(x) = \frac{1}{2\pi}\int e^{-itx}\widehat{h}_\theta(t)\,dt$, with $\|h_\theta\|_\infty \le C_1 = \frac{1}{2\pi}\int\Phi$ and Lipschitz constant $C_2 = \frac{1}{2\pi}\int|t|\Phi(t)\,dt$, and h_θ vanishes at infinity by the Riemann–Lebesgue lemma. This gives (a). Also $\|h_\theta - h_{\theta'}\|_\infty \le \frac{1}{2\pi}\int|\widehat{h}_\theta - \widehat{h}_{\theta'}|$, and dominated convergence with the bound 2Φ gives the continuity in (c). If $N'/N \to 1$, then $a_\sigma(N'/s)/a_\sigma(N/s) \to 1$ by Lemma 2.1(i), so $\theta_{N'} - \theta_N \to 0$, and the uniform continuity of $\theta \mapsto h_\theta$ on the compact Θ gives the last claim of (c).

Tails. The set $\{h_\theta : \theta \in \Theta\}$ is compact. Given $\varepsilon > 0$ take a finite ε-net $h_{\theta_1}, \dots, h_{\theta_q}$ of it and R with $h_{\theta_i} < \varepsilon$ on $|x| \ge R$ for all i. Then $h_N < 2\varepsilon$ on $|x| \ge R$ for every N, so $\varepsilon_h(R) \to 0$. If $\varepsilon'_h(R) \not\to 0$, there are $\varepsilon > 0$, $R_i \to \infty$ and N_i with $\int_{|x|>R_i} h_{N_i} \ge \varepsilon$, and by compactness we may assume $\theta_{N_i} \to \theta$. Then $h_{N_i} \to h_\theta$ uniformly, so in L^1 by Scheffé's lemma, and $\int_{|x|>R_i} h_{N_i} \le \|h_{N_i} - h_\theta\|_1 + \int_{|x|>R_i} h_\theta \to 0$, a contradiction. This proves (b).

Positivity. Fix a sign, write $Z = Z_\sigma$, $g = g_\sigma$ and $\alpha = \alpha_\sigma$, and let $Z_1, Z_2, \dots$ be independent copies of Z. The scaling of the characteristic function shows that $Z_1 + \cdots + Z_n$ has the law of $n^{1/\alpha}Z$. If $Z \ge 0$ almost surely, then $\delta = \mathbb{E}\min(Z,1) > 0$, and by the strong law of large numbers $n^{-1/\alpha}(Z_1 + \cdots + Z_n) \ge n^{1-1/\alpha} \cdot \frac{1}{n}\sum_{i\le n}\min(Z_i,1) \to \infty$ almost surely, which contradicts the tightness of a sequence with the law of Z. The same holds for $-Z$. So the open set $U = \{g > 0\}$ meets both half lines. The density g^{*n} of $Z_1 + \cdots + Z_n$ is continuous and positive at $y_1 + \cdots + y_n$ whenever $y_1, \dots, y_n \in U$, and

it equals $n^{-1/\alpha} g(n^{-1/\alpha}\,\cdot)$. Hence $y_1, \dots, y_n \in U$ implies $n^{-1/\alpha}(y_1 + \cdots + y_n) \in U$. Take $I = (a-\varepsilon, a+\varepsilon) \subset U$ with $0 < \varepsilon < a$ and put $c_n = n^{1-1/\alpha}$. Then $c_n I \subset U$ for every n. Since $c_n \to \infty$ and $c_{n+1}/c_n \to 1$, the intervals $c_n I$ with $n \geq n_0$ overlap in turn, so U contains a half line (x_0, ∞). In the same way U contains $(-\infty, -x_0)$. Every real number is the sum of a point of each half line, so the case $n = 2$ gives $U = \mathbb{R}$. Now let $\theta \in \Theta$ and $\theta_\sigma = 1$. Then $W_\theta = aZ_\sigma + Y$ with $a \in \{1-p, -p\}$ and Y independent of Z_σ, so $x \mapsto \mathbb{E}[|a|^{-1} g_\sigma((x-Y)/a)]$ is a continuous positive density of W_θ, and it equals h_θ. The maps $\theta \mapsto \sup_x h_\theta(x)$ and $\theta \mapsto \min_{|x| \leq K} h_\theta(x)$ are continuous and positive on the compact Θ, so their minima are positive. This proves (d).

The formula. Let $\tau = +$. Given $Z_- = y$, the variable $(1-p)Z_+ - p\theta_{-,N} Z_-$ has the density $(1-p)^{-1} g_+((x + p\theta_{-,N} y)/(1-p))$ at x. Since $1-p = \mu_-/s$ and $p/(1-p) = \mu_+/\mu_-$, integrating against $g_-(y)\,dy$ gives (2.6). For $\tau = -$ we condition on Z_+. □

2.4 Renewal kernels

For each sign let u_σ be the renewal mass function of the runs of sign σ,

$$u_\sigma(0) = 1, \qquad u_\sigma(j) = \sum_{m \geq 1} f_\sigma^{*m}(j) \quad (j \geq 1),$$

the probability that some partial sum of independent copies of X_σ equals j. We also need

$$v_\sigma(j) = \sum_{m \geq 1} m\, f_\sigma^{*m}(j),$$

and the iterated kernels $u_m^\sigma = f_\sigma^{*m}$ and $q_m^\sigma = T_\sigma * u_{m-1}^\sigma$ for $m \geq 1$, with $u_0^\sigma = \delta_0$. Let $X_1^\sigma, X_2^\sigma, \dots$ be independent copies of X_σ and $S_m^\sigma = X_1^\sigma + \cdots + X_m^\sigma$, with $S_0^\sigma = 0$. Then S_m^σ has the law u_m^σ, and since X_m^σ is independent of S_{m-1}^σ, for $m, k \geq 1$

$$\mathbb{P}\big(S_{m-1}^\sigma < k \leq S_m^\sigma\big) = \sum_{r<k} u_{m-1}^\sigma(r)\, T_\sigma(k-r) = q_m^\sigma(k). \tag{2.7}$$

So $q_m^\sigma(k)$ is the probability that the m-th run covers level k in the time of its sign. Every run has length at least 1, so S_m^σ is strictly increasing in m, and for each $k \geq 1$ exactly one $m \geq 1$ has $S_{m-1}^\sigma < k \leq S_m^\sigma$. Hence, for $k \geq 1$,

$$\sum_{m \geq 1} u_m^\sigma(k) = u_\sigma(k), \qquad \sum_{m \geq 1} q_m^\sigma(k) = 1, \qquad \sum_{m \geq 1} q_{m+1}^\sigma(k) = 1 - T_\sigma(k), \tag{2.8}$$

where the third sum is the second minus its term $q_1^\sigma = T_\sigma$. Summing the middle expression of (2.7) over m gives the covering identity

$$\sum_{m=1}^{k} T_\sigma(m)\, u_\sigma(k-m) = 1 \qquad (k \geq 1). \tag{2.9}$$

Under (RV) and (SPAN) for the sign σ the renewal theorem holds,

$$u_\sigma(j) \to 1/\mu_\sigma \qquad (j \to \infty), \tag{2.10}$$

and we prove it in local form in Lemma 3.4 (see also [17, Section XIII.3]).

The boundary kernel at the plus end records where a plus run can begin. For a start of sign σ_0 let

$$\kappa_+^{(\sigma_0)}(j) = \mathbb{P}\big(\text{a plus run starts immediately after the } j\text{-th minus slot}\big),$$

where for $j = 0$ the event is that a plus run starts at slot 1. Count time in minus slots only. In both cases the ends of the minus runs form a renewal process with interarrival

law X_-, started at 0. A plus run starts right after the j-th minus slot exactly when j is a renewal epoch, with one exception. For a minus start the epoch $j = 0$ is followed by the first minus run. Hence

$$\kappa_+^{(+)}(j) = u_-(j) \quad (j \geq 0), \qquad \kappa_+^{(-)}(j) = u_-(j)\mathbf{1}_{\{j\geq 1\}}. \tag{2.11}$$

The kernel at the minus end is defined in the same way with the signs exchanged. Every statement below is made at the plus end and holds at the minus end with the signs exchanged.

We refine $\kappa_+^{(\sigma_0)}(i)$ by the number of plus slots before the run. For $i \geq 0$ and $x \geq 0$ let $\rho_i^{(\sigma_0)}(x)$ be the probability that a plus run starts immediately after the i-th minus slot (at slot 1 if $i = 0$) and that exactly x plus slots precede it. At most one x is possible for each realization, so

$$\sum_{x\geq 0} \rho_i^{(\sigma_0)}(x) = \kappa_+^{(\sigma_0)}(i). \tag{2.12}$$

A plus run that starts after the i-th minus slot is preceded by m complete minus runs with $S_m^- = i$, and by m plus runs for a plus start and $m - 1$ plus runs for a minus start. The two families of runs are independent, so

$$\rho_i^{(+)}(x) = \sum_{m\geq 0} u_m^-(i)\, u_m^+(x), \qquad \rho_i^{(-)}(x) = \sum_{m\geq 1} u_m^-(i)\, u_{m-1}^+(x). \tag{2.13}$$

2.5 Exact identities

The following identities hold for every N and need no hypothesis.

Proposition 2.3 (the fresh-plus identity)**.** *For $k, j \geq 1$ with $k + j = N$,*

$$F_N^+(k) = \sum_{m\geq 1} \big[u_m^-(j)\, q_{m+1}^+(k) + u_m^+(k)\, q_m^-(j)\big],$$

and $F_N^+(N) = T_+(N)$, $F_N^+(0) = 0$. Exchanging the signs gives the fresh-minus identity $F_N^-(k) = \sum_m [u_m^+(k) q_{m+1}^-(j) + u_m^-(j) q_m^+(k)]$ for $k, j \geq 1$, with $F_N^-(0) = T_-(N)$ and $F_N^-(N) = 0$.

Proof. We split according to the run that contains slot N. For a fresh plus start the run lengths $X_1^+, X_2^+, \dots$ and $X_1^-, X_2^-, \dots$ are independent, and the runs alternate starting with a plus run. So the i-th plus run fills the slots in $(S_{i-1}^+ + S_{i-1}^-,\, S_i^+ + S_{i-1}^-]$, the i-th minus run fills those in $(S_i^+ + S_{i-1}^-,\, S_i^+ + S_i^-]$, and these intervals partition $\{1, 2, \dots\}$. Let $k + j = N$ with $0 \leq k \leq N$.

If slot N lies in the i-th plus run, the first N slots contain exactly the S_{i-1}^- minus slots of the first $i - 1$ minus runs, so $A_N = k$ means $S_{i-1}^- = j$, and then slot $N = k + j$ lies in that run exactly when $S_{i-1}^+ < k \leq S_i^+$. The two events concern independent sequences, so by (2.7) this case has probability $u_{i-1}^-(j)\, q_i^+(k)$. If slot N lies in the i-th minus run, then $A_N = S_i^+$, so $A_N = k$ means $S_i^+ = k$, and slot N lies in that run exactly when $S_{i-1}^- < j \leq S_i^-$. This case has probability $u_i^+(k)\, q_i^-(j)$.

The cases are disjoint. For $k, j \geq 1$ the plus case with $i = 1$ vanishes because $u_0^-(j) = 0$, and $m = i - 1$ in the plus case and $m = i$ in the minus case give the formula. For $k = N$ only the plus case with $i = 1$ survives, with probability $q_1^+(N) = T_+(N)$. For $k = 0$ the plus case is impossible because slot N is a plus slot, and the minus case is impossible because $S_i^+ \geq 1$. The fresh minus start is the same with the signs exchanged. □

Proposition 2.4 (the first-run decomposition)**.** *For every preparation and every* $0 \le k \le N$*,*

$$P_N(k) = Q_+(N)\mathbf{1}_{\{k=N\}} + Q_-(N)\mathbf{1}_{\{k=0\}} + \sum_{d<N} \nu_+(d)\, F^-_{N-d}(k-d) + \sum_{d<N} \nu_-(d)\, F^+_{N-d}(k).$$

In particular $P_N(N) = Q_+(N)$ *and* $P_N(0) = Q_-(N)$*.*

Proof. We condition on the first run. If $\Sigma = +$ and $D \ge N$, then $A_N = N$, and this has probability $Q_+(N)$. If $\Sigma = +$ and $D = d < N$, the first d slots are plus slots and slots $d+1, \dots, N$ are the first $N-d$ slots of a fresh walk of sign $-$, independent of (Σ, D). So $A_N = d + A'$, where A' has the law F^-_{N-d}. The case $\Sigma = -$ is the same with $A_N = 0$ when $D \ge N$ and A_N of law F^+_{N-d} when $D = d < N$. This gives the formula. For $d < N$, Proposition 2.3 gives $F^-_{N-d}(N-d) = 0$ and $F^+_{N-d}(0) = 0$, and $F^+_{N-d}(N) = F^-_{N-d}(-d) = 0$ trivially, so only the atoms survive at $k = N$ and $k = 0$. □

Lemma 2.5 (the Green identity)**.** *For each sign and every* $b \ge 1$*,*

$$\sum_{m \ge 1} \Pi^\sigma_m(b) = 1 + \mu_{\bar\sigma}\, u_\sigma(b).$$

Proof. Take a fresh walk of sign σ continued forever, with runs of sign σ of lengths $X^\sigma_1, X^\sigma_2, \dots$ and runs of sign $\bar\sigma$ of lengths $X^{\bar\sigma}_1, X^{\bar\sigma}_2, \dots$, all independent, the i-th run of sign $\bar\sigma$ coming right after the i-th run of sign σ. Let C_m be the number of slots of sign σ among the first m. By Tonelli's theorem $\sum_m \Pi^\sigma_m(b) = \mathbb{E}\,\#\{m \ge 1 : C_m = b\}$. Let $t_b < t_{b+1}$ be the b-th and $(b+1)$-th slots of sign σ. Then $C_m = b$ exactly for $t_b \le m < t_{b+1}$. If slot t_b is not the last slot of its run, then $t_{b+1} = t_b + 1$. If it is, it ends the i-th run of sign σ for the unique i with $S^\sigma_i = b$, and then $t_{b+1} - t_b = 1 + X^{\bar\sigma}_i$. Hence $\#\{m : C_m = b\} = 1 + \sum_{i \ge 1} \mathbf{1}_{\{S^\sigma_i = b\}} X^{\bar\sigma}_i$. Since $X^{\bar\sigma}_i$ is independent of S^σ_i, taking expectations and using (2.8) gives $1 + \mu_{\bar\sigma} u_\sigma(b)$. □

The next identity isolates the final run, which is the run that contains slot N.

Proposition 2.6 (the final-run identity)**.** *For a fresh plus start,* $N \ge 1$ *and* $0 \le j < N$*,*

$$\begin{aligned} P_N(N-j) = &\sum_{r \ge 0} u^-_r(j)\, \mathbb{E}\big[T_+(N-j-S^+_r);\, S^+_r \le N-j-1\big] \\ &+ \sum_{m=1}^{j} T_-(m) \sum_{r \ge 0} u^-_r(j-m)\, u^+_{r+1}(N-j). \end{aligned} \tag{2.14}$$

The first sum is the event that the final run is a plus run, and the second the event that it is a minus run of length at least m *that starts at slot* $N-m+1$*. Only* $r \le j$ *contributes, and for* $j = 0$ *the identity reads* $P_N(N) = T_+(N)$*.*

Proof. We use the notation of the proof of Proposition 2.3. The $(r+1)$-th plus run starts at slot $S^+_r + S^-_r + 1$ with S^+_r plus slots before it. It is the final run and $A_N = N - j$ exactly when the slots before it contain j minus slots, that is $S^-_r = j$, and its length satisfies $X^+_{r+1} \ge N - j - S^+_r \ge 1$. These conditions involve independent variables, which gives the term r of the first sum. The $(r+1)$-th minus run starts at slot $S^+_{r+1} + S^-_r + 1$ with S^+_{r+1} plus slots before it. If it is the final run and starts at slot $N-m+1$, then $A_N = N - j$ means $S^+_{r+1} = N - j$, hence $S^-_r = N - m - (N-j) = j - m$, which needs $1 \le m \le j$, and its length satisfies $X^-_{r+1} \ge m$. This gives the second sum. Exactly one run is final, so the cases are disjoint. Finally $u^-_r(i) = 0$ for $r > i$ and $u^-_r(0) = \mathbf{1}_{\{r=0\}}$. □

By (2.13) the first sum of (2.14) equals $\sum_{x \le N-j-1} \rho^{(+)}_j(x) T_+(N-j-x)$. Section 8 describes how these identities compute the exact law.

Lemma 2.7 (the counting lemma)**.** *Fix $j \geq 0$ and $N > j$. On the event $\{A_N = N - j\}$ at most $j + 1$ plus runs meet the first N slots (at most j if the first run is a minus run), and at least one of them has at least $(N - j)/(j + 1)$ slots among the first N. For a fresh start of either sign and every $L \geq 1$, the probability that $A_N = N - j$ and two plus runs that meet the first N slots have length at least L is at most $(j + 1)^2 T_+(L)^2$.*

Proof. On $\{A_N = N - j\}$ the first N slots contain j minus slots. Let r be the number of plus runs that meet them. Between two consecutive such runs lies a minus run inside the first N slots, so $r - 1 \leq j$, and if the first run is a minus run it adds one more minus slot, so $r \leq j$. The $N - j$ plus slots are shared among $r \leq j + 1$ runs, so one of them has at least $(N - j)/(j + 1)$ slots. For a fresh start the plus runs have independent lengths $X_1^+, X_2^+, \ldots$, and on the event of the lemma there are $1 \leq a < b \leq j + 1$ with $X_a^+ \geq L$ and $X_b^+ \geq L$. The union bound gives at most $\binom{j+1}{2} T_+(L)^2 \leq (j + 1)^2 T_+(L)^2$, and counting only the slots among the first N makes the event smaller. □

3 The bulk

The bulk is the window $|k - pN| = O(B_N)$, where the law of A_N is stable. The theorem of this section is the whole-row statement for a fresh start. Outside the bulk it says only that every bin is $o(1/B_N)$, and Sections 4 and 5 refine the outside and add the preparation.

Theorem A (the bulk, fresh start)**.** *Assume (RV) and (SPAN). For each starting sign σ_0,*

(a) $\sup_{k \in \mathbb{Z}} \big| B_N F_N^{\sigma_0}(k) - h_N(x_{N,k}) \big| \to 0$.

(b) $\sup_{N \geq 1} \sup_k B_N F_N^{\sigma_0}(k) < \infty$.

(c) for every $K > 0$, $\displaystyle\sup_{|k - pN| \leq K B_N} \Big| \frac{B_N F_N^{\sigma_0}(k)}{h_N(x_{N,k})} - 1 \Big| \to 0$.

The convergence covers every relation between the two scales $b_{+,N}$ and $b_{-,N}$, including ratios that oscillate.

We prove (a) in Section 3.3, and (b) and (c) at its end.

3.1 Classical inputs

We use four facts about the sums S_m^σ, stated with the normalization of Section 2.3. We write $\bar{T}_\sigma(x) = \mathbb{P}(X_\sigma > x)$, which is regularly varying with $\bar{T}_\sigma(x) \sim T_\sigma(x)$ under (RV).

Lemma 3.1 (stable and local limits)**.** *Assume (RV) and (SPAN) for the sign σ, and write $\alpha = \alpha_\sigma$, $\mu = \mu_\sigma$, $a = a_\sigma$, $g = g_\sigma$, $T = T_\sigma$ and $S_m = S_m^\sigma$. Parts (i), (iii) and (iv) need only (RV).*

(i) $Z_m = (S_m - m\mu)/a(m)$ converges in law to Z_σ, whose density g is bounded, uniformly continuous, and tends to zero at infinity. Moreover $m\big(\mathbb{E}e^{-(X_\sigma - \mu)/a(m)} - 1\big) \to K_\alpha$.

(ii) (Gnedenko's local limit theorem) $\sup_{k \in \mathbb{Z}} \big| a(m)\mathbb{P}(S_m = k) - g\big((k - m\mu)/a(m)\big) \big| \to 0$ *as $m \to \infty$, and $\bar{u}_\sigma = \sup_{m \geq 1} \sup_k a(m)\mathbb{P}(S_m = k) < \infty$.*

(iii) (local large deviations) There is a constant $C_{\mathrm{LD}} = C_{\mathrm{LD}}(\sigma)$ such that $\mathbb{P}(S_m = j) \leq C_{\mathrm{LD}}\, m\, T(j - m\mu)/a(m)$ for all $m \geq 1$ and all integers j with $j - m\mu \geq 1$.

(iv) (left deviations) The constant $C_- = C_-(\sigma) = \sup_{m \geq 1} m \log \mathbb{E}e^{-(X_\sigma - \mu)/a(m)}$ is finite, and $\mathbb{P}(S_m \leq m\mu - y) \leq e^{C_-}\, e^{-y/a(m)}$ for all $m \geq 1$ and $y \geq 0$.

Proof. (i) The domain of attraction is classical [18, Section XVII.5], and we check the normalization, since the constant K_α is ours. Let $X = X_\sigma$ and, for $\operatorname{Re}\xi \ge 0$, $\Lambda(\xi) = \mathbb{E}[e^{-\xi X} - 1 + \xi X]$. Since $e^{-\xi x} - 1 + \xi x = \int_0^x \xi(1 - e^{-\xi y})\,dy$ for $x \ge 0$, Fubini's theorem gives, with $\xi = te^{i\vartheta}$, $t > 0$ and $|\vartheta| \le \pi/2$,

$$\Lambda(\xi) = \int_0^\infty \xi\bigl(1 - e^{-\xi y}\bigr)\bar T(y)\,dy = \int_0^\infty e^{i\vartheta}\bigl(1 - e^{-e^{i\vartheta}v}\bigr)\bar T(v/t)\,dv.$$

The integrand is at most $\min(2, v)\bar T(v/t)$ in modulus. As $t \to 0$, $\bar T(v/t)/\bar T(1/t) \to v^{-\alpha}$, and Lemma 2.1(ii) dominates the ratio by $2\max(v^{-\alpha-\delta}, v^{-\alpha+\delta})$ for $v \ge tx_\delta$, with $1 < \alpha \pm \delta < 2$. The part $v < tx_\delta$ contributes at most $t^2 x_\delta^2/\bar T(1/t) \to 0$ to $\Lambda(\xi)/\bar T(1/t)$, because $\alpha < 2$. Hence, uniformly in ϑ,

$$\Lambda(te^{i\vartheta}) = \bar T(1/t)\bigl(F(e^{i\vartheta}) + o(1)\bigr), \qquad F(\xi) = \int_0^\infty \xi\bigl(1 - e^{-\xi v}\bigr)v^{-\alpha}\,dv.$$

The function F is continuous on $\operatorname{Re}\xi \ge 0$, analytic inside, and $F(t) = t^\alpha F(1)$ for $t > 0$. An integration by parts gives $F(1) = \Gamma(2-\alpha)/(\alpha - 1) = K_\alpha$, so $F(\xi) = K_\alpha \xi^\alpha$ with the principal branch. Next $\mathbb{E}e^{-\xi(X-\mu)} - 1 = \Lambda(\xi) + O(|\xi|^2)$ as $\xi \to 0$. With $\xi = -i\theta/a(m)$, $m\bar T(a(m)/|\theta|) \to |\theta|^\alpha$ and $m/a(m)^2 \to 0$ give $m(\mathbb{E}e^{i\theta(X-\mu)/a(m)} - 1) \to K_\alpha(-i\theta)^\alpha$, and Lévy's continuity theorem gives $Z_m \to Z_\sigma$. The characteristic function of Z_σ has modulus $\exp(K_\alpha \cos(\pi\alpha/2)|\theta|^\alpha)$, which is integrable, so Fourier inversion and the Riemann–Lebesgue lemma give the properties of g. With $\xi = 1/a(m)$ the same expansion gives the last claim.

(ii) The support of $X_\sigma - \mu$ lies in $\mathbb{Z} - \mu$, and by (SPAN) its maximal lattice span is 1. Given (i), Gnedenko's local limit theorem [23, Theorem 4.2.1] (see also [5, Theorem 8.4.1]) gives the uniform limit. The norming sequence there may be any $a_m \sim a(m)$, since g is bounded, uniformly continuous and tends to zero at infinity. For $\bar u_\sigma$, the limit gives an m_0 with $a(m)\mathbb{P}(S_m = k) \le \|g\|_\infty + 1$ for $m \ge m_0$, and $a(m)\mathbb{P}(S_m = k) \le a(m_0)$ for $m < m_0$.

(iii) Caravenna and Doney [6, Theorem 1.1] assume right and left tails $p/A(x)$ and $q/A(x)$ (or $o(1/A(x))$) with A regularly varying of index in $(0,1) \cup (1,2)$, $p > 0$, and mean zero when the index exceeds one. Their bound (1.6) then holds with no lattice assumption, for all $m \ge 1$, $x \ge 0$ and bounded intervals $\mathcal{I}$, and reads $\mathbb{P}(S_m - m\mu \in x + \mathcal{I}) \le C_0'(\mathcal{I})\,m/(a_m A(x))$ with $a_m = A^{-1}(m)$. Under (RV) the variable $X_\sigma - \mu$ satisfies these hypotheses with $p = 1$, $q = 0$ and $A(x) \sim 1/T(x)$. We take $\mathcal{I} = (-1, 0]$ and $x = j - m\mu$, so that the event contains $\{S_m = j\}$. The asymptotic inverses a_m and $a(m)$ satisfy $a_m/a(m) \to 1$ [5, Theorem 1.5.12], so $a(m) \le c\,a_m$ for all m, and $1/A(x) \le cT(x)$ for $x \ge 1$. This gives (iii).

(iv) By (i), $m(\mathbb{E}e^{-(X_\sigma-\mu)/a(m)} - 1)$ is bounded, and $\log(1+z) \le z$ makes C_- finite. Markov's inequality for $e^{-(S_m - m\mu)/a(m)}$ and independence give $\mathbb{P}(S_m - m\mu \le -y) \le e^{-y/a(m)}(\mathbb{E}e^{-(X_\sigma-\mu)/a(m)})^m \le e^{C_- - y/a(m)}$. This is the exponential bound behind the Fuk–Nagaev inequalities [31]. □

3.2 Kernels of one sign

For each sign σ the fresh identities use three kernels,

$$\mathsf{k}_m^{\sigma,1} = u_m^\sigma, \qquad \mathsf{k}_m^{\sigma,2} = q_m^\sigma, \qquad \mathsf{k}_m^{\sigma,3} = q_{m+1}^\sigma \qquad (m \ge 1),$$

with weights $c_{\mathsf{k}^{\sigma,1}} = 1$ and $c_{\mathsf{k}^{\sigma,2}} = c_{\mathsf{k}^{\sigma,3}} = \mu_\sigma$. A kernel of sign σ is one of these three sequences, and we write k_m and c_k when the choice does not matter. We use four facts.

(K1) $\sum_{m\ge1} u_m^\sigma(j) \le 1$, $\sum_{m\ge1} q_m^\sigma(j) = 1$ and $\sum_{m\ge1} q_{m+1}^\sigma(j) \le 1$ for $j \ge 1$, by (2.8).

(K2) $\mathsf{k}_m(j) = 0$ for $m > j$, because $S^\sigma_m \geq m$.

(K3) Let $J_\sigma(j) = \min\{m \geq 1 : S^\sigma_m \geq j\}$. Then $\mathbb{P}(J_\sigma(j) = m) = q^\sigma_m(j)$ by (2.7), and $u^\sigma_m(j) \leq q^\sigma_m(j)$, since $\{S^\sigma_m = j\} \subset \{J_\sigma(j) = m\}$.

(K4) $\sum_j q^\sigma_m(j) = \mu_\sigma$ and $\sum_j u^\sigma_m(j) = 1$.

Lemma 3.2 (kernel local limits)**.** *Assume (RV) and (SPAN). For each sign σ and each kernel k of sign σ,*

$$\delta_{\mathsf{k}}(m) = \sup_{j\in\mathbb{Z}} \big|a_\sigma(m)\mathsf{k}_m(j) - c_{\mathsf{k}}\, g_\sigma\big((j - m\mu_\sigma)/a_\sigma(m)\big)\big| \to 0 \qquad (m \to \infty),$$

and $\mathsf{k}_m(j) \leq C_3/a_\sigma(m)$ for all $m \geq 1$ and j, where $C_3 = C_3(\sigma) = \max\big(a_\sigma(1),\, \mu_\sigma r_\sigma \bar{u}_\sigma\big)$ and $r_\sigma = \sup_{m\geq 2} a_\sigma(m)/a_\sigma(m-1) < \infty$.

Proof. For u^σ_m this is Lemma 3.1(ii). The kernel q^σ_m is a sum of shifts of u^σ_{m-1} with weights $T_\sigma(v)$ of total mass μ_σ, and q^σ_{m+1} is the same sum built on u^σ_m. This gives the bounds, with $q^\sigma_1 = T_\sigma \leq 1$ at $m = 1$. For the limit of q^σ_m write $a = a_\sigma$, $g = g_\sigma$, $z = (j - m\mu)/a(m)$, and fix $v_0 \geq 1$ and $Z > 0$. The shifts $v > v_0$ contribute at most $(\bar{u}_\sigma + \|g\|_\infty)\sum_{v>v_0} T(v)$ to $|a(m-1)q_m(j) - \mu g(z)|$. For $v \leq v_0$ the point $(j - v - (m-1)\mu)/a(m-1) = z\,a(m)/a(m-1) + (\mu - v)/a(m-1)$ lies within $\epsilon_m = Z|a(m)/a(m-1) - 1| + (v_0 + \mu)/a(m-1)$ of z when $|z| \leq Z$, and has modulus at least $Z/2$ for large m when $|z| > Z$. So for large m

$$\big|a(m-1)q_m(j) - \mu g(z)\big| \leq \mu\big(\delta_u(m-1) + \omega_g(\epsilon_m) + \bar{g}(Z/2)\big) + (\bar{u}_\sigma + \|g\|_\infty)\sum_{v>v_0} T(v),$$

where ω_g is the modulus of continuity of g and $\bar{g}(r) = \sup_{|y|\geq r} g(y)$. Let $m \to \infty$, then $v_0, Z \to \infty$, and use $a(m)/a(m-1) \to 1$. The kernel q^σ_{m+1} is the same with the shift $-v/a(m)$ in place of $(\mu - v)/a(m-1)$. □

Lemma 3.3 (small counts)**.** *Assume (RV). For each sign σ there is $C_4 = C_4(\sigma)$ such that every kernel of sign σ satisfies*

$$\mathsf{k}_m(j) \leq C_4\big(1 + m/a_\sigma(m)\big)T_\sigma(j) \qquad \textit{whenever } 8m\mu_\sigma \leq j.$$

Consequently, for $0 < \beta \leq 1$ there is $C_5 = C_5(\sigma, \beta)$ with

$$\mathsf{k}_m(j) \leq C_5\, e_{\sigma,N}, \qquad e_{\sigma,N} = \big(1 + N/a_\sigma(N)\big)T_\sigma(N), \tag{3.1}$$

for all $N \geq 1$, all $\beta N \leq j \leq N$ and all m with $8m\mu_\sigma \leq \beta N$. Moreover $B_N e_{\sigma,N} \to 0$.

Proof. Write $T = T_\sigma$, $a = a_\sigma$, $\mu = \mu_\sigma$, and let C_T be the constant of Lemma 2.1(ii) with $T(x/2) \leq C_T T(x)$. If $4m\mu \leq i$, then $i - m\mu \geq i/2 \geq 1$, and Lemma 3.1(iii) gives

$$u_m(i) \leq C_{\mathrm{LD}}\, m\, T(i/2)/a(m) \leq C_{\mathrm{LD}} C_T\, m\, T(i)/a(m). \tag{3.2}$$

This is the claim for u_m. Let $8m\mu \leq j$. For $q_1 = T$ there is nothing to prove. For q_m with $m \geq 2$ we split $\sum_v T(v)u_{m-1}(j - v)$ at $v = j/2$. If $v \leq j/2$, then $j - v \geq j/2 \geq 4(m-1)\mu$, so (3.2) bounds $u_{m-1}(j - v)$ by $C_{\mathrm{LD}}C_T^2 r_\sigma\, m\, T(j)/a(m)$, and the weights $T(v)$ add up to at most μ. If $v > j/2$, then $T(v) \leq C_T T(j)$, and the values $u_{m-1}(j - v)$ add up to at most 1 by (K4). So $q_m(j) \leq (\mu C_{\mathrm{LD}} C_T^2 r_\sigma\, m/a(m) + C_T)T(j)$. The kernel q_{m+1} is the same with u_m in place of u_{m-1}. This proves the first claim with $C_4 = C_T + \mu C_{\mathrm{LD}} C_T^2 r_\sigma$.

For (3.1), $m/a(m)$ has the positive index $1 - 1/\alpha_\sigma$, so Lemma 2.1(iii) gives $m/a(m) \leq C_P N/a(N)$ for $m \leq N$, and $T(j) \leq T(\beta N) \leq C_{T,\beta} T(N)$ by Lemma 2.1(ii). Finally $B_N N T_\sigma(N)/a_\sigma(N)$ has the index

$$\frac{1}{\alpha_*} - \frac{1}{\alpha_\sigma} + 1 - \alpha_\sigma \leq 1 - \alpha_* < 0,$$

because $1/\alpha_* - 1/\alpha_\sigma = (\alpha_\sigma - \alpha_*)/(\alpha_*\alpha_\sigma) \le \alpha_\sigma - \alpha_*$, and $B_N T_\sigma(N) \to 0$ by Lemma 2.1(v). □

The next lemma is the renewal theorem in local form, with the index of the run that covers a level resolved on its natural scale. For $j \ge 1$ and $K > 0$ let

$$I_j^\sigma(K) = \big\{m \ge 1 :\ |j - m\mu_\sigma| \le K\, a_\sigma(j/\mu_\sigma)\big\}, \qquad y_m^\sigma(j) = \frac{j - m\mu_\sigma}{a_\sigma(j/\mu_\sigma)}.$$

We call $I_j^\sigma(K)$ the index window. For a modulus ω (a function with $\omega(r) \to 0$ as $r \to 0$) let $\mathcal{F}(\omega)$ be the set of functions ψ on $\mathbb{R}$ with $|\psi| \le 1$ and $|\psi(y) - \psi(y')| \le \omega(|y - y'|)$.

Lemma 3.4 (the index of the covering run)**.** *Assume (RV) and (SPAN), and let* k *be a kernel of sign* σ*.*

(a) For every $K \ge 2$*,*

$$\limsup_{j\to\infty} \sum_{m \notin I_j^\sigma(K)} \mathsf{k}_m(j) \le \eta_\sigma(K) = \mathbb{P}(Z_\sigma \ge K - 1) + e^{C_-(\sigma)+1-K/2}.$$

(b) For every modulus ω*,*

$$\sup_{\psi\in\mathcal{F}(\omega)} \Big| \sum_{m\ge 1} \mathsf{k}_m(j)\, \psi\big(y_m^\sigma(j)\big) - \frac{c_\mathsf{k}}{\mu_\sigma} \int_{\mathbb{R}} g_\sigma(y)\psi(y)\, dy \Big| \to 0 \qquad (j \to \infty).$$

In particular $\sum_m u_m^\sigma(j) \to 1/\mu_\sigma$*, which is* (2.10)*.*

Proof. Write $\mu = \mu_\sigma$, $a = a_\sigma$, $J = J_\sigma(j)$ and $m_\pm = (j \pm Ka(j/\mu))/\mu$. Note that $a(x)/x \to 0$.

(a) Write $I = I_j^\sigma(K)$. By (K3), $\sum_{m\notin I} u_m(j)$ and $\sum_{m\notin I} q_m(j)$ are at most $\mathbb{P}(J \notin I)$, and $\sum_{m\notin I} q_{m+1}(j) \le \mathbb{P}(J - 1 \notin I)$. All three are at most $\mathbb{P}(J \le m_- + 1) + \mathbb{P}(J > m_+)$. A count of runs below its typical value forces the sum of the runs above its mean, a right deviation, and a count above its typical value forces the sum below its mean, a left deviation.

Too few runs. Let $m' = \lfloor m_- + 1 \rfloor$. If $J \le m'$, then $S_{m'} \ge j$. Now $j - m'\mu \ge Ka(j/\mu) - \mu$, and $a(m') \le a(j/\mu)$ for large j. So

$$\mathbb{P}(J \le m_- + 1) \le \mathbb{P}\Big(\frac{S_{m'} - m'\mu}{a(m')} \ge K - \frac{\mu}{a(j/\mu)}\Big) \le \mathbb{P}(Z_{m'} \ge K - 1)$$

for large j. Since $m' \to \infty$, Lemma 3.1(i) gives $\limsup_j \mathbb{P}(Z_{m'} \ge K - 1) \le \mathbb{P}(Z_\sigma \ge K - 1)$.

Too many runs. Let $m'' = \lfloor m_+ \rfloor$. If $J > m_+$, then $S_{m''} < j$, and $m''\mu - j \ge Ka(j/\mu) - \mu$. The local bound Lemma 3.1(iii) does not apply to this left deviation, and Lemma 3.1(iv) gives

$$\mathbb{P}(J > m_+) \le \exp\Big(C_- - \frac{Ka(j/\mu) - \mu}{a(m'')}\Big) \le e^{C_-+1-K/2}$$

for large j, since $m'' \le 2j/\mu$, $a(2j/\mu) \le 2a(j/\mu)$ and $\mu/a(j/\mu) \le 1$ eventually.

(b) Fix $K \ge 2$. The terms with $m \notin I_j^\sigma(K)$ contribute at most $\eta_\sigma(K) + o(1)$ by (a), and $\int_{|y|>K} g_\sigma \le \mathbb{P}(|Z_\sigma| > K)$. For $m \in I_j^\sigma(K)$ we have $m\mu/j \to 1$ uniformly, so $a(m)/a(j/\mu) \to 1$ uniformly. With $y_m = y_m^\sigma(j) \in [-K, K]$ the argument $(j - m\mu)/a(m) = y_m\, a(j/\mu)/a(m)$ differs from y_m by at most $K|a(j/\mu)/a(m) - 1|$. Lemma 3.2 and the uniform continuity of g_σ give, uniformly in $m \in I_j^\sigma(K)$,

$$\mathsf{k}_m(j) = \frac{c_\mathsf{k}\, g_\sigma(y_m) + o(1)}{a(j/\mu)}.$$

The points y_m form an arithmetic progression with step $\Delta_j = \mu/a(j/\mu) \to 0$ in $[-K, K]$, with at most $2K/\Delta_j + 1$ points, so the $o(1)$ terms add up to $o(1)$. What remains is c_{k}/μ times a Riemann sum of $g_\sigma\psi$ over $[-K, K]$ with mesh Δ_j, whose distance from $\int_{-K}^{K} g_\sigma\psi$ is at most $2K(\omega_{g_\sigma}(\Delta_j) + \|g_\sigma\|_\infty\omega(\Delta_j)) + 2\Delta_j\|g_\sigma\|_\infty$, uniformly in $\psi \in \mathcal{F}(\omega)$. Hence the $\limsup$ of the supremum in (b) is at most $\eta_\sigma(K) + (c_{\mathsf{k}}/\mu)\mathbb{P}(|Z_\sigma| > K)$, and we let $K \to \infty$. With $\psi \equiv 1$ and $\mathsf{k} = u^\sigma$ we get (2.10). □

3.3 Proof of Theorem A

We write $F_N^{\sigma_0}(k)$ through the fresh identity as a sum over the index m of products of a kernel of sign $+$ and a kernel of sign $-$. In each product we resolve the factor of the dominant sign τ by its local limit and sum the factor of the other sign through Lemma 3.4. The bins near the ends, the indices far from the index window of the other sign, the remote bins and the central bins are treated in turn, and no assumption on the ratio of the two scales enters.

Proof of Theorem A(a). Throughout we fix

$$\varepsilon = \min\Big\{\frac{p}{2},\ \frac{1-p}{2},\ \frac{1}{16\mu_+},\ \frac{1}{16\mu_-}\Big\},$$

and we write $\mu_{\max} = \max_\sigma \mu_\sigma$, $\mu_{\min} = \min_\sigma \mu_\sigma$, $\eta(K) = \max_\sigma \eta_\sigma(K)$ and $\delta(n) = \sup_{m\ge n} \max_{\sigma,\mathsf{k}} \delta_{\mathsf{k}}(m)$, the maximum over the six kernels. Then $\eta(K) \to 0$ as $K \to \infty$ and $\delta(n) \to 0$ as $n \to \infty$. A constant called C_i or c_i depends only on the laws of the runs and on ε, never on N, k, K or R.

Step 1. The fresh identity as a sum of products. We write $\mathcal{E}_N(k) = |B_N F_N^{\sigma_0}(k) - h_N(x_{N,k})|$ for $k \in \mathbb{Z}$. Let $1 \le k \le N-1$ and write $k_+ = k$ and $k_- = N - k$. By Proposition 2.3,

$$F_N^{\sigma_0}(k) = \sum_{i=1}^{2} \sum_{m\ge 1} \mathsf{k}_m^{+,i}(k_+)\, \mathsf{k}_m^{-,i}(k_-), \tag{3.3}$$

where $(\mathsf{k}^{+,1}, \mathsf{k}^{-,1}, \mathsf{k}^{+,2}, \mathsf{k}^{-,2})$ is $(q_{m+1}^+, u_m^-, u_m^+, q_m^-)$ for $\sigma_0 = +$ and $(u_m^+, q_{m+1}^-, q_m^+, u_m^-)$ for $\sigma_0 = -$. Each is a kernel of the sign it carries, and in both cases

$$c_{\mathsf{k}^{+,1}} c_{\mathsf{k}^{-,1}} + c_{\mathsf{k}^{+,2}} c_{\mathsf{k}^{-,2}} = \mu_+ + \mu_- = s. \tag{3.4}$$

The endpoint bins are $T_\sigma(N)$ or 0, and $B_N T_\sigma(N) \to 0$ by Lemma 2.1(v). The profile at $x_{N,k}$ for $k \le 0$ or $k \ge N$ is at most $\varepsilon_h(\min(p, 1-p)N/B_N)$, which tends to zero by Lemma 2.2(b). So it remains to treat $1 \le k \le N-1$.

We also need a uniform bound for one factor. Let N_1 be so large that for $N \ge N_1$ and both signs $C_5(\sigma,\varepsilon) b_{\sigma,N} e_{\sigma,N} \le 1$ and $a_\sigma(N/s) \le (8/\varepsilon) a_\sigma(\varepsilon N/(8\mu_\sigma))$. The second inequality holds eventually because the ratio tends to $(8\mu_\sigma/(\varepsilon s))^{1/\alpha_\sigma} < 8/\varepsilon$. Then for $N \ge N_1$, $\varepsilon N \le j \le N$, all $m \ge 1$ and every kernel k of sign σ,

$$\mathsf{k}_m(j) \le \frac{C_6}{b_{\sigma,N}}, \qquad C_6 = 1 + 8C_3/\varepsilon. \tag{3.5}$$

Indeed, for $8m\mu_\sigma \le \varepsilon N$ this is (3.1) with $\beta = \varepsilon$, and for larger m Lemma 3.2 gives $\mathsf{k}_m(j) \le C_3/a_\sigma(\varepsilon N/(8\mu_\sigma)) \le 8C_3/(\varepsilon b_{\sigma,N})$. For the dominant sign the bound is C_6/B_N. For the other sign it may be much larger, and we never use it there. In every product of (3.3) we bound the factor of the dominant sign and sum the factor of the other sign.

Step 2. Small counts and the ends of the row. Let $k_- < \varepsilon N$. By (K2) every nonzero product in (3.3) has $m \le k_- < \varepsilon N$. The plus factor sits at $k_+ > (1-\varepsilon)N \ge N/2$, and $8m\mu_+ \le N/2$, so Lemma 3.3 with $\beta = 1/2$ bounds it by $C_5(+, \frac{1}{2})e_{+,N}$. The minus factors add up to at most 1 by (K1). Hence $B_N F_N^{\sigma_0}(k) \le 2C_5(+, \frac{1}{2})B_N e_{+,N} \to 0$ uniformly in these k, and the case $k_+ < \varepsilon N$ is the same with the signs exchanged. In both cases $|k - pN| \ge \varepsilon N$, so the profile is at most $\varepsilon_h(\varepsilon N/B_N) \to 0$. So

$$\sup_{\min(k, N-k) < \varepsilon N} \mathcal{E}_N(k) \to 0. \tag{3.6}$$

From now on $\varepsilon N \le k_+, k_- \le N$, and (3.5) applies to both factors.

Step 3. Indices far from the index window of the other sign. Fix $K \ge 2$ and write $I = I^{\bar\tau}_{k_{\bar\tau}}(K)$. By (3.5) for the dominant factor and Lemma 3.4(a) for the other factor at the level $k_{\bar\tau} \ge \varepsilon N$,

$$\limsup_{N\to\infty} \sup_{\varepsilon N \le k \le (1-\varepsilon)N} B_N \sum_{i=1}^{2} \sum_{m \notin I} \mathsf{k}_m^{+,i}(k_+)\mathsf{k}_m^{-,i}(k_-) \le 2C_6\, \eta(K). \tag{3.7}$$

For $m \in I$ we record three facts. Let $N \ge N_2(K)$, where $N_2(K) \ge N_1$ is so large that $K a_\sigma(N/\mu_\sigma) \le \varepsilon N/2$ for both signs. Then every $m \in I$ satisfies

$$\frac{\varepsilon N}{2\mu_{\bar\tau}} \le m \le \frac{2N}{\mu_{\bar\tau}}, \tag{3.8}$$

so Lemma 3.2 applies to the dominant factor with the error $\delta(\lceil \varepsilon N/(2\mu_{\max})\rceil) \to 0$. Next, the definitions of p and s give the exact identity

$$k_\tau - m\mu_\tau = \varsigma_\tau \frac{s(k-pN)}{\mu_{\bar\tau}} + \frac{\mu_\tau}{\mu_{\bar\tau}}\big(k_{\bar\tau} - m\mu_{\bar\tau}\big). \tag{3.9}$$

For $\tau = +$ the right side is $(sk - \mu_+ N + \mu_+ k_-)/\mu_- - m\mu_+ = k - m\mu_+$, and $\tau = -$ is the same computation. Finally, by Lemma 2.1(ii) there is $N_3 \ge N_2(K)$ such that for $N \ge N_3$

$$a_\sigma(N/\mu_\sigma) \le c_0\, b_{\sigma,N}, \qquad a_\sigma(2N/\mu_{\min}) \le c_1 b_{\sigma,N}, \qquad B_N \le c_2\, a_\tau(\varepsilon N/(2\mu_{\max})), \tag{3.10}$$

with $c_0 = s/\mu_{\min}$, $c_1 = 2s/\mu_{\min}$ and $c_2 = 2/\varepsilon$, because the ratios tend to $(s/\mu_\sigma)^{1/\alpha_\sigma}$, $(2s/\mu_{\min})^{1/\alpha_\sigma}$ and $(2\mu_{\max}/(\varepsilon s))^{1/\alpha_\tau}$.

Step 4. Remote bins. Let $R \ge 4\mu_{\max}/\mu_{\min}$ and put $K = K(R) = sR/(2c_0\mu_{\max}) = \mu_{\min}R/(2\mu_{\max}) \ge 2$. Let $N \ge N_3$ with N_3 taken for this K, and let $\varepsilon N \le k \le (1-\varepsilon)N$ and $|k - pN| > RB_N$. For $m \in I$ we have $|k_{\bar\tau} - m\mu_{\bar\tau}| \le K a_{\bar\tau}(N/\mu_{\bar\tau}) \le c_0 K B_N$ by (3.10), so (3.9) gives

$$|k_\tau - m\mu_\tau| \ge \frac{sRB_N}{\mu_{\bar\tau}} - \frac{\mu_\tau}{\mu_{\bar\tau}} c_0 K B_N \ge \frac{sRB_N}{2\mu_{\bar\tau}}.$$

By (3.8) and (3.10), $a_\tau(m) \le c_1 B_N$, so the scaled argument $z = (k_\tau - m\mu_\tau)/a_\tau(m)$ of the dominant factor has modulus at least $c_8 R$ with $c_8 = s/(2c_1\mu_{\max})$. Lemma 3.2 and (3.10) give

$$B_N \mathsf{k}_m^{\tau,i}(k_\tau) \le \frac{B_N}{a_\tau(m)}\big(\mu_{\max}\bar g_\tau(c_8 R) + \delta(m)\big) \le c_2\big(\mu_{\max}\bar g_\tau(c_8 R) + \delta(\lceil \varepsilon N/(2\mu_{\max})\rceil)\big).$$

The factors of sign $\bar\tau$ add up to at most 1 for each i by (K1). Together with (3.7) and Lemma 2.2(b),

$$\limsup_{N\to\infty} \sup_{\substack{\varepsilon N \le k \le (1-\varepsilon)N \\ |k-pN| > RB_N}} \mathcal{E}_N(k) \le e(R) = 2c_2\mu_{\max} \max_\sigma \bar g_\sigma(c_8 R) + 2C_6\eta(K(R)) + \varepsilon_h(R), \tag{3.11}$$

and $e(R) \to 0$ as $R \to \infty$.

Step 5. The central sum. Fix $R \geq 1$ and $K \geq 2$, and let $|k-pN| \leq RB_N$, so $x = x_{N,k} \in [-R, R]$. For large N these bins satisfy $\varepsilon N \leq k \leq (1-\varepsilon)N$, since $B_N = o(N)$. Let $m \in I$ and $y_m = y^{\bar\tau}_m(k_{\bar\tau}) \in [-K, K]$. Since $k_+/\mu_+ = N/s + xB_N/\mu_+$ and $k_-/\mu_- = N/s - xB_N/\mu_-$, we have $k_{\bar\tau}/\mu_{\bar\tau} = (N/s)(1+o(1))$ and $m = (N/s)(1+o(1))$, uniformly in $|x| \leq R$ and $m \in I$. Lemma 2.1(i), applied to a_+ and to a_- separately, gives

$$a_{\bar\tau}(k_{\bar\tau}/\mu_{\bar\tau}) = b_{\bar\tau,N}(1+o(1)), \qquad a_\tau(m) = B_N(1+o(1)),$$

uniformly in $|x| \leq R$ and $m \in I$, whichever sign is dominant. Since $b_{\bar\tau,N} = \theta_{\bar\tau,N} B_N$, the identity (3.9) gives for the argument of the dominant factor

$$z_m = \frac{k_\tau - m\mu_\tau}{a_\tau(m)} = \varsigma_\tau \frac{sx}{\mu_{\bar\tau}}\, r_1 + \frac{\mu_\tau}{\mu_{\bar\tau}} \theta_{\bar\tau,N}\, y_m\, r_2, \qquad r_1 = \frac{B_N}{a_\tau(m)}, \quad r_2 = \frac{a_{\bar\tau}(k_{\bar\tau}/\mu_{\bar\tau})}{b_{\bar\tau,N}} \frac{B_N}{a_\tau(m)},$$

where $r_1, r_2 \to 1$ uniformly. Since $|x| \leq R$, $|y_m| \leq K$ and $\theta_{\bar\tau,N} \leq 1$,

$$\sup_{|x|\leq R} \sup_{m\in I} \big| z_m - \zeta_{N,x}(y_m) \big| \to 0, \qquad \zeta_{N,x}(y) = \varsigma_\tau \frac{sx}{\mu_{\bar\tau}} + \frac{\mu_\tau}{\mu_{\bar\tau}} \theta_{\bar\tau,N}\, y.$$

Lemma 3.2, $B_N/a_\tau(m) \to 1$ and the uniform continuity of g_τ give, uniformly in $|x| \leq R$ and $m \in I$,

$$B_N \mathsf{k}^{\tau,i}_m(k_\tau) = c_{\mathsf{k}^{\tau,i}}\, g_\tau\big(\zeta_{N,x}(y_m)\big) + o(1). \tag{3.12}$$

Let $\psi_{N,x} = g_\tau \circ \zeta_{N,x} / \|g_\tau\|_\infty$. The slope of $\zeta_{N,x}$ is at most $\mu_{\max}/\mu_{\min}$, so $\psi_{N,x} \in \mathcal{F}(\omega)$ for all N and x, with $\omega(r) = \max_\sigma \omega_{g_\sigma}(\mu_{\max} r/\mu_{\min})/\|g_\sigma\|_\infty$. We multiply (3.12) by $\mathsf{k}^{\bar\tau,i}_m(k_{\bar\tau})$ and sum over $m \in I$. The $o(1)$ terms add up to $o(1)$ by (K1). Then we add back the indices $m \notin I$ in the sum of $\mathsf{k}^{\bar\tau,i}_m \psi_{N,x}(y_m)$, at a cost of at most $\eta(K) + o(1)$ by Lemma 3.4(a), and apply Lemma 3.4(b) at the level $k_{\bar\tau} \geq \varepsilon N$. This gives, uniformly in $|x| \leq R$,

$$B_N \sum_{m\in I} \mathsf{k}^{+,i}_m(k_+) \mathsf{k}^{-,i}_m(k_-) = \frac{c_{\mathsf{k}^{\tau,i}} c_{\mathsf{k}^{\bar\tau,i}}}{\mu_{\bar\tau}} \int_{\mathbb{R}} g_{\bar\tau}(y)\, g_\tau\big(\zeta_{N,x}(y)\big) dy + O\big(\eta(K)\big) + o(1),$$

with the constant $\mu_{\max} \max_\sigma \|g_\sigma\|_\infty$ in $O(\eta(K))$. We sum over i and use (3.4). By Lemma 2.2(e) the integral with the weight $s/\mu_{\bar\tau}$ is exactly $h_N(x)$. Adding (3.7) for the indices outside I,

$$\limsup_{N\to\infty} \sup_{|k-pN|\leq RB_N} \mathcal{E}_N(k) \leq C_7\, \eta(K), \tag{3.13}$$

with $C_7 = 2C_6 + 2\mu_{\max} \max_\sigma \|g_\sigma\|_\infty$. Since K is arbitrary, the left side is zero.

Combination. Let $R \geq 4\mu_{\max}/\mu_{\min}$. Every $k \in \mathbb{Z}$ is an endpoint or lies outside the row (Step 1), or within εN of an end (Step 2), or is a remote bin with $|k - pN| > RB_N$ (Step 4), or a central bin with $|k - pN| \leq RB_N$ (Step 5). By (3.6), (3.11) and (3.13), $\limsup_N \sup_{k\in\mathbb{Z}} \mathcal{E}_N(k) \leq e(R)$, and we let $R \to \infty$. No step assumed that θ_N converges, since every error term is a maximum over the two signs of quantities that do not involve θ_N, and θ_N enters only through the slope of $\zeta_{N,x}$, which is at most $\mu_{\max}/\mu_{\min}$. □

Proof of Theorem A(b) and (c). For (b), part (a) and Lemma 2.2(a) give an N_4 with $B_N F^{\sigma_0}_N(k) \leq C_1 + 1$ for all $N \geq N_4$ and all k, and $B_N F^{\sigma_0}_N(k) \leq B_N \leq B_{N_4}$ for $N < N_4$, since B_N is nondecreasing. For (c), Lemma 2.2(d) gives $c_K > 0$ with $h_N \geq c_K$ on $[-K, K]$ for all N, and dividing the uniform error of (a) by c_K gives (c). □

The proof used (SPAN) only through Lemma 3.1(ii), and it did not use (LR).

4 Big jumps and the boundary layer

Outside the bulk each bin is $o(1/B_N)$ by Theorem A, and the truth is smaller by a power of N. This section describes these bins and the boundary layer with one kernel.

Throughout we write $j = N - k$ for the number of minus slots. For $k > pN$ put

$$L_k = \frac{k - pN}{1 - p}, \qquad \text{so that} \qquad N - L_k = \frac{j}{1-p} = \frac{js}{\mu_-}. \tag{4.1}$$

This is the length of the plus run that moves a typical configuration to the plus count k. For a start of sign σ_0 and $j \geq 1$ put

$$\gamma_+^{(\sigma_0)}(j) = \sum_{i=0}^{j-1} \kappa_+^{(\sigma_0)}(i)\big(1 + \mu_+ u_-(j-i)\big), \qquad \gamma_+^{(\sigma_0)}(0) = 0, \tag{4.2}$$

and define the big-jump kernel by

$$G_N^{(\sigma_0)}(k) = \gamma_+^{(\sigma_0)}(N-k)\, f_+(\lfloor L_k \rfloor) + T_+(\lfloor L_k \rfloor)\, \kappa_+^{(\sigma_0)}(N-k), \qquad pN < k \leq N. \tag{4.3}$$

The first term is a complete plus run of length about L_k. The coefficient $\gamma_+^{(\sigma_0)}(j)$ counts its admissible positions. The run starts right after the i-th minus slot, which has probability $\kappa_+^{(\sigma_0)}(i)$, and the fresh minus walk that follows it must hold the other $j - i$ minus slots. By the Green identity, Lemma 2.5, the expected number of lengths of that walk that do so is $1 + \mu_+ u_-(j-i)$. The second term is a censored final plus run of length at least L_k. By (2.10) and (2.11), $\kappa_+^{(\sigma_0)}(i) \to 1/\mu_-$ and $u_-(i) \to 1/\mu_-$, so a Cesàro average gives

$$\gamma_+^{(\sigma_0)}(j) \sim \frac{js}{\mu_-^2} = \frac{N - L_k}{\mu_-} \qquad (j \to \infty). \tag{4.4}$$

At $k = N - j$ with j fixed the first term is of order $T_+(N)/N$ and the second is $T_+(N)\kappa_+^{(\sigma_0)}(j)$ up to a factor $1 + o(1)$, so the kernel reduces to the boundary layer. As $j \to \infty$ the coefficient of the second term tends to $1/\mu_-$, its value in the interior. So one kernel covers the big-jump zone and the boundary layer.

The zone starts at a multiple of B_N when the plus scale is the larger one. Otherwise it starts a little later, and we put

$$\lambda_N = 1 + \log \frac{B_N}{b_{+,N}} = 1 + \log \frac{1}{\theta_{+,N}} \ \geq 1. \tag{4.5}$$

So $\lambda_N = 1$ when $b_{+,N} = B_N$, for instance for all large N when $\alpha_+ < \alpha_-$, and $\lambda_N = O(\log N)$ in all cases.

Theorem B (big jumps and the boundary layer)**.** *Let the start be fresh with sign σ_0, and write $\kappa = \kappa_+^{(\sigma_0)}$ and $G_N = G_N^{(\sigma_0)}$.*

(a) (The boundary layer.) Assume (RV) for the plus sign. For every fixed $j \geq 0$,

$$\frac{P_N(N-j)}{T_+(N)} \to \kappa(j).$$

(b) (The big-jump zone.) Assume (RV) and (SPAN) for both signs and (LR) for the plus sign. Then

$$\lim_{K\to\infty} \limsup_{N\to\infty} \sup_{pN + K\lambda_N B_N \leq k \leq N} \Big| \frac{P_N(k)}{G_N(k)} - 1 \Big| = 0,$$

where $G_N(k) = 0$ only if $P_N(k) = 0$, and then the ratio is read as 1. In particular, with the size function

$$\widehat{G}_N(k) = \frac{N}{L_k}\, T_+(\lfloor L_k \rfloor), \tag{4.6}$$

we have

$$\lim_{K\to\infty}\ \limsup_{N\to\infty}\ \sup_{pN+K\lambda_N B_N\le k\le N}\frac{|P_N(k)-G_N(k)|}{\widehat{G}_N(k)}=0,$$

and this additive form also holds for the kernel $\overline{G}_N=\overline{G}_N^{(\sigma_0)}$ *obtained from* (4.3) *by replacing* $\gamma_+^{(\sigma_0)}(N-k)$ *with* $(N-L_k)/\mu_-$.

(c) (Uniform boundary layer.) Under the hypotheses of (b), for every $\delta_N\to 0$,

$$\sup_{0\le j\le\delta_N N}\Big|\frac{P_N(N-j)}{T_+(N)}-\kappa(j)\Big|\to 0$$

and

$$\sup\Big\{\Big|\frac{P_N(N-j)}{T_+(N)\kappa(j)}-1\Big|\ \Big|\ 0\le j\le\delta_N N,\ \kappa(j)>0\Big\}\to 0.$$

The same holds at the minus end with the signs exchanged.

The exact coefficient $\gamma_+^{(\sigma_0)}(j)$ matters for the ratio form at bounded j with $\kappa(j)=0$. If $f_-(1)=0$, then for a plus start $P_N(N-1)=f_+(N-1)$ and $G_N(N-1)=f_+(\lfloor N-s/\mu_-\rfloor)$, whose ratio tends to one by (LR), while the limit coefficient s/μ_-^2 would not. In the additive form the two coefficients agree. In (c) the condition $\kappa(j)>0$ excludes nothing but $j=0$ for a minus start when $f_-(1)>0$.

The double limit in (b) is forced. Fix K and let $k=pN+KB_N$. Theorem A(c) gives $B_N P_N(k)=h_N(x_{N,k})+o(1)$. By Remark 4.9 the ratio of $h_N(K)$ to $B_N G_N(k)$ tends to one only as $K\to\infty$. When $\theta_{+,N}\to 0$ a factor in front of B_N that tends to infinity is also needed (Remark 4.10). Then the window $KB_N\le k-pN\le K\lambda_N B_N$ is described only by the additive bound of Theorem A(a) and the lower bound $P_N(k)\ge(1-o(1))G_N(k)$ of Corollary 4.7, and we prove no upper bound of relative order there. A natural conjecture is $P_N(k)=(1+o(1))(B_N^{-1}h_N(x_{N,k})+G_N(k))$ on that window.

4.1 Giant runs

Every bin in this section is decomposed according to the plus runs that occupy many slots. We use the weights $\rho_i^{(\sigma_0)}(x)$ of Section 2.4, with the representation (2.13), and for $i\ge 1$ and $x\ge 0$ we put

$$\beta_i(x)=\Pi^-_{i+x}(i),$$

the probability that a fresh minus walk on $i+x$ slots has exactly i minus slots. Then

$$\sum_{x\ge 0}\rho_i^{(\sigma_0)}(x)=\kappa_+^{(\sigma_0)}(i),\qquad\sum_{x\ge 0}\beta_i(x)=1+\mu_+u_-(i),\tag{4.7}$$

by (2.12) and by the Green identity, Lemma 2.5.

We need β_i in terms of sums of runs. Take a fresh minus walk with minus runs $X_1^-,X_2^-,\dots$ and plus runs $X_1^+,X_2^+,\dots$ in the order $X_1^-X_1^+X_2^-X_2^+\cdots$. Let $J=J_-(i)$ be the index of the minus run that contains the i-th minus slot, in the notation of (K3) in Section 3.2, and put $\xi_i=X_J^+\mathbf{1}_{\{S_J^-=i\}}$. The i-th minus slot is slot $i+S_{J-1}^+$. The next minus slot follows it at once unless the run X_J^- ends there, and then it follows after X_J^+ plus slots. Hence

$$\beta_i(x)=\mathbb{P}\big(S_{J-1}^+\le x\le S_{J-1}^++\xi_i\big).\tag{4.8}$$

Here J depends only on the minus runs, and X_J^+ is independent of $(J,S_{J-1}^+,X_1^-,\dots,X_J^-)$.

For an integer $g\ge 1$ let $\mathcal{G}_g$ be the number of plus runs that have at least g slots among the first N slots. We call these runs giant. A complete run counts with its length, and the final run counts with the number of its slots up to slot N. Let M_N be the largest number of slots that a plus run has among the first N slots, so $\{M_N<g\}=\{\mathcal{G}_g=0\}$.

Lemma 4.1 (the giant identity)**.** *For every start of sign σ_0, every N, every $0 \le j < N$ with $k = N - j$, and every integer $g \ge 1$,*

$$\mathbb{E}\big[\mathcal{G}_g \mathbf{1}_{\{A_N = k\}}\big] = \mathrm{Cen}_g(k) + \mathrm{Int}_g(k), \tag{4.9}$$

where, with $\rho_i = \rho_i^{(\sigma_0)}$,

$$\mathrm{Cen}_g(k) = \sum_{x \ge 0,\ N-j-x \ge g} \rho_j(x)\, T_+(N-j-x),$$

$$\mathrm{Int}_g(k) = \sum_{i=0}^{j-1} \sum_{x,x' \ge 0,\ N-j-x-x' \ge g} \rho_i(x)\, \beta_{j-i}(x')\, f_+(N-j-x-x').$$

Consequently

$$P_N(k) = \mathbb{P}(A_N = k,\ M_N < g) + \mathrm{Cen}_g(k) + \mathrm{Int}_g(k) - \mathbb{E}\big[(\mathcal{G}_g - 1)^+ \mathbf{1}_{\{A_N = k\}}\big], \tag{4.10}$$

and, with $f_+^*(n) = \sup_{m \ge n} f_+(m)$,

$$\mathrm{Cen}_g(k) \le \kappa_+^{(\sigma_0)}(j)\, T_+(g), \qquad \mathrm{Int}_g(k) \le \gamma_+^{(\sigma_0)}(j)\, f_+^*(g). \tag{4.11}$$

Proof. By linearity, the left side of (4.9) is the sum over all plus runs of the probability that the run is giant and $A_N = k$. We split the runs into the final run and the complete runs.

Let the final run be a plus run that starts at slot $t+1$. All j minus slots precede it, so $t = j + x$, where x is the number of plus slots before it. The run has $N - j - x$ slots up to slot N, and its length is at least $N - j - x$. The length is independent of the runs before it. So this case contributes $\rho_j(x) T_+(N-j-x)$ for each x with $N - j - x \ge g$, which is $\mathrm{Cen}_g(k)$.

Let a complete plus run start right after the i-th minus slot, with x plus slots before it, and let its length be y. It occupies slots $i + x + 1$ to $i + x + y$, and $i + x + y \le N - 1$. The runs after it form a fresh minus walk, independent of everything before. On $\{A_N = k\}$ this walk holds the remaining $j - i$ minus slots among its $N - i - x - y \ge 1$ slots. Since its first slot is a minus slot, $j - i \ge 1$. Write $N - i - x - y = (j - i) + x'$. Then $y = N - j - x - x'$, and this case contributes $\rho_i(x) f_+(y) \beta_{j-i}(x')$. A fresh plus start has $\rho_0 = \delta_0$, and then the first run has law f_+, as every other plus run. Summing over $i < j$, x and x' with $y \ge g$ gives $\mathrm{Int}_g(k)$.

For (4.10) note that $\mathbf{1}_{\{\mathcal{G}_g \ge 1\}} = \mathcal{G}_g - (\mathcal{G}_g - 1)^+$. For (4.11) bound T_+ by $T_+(g)$ and f_+ by $f_+^*(g)$ on the ranges of summation, and use (4.7). □

4.2 The boundary layer

We use two consequences of (RV) for the plus sign. First, $T_+(n+1)/T_+(n) \to 1$, so $f_+(n)/T_+(n) \to 0$ and

$$f_+^*(n) \le \sup_{m \ge n} \frac{f_+(m)}{T_+(m)}\, T_+(n) = o\big(T_+(n)\big). \tag{4.12}$$

Second, a sum of $m' \ge 1$ plus runs equal to x has a summand of size at least x/m'. Conditioning on the other summands gives

$$\mathbb{P}(S_{m'}^+ = x) \le m'\, f_+^*(x/m'). \tag{4.13}$$

Proof of Theorem B(a). Fix $j \ge 0$. If $j = 0$ then $P_N(N)$ is $T_+(N)$ for a plus start and 0 for a minus start, which is $\kappa(0) T_+(N)$. Let $j \ge 1$ and take $g = g'_N = \lceil (N-j)/(j+1) \rceil$ in

Lemma 4.1. By Lemma 2.7, on $\{A_N = N-j\}$ some plus run has at least g'_N slots, so the first term of (4.10) vanishes. Since $g'_N \geq N/(2j+2)$ for large N, Lemma 2.1(ii) gives $T_+(g'_N) \leq C_j T_+(N)$.

The last term of (4.10). There are at most $j+1$ plus runs, so $(\mathcal{G}_g - 1)^+ \leq (j+1)\mathbf{1}_{\{\mathcal{G}_g \geq 2\}}$. The second part of Lemma 2.7 gives

$$\mathbb{E}\big[(\mathcal{G}_g - 1)^+ \mathbf{1}_{\{A_N = N-j\}}\big] \leq (j+1)^3\, T_+(g'_N)^2 = O\big(T_+(N)^2\big) = o\big(T_+(N)\big).$$

The term Int_g. It contains the configurations of the second sum of (2.14), a complete giant plus run followed by a final minus run of length $m \leq j$, which are the leading term of $P_N(N-j)$ when $\kappa(j) = 0$. By (4.11) and (4.12), $\mathrm{Int}_g(N-j) \leq \gamma_+^{(\sigma_0)}(j) f_+^*(g'_N) = o(T_+(g'_N)) = o(T_+(N))$.

The term Cen_g. Split the sum at $x = N/4$. For $x \leq N/4$ and large N we have $N - j - x \geq N/2 \geq g'_N$, and $T_+(N-j-x)/T_+(N)$ is at most $T_+(\lfloor N/2 \rfloor)/T_+(N)$, which is bounded by Lemma 2.1(ii). For each fixed x the ratio tends to one. Since ρ_j is a finite measure that does not depend on N, dominated convergence and (4.7) give

$$\frac{1}{T_+(N)} \sum_{x \leq N/4} \rho_j(x)\, T_+(N-j-x) \to \sum_{x \geq 0} \rho_j(x) = \kappa(j).$$

For $x > N/4$ we use (2.13). The plus sums there have at most j terms, and $\sum_m u_m^-(j) \leq 1$. So (4.13) gives $\rho_j(x) \leq j\, f_+^*(N/(4j))$. Since $\sum_x T_+(N-j-x) \leq \sum_{n \geq 1} T_+(n) = \mu_+$,

$$\sum_{x > N/4} \rho_j(x)\, T_+(N-j-x) \leq j\mu_+ f_+^*\big(N/(4j)\big) = o\big(T_+(N)\big),$$

by (4.12) and Lemma 2.1(ii). Adding the four terms gives $P_N(N-j) = \kappa(j) T_+(N) + o(T_+(N))$. □

At fixed j the giant run is censored and its law enters only through T_+, which is why (a) needs no local hypothesis.

4.3 The big-jump zone

In the zone we take $g = \lceil \eta L_k \rceil$ in Lemma 4.1 with a small fixed η. Then (4.10) has four terms, and we treat them in three lemmas. Lemma 4.3 bounds the bins with no giant run, Lemma 4.4 the bins with two, and Lemma 4.6 evaluates Cen_g and Int_g. Note that $k \geq L_k$ for $pN < k \leq N$, since $k - L_k = p(N-k)/(1-p)$.

We also need a bound for right deviations without big summands. It is the elementary exponential estimate behind the inequalities of Fuk and Nagaev [20] and [31, Theorems 1.1 and 1.2], and we include the short proof.

Lemma 4.2 (truncated right deviations)**.** *Assume (RV) for the plus sign. For all $m \geq 1$, $w \geq 1$ and $z > 0$,*

$$\mathbb{P}\Big(S_m^+ - m\mu_+ \geq z,\ \max_{i \leq m} X_i^+ < w\Big) \leq e^{C_{\mathrm{K}}} \big(m\, T_+(w)\big)^{z/(2w)}.$$

Proof. If $mT_+(w) \geq 1$ there is nothing to prove. Otherwise let $\vartheta = \frac{1}{2}\log(1/(mT_+(w))) > 0$. For $0 \leq v \leq \vartheta$ we have $e^v \leq 1 + v + \frac{1}{2}v^2 e^\vartheta$. We use this with $v = \vartheta X/w$ on $\{X < w\}$, where $X = X_+$. By summation by parts and (2.3), $\mathbb{E}[X^2 \mathbf{1}_{\{X<w\}}] \leq 2\sum_{n<w} nT_+(n) \leq 2C_{\mathrm{K}} w^2 T_+(w)$. With $1 + v \leq e^v$ this gives $\mathbb{E}[e^{\vartheta X/w} \mathbf{1}_{\{X<w\}}] \leq \exp(\vartheta\mu_+/w + C_{\mathrm{K}}\vartheta^2 e^\vartheta T_+(w))$. Markov's inequality and independence give the bound $\exp(-\vartheta z/w + C_{\mathrm{K}} m\vartheta^2 e^\vartheta T_+(w))$. Here $m\vartheta^2 e^\vartheta T_+(w) = \vartheta^2 e^{-\vartheta} \leq 1$, and $e^{-\vartheta z/w} = (mT_+(w))^{z/(2w)}$. □

Lemma 4.3 (no giant run)**.** *Assume (RV) and (SPAN). Let $0 < \eta \le 1/16$ and $r = 1/(16\eta)$, so $r \ge 1 > 1/\alpha_+$. There are constants $C, c, C_0 > 0$ such that, for every fresh start, all large N and all $pN < k \le N$ with $L_k \ge C_0 B_N$,*

$$\mathbb{P}\big(A_N = k,\ M_N < \eta L_k\big) \le C\,\frac{(NT_+(L_k))^r}{b_{+,N}} + \frac{C}{b_{-,N}}\,e^{-cL_k/b_{-,N}} + C\,e^{-N^c}.$$

Proof. Write $L = L_k$, $j = N - k$ and P_0 for the probability on the left. If $j = 0$ then the whole walk is one plus run and $P_0 = 0$. Let $j \ge 1$, and take first a plus start. On $\{A_N = k\}$ slot N lies either in the $(m+1)$-th plus run, and then the first m minus runs sum to j and $S^+_m < k \le S^+_{m+1}$, or in the m-th minus run, and then $S^+_m = k$ and $S^-_{m-1} < j \le S^-_m$, as in the proof of Proposition 2.3. On $\{M_N < \eta L\}$ the plus runs that lie inside the first N slots are shorter than ηL, and in the first case the final run has $k - S^+_m < \eta L$ slots. The plus and minus runs are independent. Hence $P_0 \le E_1 + E_2$ with

$$E_1 = \sum_{m\ge 0} u^-_m(j)\,q^\eta_{m+1}, \qquad q^\eta_{m+1} = \sum_{1\le v<\eta L} \mathbb{P}\Big(S^+_m = k - v,\ \max_{i\le m} X^+_i < \eta L\Big) T_+(v),$$

$$E_2 = \sum_{m\ge 1} \mathbb{P}\Big(S^+_m = k,\ \max_{i\le m} X^+_i < \eta L\Big)\, q^-_m(j).$$

For a minus start the plus run that contains slot N is the m-th one, preceded by m minus runs and $m-1$ plus runs, and the minus run that contains slot N is the $(m+1)$-th one, preceded by m plus runs. So E_1 becomes $\sum_{m\ge1} u^-_m(j) q^\eta_m$, where q^η_m is built on $\mathbb{P}(S^+_{m-1} = k - v, \dots)$, and E_2 becomes $\sum_{m\ge1} \mathbb{P}(S^+_m = k, \dots)\, q^-_{m+1}(j)$. We note below the two places where this changes the argument.

We split the indices. Put $d_1 = k - m\mu_+$ and $d_2 = m\mu_- - j$. Then $\mu_- d_1 + \mu_+ d_2 = \mu_- k - \mu_+ j = s(k - pN) = \mu_- L$. Let $\mathcal{A}$ be the set of m with $d_1 \ge L/2$ and $\mathcal{B}$ its complement. For $m \in \mathcal{B}$ we get $d_2 \ge y_2 = \mu_- L/(2\mu_+)$. The kernel sums $\sum_m u^-_m(j)$, $\sum_m q^-_m(j)$, $\sum_m q^-_{m+1}(j)$, $\sum_m q^+_m(k)$ and $\sum_m u^+_m(k)$ are at most 1 by (2.8). So $E_1 + E_2$ is at most the sum of the suprema over $m \in \mathcal{A}$ of the plus factors and the suprema over $m \in \mathcal{B}$ of the minus factors.

The indices in $\mathcal{A}$. Let $m \in \mathcal{A}$ and let v' be an integer with $v' - m\mu_+ \ge L/4$. Every $k - v$ in q^η_{m+1} is such a v', since $k - v - m\mu_+ \ge L/2 - \eta L$, and so is k. For a minus start the relevant sum S^+_{m-1} has one term fewer, so its deviation $k - v - (m-1)\mu_+$ is larger by μ_+, and the bound below applies with $m-1$ in place of m. For $m \le 1$ the probability $\mathbb{P}(S^+_m = v', \max_i X^+_i < \eta L)$ vanishes, because $v' \ge L/4 > \eta L$. For $m \ge 2$ split the indices into two halves of sizes $\lfloor m/2 \rfloor$ and $\lceil m/2 \rceil$. On $\{S^+_m = v'\}$ one half exceeds its mean by at least $L/8$, and then the other half takes one exact value. By Lemma 4.2 with $w = \eta L$ and $z = L/8$, and by Lemma 3.1(ii),

$$\mathbb{P}\Big(S^+_m = v',\ \max_{i\le m} X^+_i < \eta L\Big) \le 2e^{C_{\mathrm{K}}} \big(mT_+(\eta L)\big)^r \min\Big(1, \frac{\bar u_+}{a_+(\lfloor m/2 \rfloor)}\Big).$$

The indices in $\mathcal{A}$ satisfy $m \le m_+ = k/\mu_+ \le N/\mu_+$, and $m_+ \ge N/s$ since $k > pN$. Since $r > 1/\alpha_+$, Lemma 2.1(ii) for a_+ shows that the right side is largest up to a constant at $m = m_+$. Also $a_+(m_+) \ge a_+(N/s) = b_{+,N}$ and $T_+(\eta L) \le C_\eta T_+(L)$. With $\sum_v T_+(v) \le 1 + \mu_+$ in q^η_{m+1} the plus factors are at most $C(NT_+(L))^r / b_{+,N}$.

The indices in $\mathcal{B}$. Let $m \in \mathcal{B}$, so $x_m = m\mu_- - j \ge y_2$ and in particular $m \ge y_2/\mu_-$, which is large. On $\{S^-_m = j\}$ one half of the minus runs falls below its mean by at least $x_m/2$, and the other half takes one exact value. Lemma 3.1(iv) for the half, with $a_-(\lceil m/2 \rceil) \le a_-(m)$, and Lemma 3.1(ii) give

$$u^-_m(j) \le 2e^{C_-} e^{-x_m/(2a_-(m))} \min\Big(1, \frac{\bar u_-}{a_-(\lfloor m/2 \rfloor)}\Big).$$

Next, $q^-_m(j) = \sum_{v\ge1} T_-(v) u^-_{m-1}(j-v) \le \mu_- \sup_{v\ge1} u^-_{m-1}(j-v)$, and $(m-1)\mu_- - (j-v) \ge x_m - \mu_- \ge x_m/2$. So $q^-_m(j)$ obeys the same bound with μ_- in front, $x_m/4$ in the exponent and $\lfloor (m-1)/2 \rfloor$ in place of $\lfloor m/2 \rfloor$. For a minus start, $q^-_{m+1}(j) = \sum_{v\ge1} T_-(v) u^-_m(j-v) \le \mu_- \sup_{v\ge1} u^-_m(j-v)$ and $m\mu_- - (j-v) \ge x_m$, so it obeys the bound for $u^-_m(j)$ with μ_- in front. Let $m \le 2N/\mu_-$. Then $t = a_-(m) \le c_3 b_{-,N}$, and Lemma 2.1(ii) gives $a_-(m) \le c_4 a_-(\lfloor (m-1)/2 \rfloor)$. So all these bounds are at most $Ct^{-1}e^{-y_2/(4t)}$. The function $t \mapsto t^{-1}e^{-y_2/(4t)}$ increases on $(0, y_2/4]$. Choose $C_0 = 8c_3\mu_+/\mu_-$. Then $L \ge C_0 B_N$ gives $c_3 b_{-,N} \le y_2/4$, and the bounds are at most $C b^{-1}_{-,N} e^{-cL/b_{-,N}}$. Let $m > 2N/\mu_-$. Then $x_m \ge m\mu_- - N \ge m\mu_-/2$, and Lemma 2.1(ii) gives $a_-(m) \le m^{1-2c}$ for large m with $c = (1 - 1/\alpha_-)/4$. So the bounds are at most $Ce^{-cm^{2c}} \le Ce^{-N^c}$.

Adding the bounds for $\mathcal{A}$ and $\mathcal{B}$ proves the lemma. □

The second term of Lemma 4.3, from minus runs falling short of their mean, is the reason for λ_N.

Lemma 4.4 (two giant runs)**.** *Assume (RV) and (LR) for the plus sign, and let $0 < \eta < 1$. There is a constant C such that, for every fresh start, all N and all $pN < k \le N$ with $\eta L_k \ge 1$ and $g = \lceil \eta L_k \rceil$,*

$$\mathbb{E}\big[(\mathcal{G}_g - 1)^+ \mathbf{1}_{\{A_N = k\}}\big] \le C\Big[(j+1)^2 \frac{T_+(L_k)^2}{L_k} + (j+1)\, T_+(L_k)^2\Big].$$

Proof. Since $(n-1)^+ \le n(n-1)/2$ for integers $n \ge 0$, it suffices to bound the expected number of pairs of giant runs on $\{A_N = k\}$. The second run of a pair is complete or final. As in the proof of Lemma 4.1, the walk between the two runs is a fresh minus walk until a plus run starts after $i_2 \ge 1$ of its minus slots, with x_2 plus slots before it, which has probability $\rho^{(-)}_{i_2}(x_2)$. For two complete runs of lengths y_1, y_2 the expected number of pairs is

$$\sum_{i_1+i_2+i_3=j} \sum_{x_1,x_2,x_3} \rho^{(\sigma_0)}_{i_1}(x_1)\, \rho^{(-)}_{i_2}(x_2)\, \beta_{i_3}(x_3) \sum_{y_1+y_2 = N-j-x_1-x_2-x_3,\ y_1,y_2 \ge g} f_+(y_1) f_+(y_2),$$

with $i_2, i_3 \ge 1$. For a complete run followed by the final run the expected number is the same with $\beta_{i_3}(x_3)$ removed, $i_1 + i_2 = j$, and $T_+(y_2)$ in place of $f_+(y_2)$. By (LR), $f_+(n) \le C_{\mathrm{LR}} T_+(n)/n$ for all $n \ge 1$, so $f^*_+(g) \le C_{\mathrm{LR}} T_+(g)/g$. Let $y_1 + y_2 = y$ with both at least g, so $y/2 \ge g$. The larger of y_1, y_2 is at least $y/2$. So

$$\sum f_+(y_1) f_+(y_2) \le 2 f^*_+(g) T_+(g), \qquad \sum f_+(y_1) T_+(y_2) \le f^*_+(g) \sum_{n\ge g} T_+(n) + T_+(g)^2,$$

and the second sum is at most $(C_{\mathrm{K}} C_{\mathrm{LR}} + 1) T_+(g)^2$ by (2.3). The weights have total mass at most $(1+\mu_+)(j+1)^2$ in the first case and $j+1$ in the second, by (4.7). Finally $T_+(g) \le C_\eta T_+(L_k)$ and $g \ge \eta L_k$. □

The terms Cen_g and Int_g need the spread of the weights. A plus run that starts after the i-th minus slot is preceded by about $\bar{x}_i = i\mu_+/\mu_-$ plus slots, and the fluctuation is of order B_N. For $z > 0$ put

$$\mathrm{sp}^\rho_i(z) = \sum_{x\,:\,|x-\bar{x}_i|>z} \rho^{(\sigma_0)}_i(x), \qquad \mathrm{sp}^\beta_i(z) = \sum_{x\,:\,|x-\bar{x}_i|>z} \beta_i(x).$$

Lemma 4.5 (spread of the weights)**.** *Assume (RV). For every $\varepsilon > 0$ there are Z and N_0 such that $\mathrm{sp}^\rho_i(ZB_N) \le \varepsilon$ and $\mathrm{sp}^\beta_i(ZB_N) \le \varepsilon$ for all $N \ge N_0$ and all $0 \le i \le N$.*

Proof. Let $\varpi_\sigma(z) = \sup_{m\ge 1}\mathbb{P}(|S^\sigma_m - m\mu_\sigma| > z\,a_\sigma(m))$. By Lemma 3.1(i) the laws of $(S^\sigma_m - m\mu_\sigma)/a_\sigma(m)$ converge, so they are tight, and $\varpi_\sigma(z)\to 0$ as $z\to\infty$. By Lemma 2.1(ii) there is c_5 with $a_\sigma(m)\le c_5 B_N$ for $m\le 4N/\mu_-$ and large N.

The number of minus runs. By (K3), $\{S^-_m = i\}\subset\{J_-(i)=m\}$, so for $w>0$

$$\sum_{m\,:\,|m\mu_- - i|>w} u^-_m(i) \le \mathbb{P}\big(|J_-(i)\mu_- - i|>w\big) \le \mathbb{P}\big(S^-_{m_1}\ge i\big) + \mathbb{P}\big(S^-_{m_2}<i\big),$$

with $m_1 = \lfloor (i-w)/\mu_-\rfloor$ (the first term is absent if $i\le w$) and $m_2 = \lfloor (i+w)/\mu_-\rfloor$. Indeed $J_-(i) > (i+w)/\mu_-$ gives $J_-(i) > m_2$, and then $S^-_{m_2}\le S^-_{J_-(i)-1} < i$. The first event is a deviation of at least w above the mean, and the second a deviation of at least $w-\mu_-$ below it. For $w\le N$ both indices are at most $3N/\mu_-$. So for $w = zB_N$ the right side is at most $2\varpi_-(z/(2c_5))$ for large N.

The bound for sp^ρ_i. Take a plus start. On $\{S^-_m = i\}$ we have $m\le i\le N$ and $|S^+_m - \bar x_i| \le |S^+_m - m\mu_+| + (\mu_+/\mu_-)|m\mu_- - i|$. By (2.13),

$$\begin{aligned}\mathrm{sp}^\rho_i(ZB_N) &\le \sup_{m\le N}\mathbb{P}\big(|S^+_m - m\mu_+| > ZB_N/2\big) + \sum_{m\,:\,|m\mu_- - i|>\mu_- ZB_N/(2\mu_+)} u^-_m(i)\\ &\le \varpi_+\Big(\frac{Z}{2c_5}\Big) + 2\varpi_-\Big(\frac{\mu_- Z}{4\mu_+ c_5}\Big).\end{aligned}$$

For a minus start (2.13) has S^+_{m-1} in place of S^+_m, and $|S^+_{m-1} - (m-1)\mu_+| > ZB_N/2 - \mu_+$ is needed for the first event. For large N this is at least $ZB_N/4$, so the same bound holds with $Z/(4c_5)$ in place of $Z/(2c_5)$ in the first term.

The bound for sp^β_i. Write $S = S^+_{J-1}$ in (4.8). Then $\mathrm{sp}^\beta_i(z)$ is the expected number of integers $x\in[S, S+\xi_i]$ with $|x-\bar x_i|>z$, which is at most $(1+\xi_i)\mathbf{1}_{\{|S-\bar x_i|>z/2\}} + \xi_i\mathbf{1}_{\{\xi_i>z/2\}}$. Since X^+_J is independent of (J,S) and $\xi_i\le X^+_J$,

$$\mathrm{sp}^\beta_i(z) \le (1+\mu_+)\mathbb{P}(|S-\bar x_i|>z/2) + \mathbb{E}\big[X_+\mathbf{1}_{\{X_+>z/2\}}\big].$$

The last term tends to zero as $z\to\infty$. For the first, J is independent of the plus runs, and $\mathbb{P}(J>3N/\mu_-)\le \mathbb{P}(S^-_{\lfloor 3N/\mu_-\rfloor}<N)\to 0$ uniformly in $i\le N$ by Lemma 3.1(iv). On $\{J\le 3N/\mu_-\}$ we bound $|S-\bar x_i|$ as for sp^ρ_i, with $J-1$ in place of m. The plus part is controlled by ϖ_+ at the index $J-1$, and the number of minus runs by ϖ_- through the display above, since $J = J_-(i)$ and $|(J-1)\mu_- - i|\le |J\mu_- - i| + \mu_-$. Taking $z = ZB_N$ with Z large and then N large proves the lemma. □

Lemma 4.6 (one giant run)**.** *Assume (RV) and (SPAN), let the start be fresh with sign σ_0, let $0<\eta\le 1/2$, and put $g = \lceil \eta L_k\rceil$. Write $\kappa = \kappa^{(\sigma_0)}_+$ and $\gamma = \gamma^{(\sigma_0)}_+$. Then $\mathrm{Cen}_g(k) = 0$ when $\kappa(N-k)=0$, $\mathrm{Int}_g(k)=0$ when $\gamma(N-k)=0$, and*

$$\lim_{K\to\infty}\ \limsup_{N\to\infty}\ \sup_{k\,:\,L_k\ge KB_N,\ \kappa(N-k)>0}\Big|\frac{\mathrm{Cen}_g(k)}{\kappa(N-k)\,T_+(\lfloor L_k\rfloor)} - 1\Big| = 0.$$

If in addition (LR) holds for the plus sign, then also

$$\lim_{K\to\infty}\ \limsup_{N\to\infty}\ \sup_{k\,:\,L_k\ge KB_N,\ \gamma(N-k)>0}\Big|\frac{\mathrm{Int}_g(k)}{\gamma(N-k)\,f_+(\lfloor L_k\rfloor)} - 1\Big| = 0.$$

Proof. The two vanishing statements follow from (4.11). By (2.10) and (2.11), $\kappa(i)\to 1/\mu_-$, and by (4.4), $\gamma(i)/i\to s/\mu_-^2$. Hence

$$\kappa_* = \inf\{\kappa(i)\mid \kappa(i)>0\} > 0, \qquad \gamma_* = \inf\{\gamma(i)/(i+1)\mid \gamma(i)>0\} > 0.$$

Write $L = L_k$, $\hat{L} = \lfloor L \rfloor$ and $j = N - k$. Note that $N - j - x = L - (x - \bar{x}_j)$ by (4.1). Fix $\varepsilon > 0$ and take Z, N_0 such that Lemma 4.5 holds with $Z/2$ in place of Z. Since $\mathrm{sp}_i^{\rho}(z)$ and $\mathrm{sp}_i^{\beta}(z)$ decrease in z, the same bounds hold at ZB_N.

The term Cen_g. Let $\kappa(j) > 0$ and write $\mathrm{Cen}_g(k) - \kappa(j)T_+(\hat{L}) = \sum_x \rho_j(x)[T_+(N-j-x)\mathbf{1}_{\{N-j-x\ge g\}} - T_+(\hat{L})]$. If $|x - \bar{x}_j| \le ZB_N$, then $|N-j-x-\hat{L}| \le ZB_N + 1 \le (Z/K + 1/L)L$. Since $Z/K \le \eta/2$ for large K, the indicator is 1, and by Lemma 2.1(i) the bracket is at most $\omega_T(Z/K)T_+(\hat{L})$, where $\omega_T(\delta) \to 0$ as $\delta \to 0$ once N is large. For the other x the bracket is at most $T_+(g) + T_+(\hat{L}) \le C_\eta T_+(\hat{L})$, and their total weight is $\mathrm{sp}_j^{\rho}(ZB_N) \le \varepsilon \le (\varepsilon/\kappa_*)\kappa(j)$. Hence

$$\big|\mathrm{Cen}_g(k) - \kappa(j)T_+(\hat{L})\big| \le \big(\omega_T(Z/K) + C_\eta \varepsilon/\kappa_*\big)\kappa(j)T_+(\hat{L}).$$

The term Int_g. Let $\gamma(j) > 0$ and define the measure χ_j by

$$\chi_j(z) = \sum_{i<j} \sum_{x+x'=z} \rho_i(x)\beta_{j-i}(x').$$

It has total mass $\gamma(j)$ by (4.7), and

$$\mathrm{Int}_g(k) = \sum_z \chi_j(z) f_+(N-j-z)\mathbf{1}_{\{N-j-z\ge g\}}.$$

If $|z - \bar{x}_j| > ZB_N$, then the split (x, x') has $|x - \bar{x}_i| > ZB_N/2$ or $|x' - \bar{x}_{j-i}| > ZB_N/2$. So Lemma 4.5 and (4.7) bound the mass of these z by

$$\sum_{i<j}\Big[(1+\mu_+)\mathrm{sp}_i^{\rho}(ZB_N/2) + \kappa(i)\mathrm{sp}_{j-i}^{\beta}(ZB_N/2)\Big] \le (2+\mu_+)\, j\, \varepsilon \le \frac{(2+\mu_+)\varepsilon}{\gamma_*}\,\gamma(j).$$

Under (LR), $x \mapsto T_+(x)/x$ is regularly varying and $f_+(n) \sim \alpha_+ T_+(n)/n$. So Lemma 2.1(i) gives $|f_+(n)/f_+(\hat{L}) - 1| \le \omega_f(Z/K)$ for $|n - \hat{L}| \le ZB_N + 1$ and large N, and Lemma 2.1(ii) gives $f_+^*(g) \le C_\eta f_+(\hat{L})$. As for Cen_g,

$$\big|\mathrm{Int}_g(k) - \gamma(j)f_+(\hat{L})\big| \le \Big(\omega_f(Z/K) + \frac{(C_\eta+1)(2+\mu_+)\varepsilon}{\gamma_*}\Big)\gamma(j)f_+(\hat{L}).$$

Let $N \to \infty$, then $K \to \infty$, then $\varepsilon \to 0$. □

Proof of Theorem B(b) and (c). Write κ, γ as in Lemma 4.6, $L = L_k$, $\hat{L} = \lfloor L \rfloor$ and $j = N - k$. Note that $T_+(\hat{L}) \ge T_+(L)$. By (2.10) and (4.4) there is $J_0 \ge 1$ with $\kappa(j) \ge 1/(2\mu_-)$ and $\gamma(j) \ge js/(2\mu_-^2)$ for $j \ge J_0$. Fix

$$\eta = \min\Big\{\frac{1}{32},\ \frac{1}{2J_0+2}\Big\}, \qquad g = \lceil \eta L \rceil,$$

so that $r = 1/(16\eta) \ge 2$. Fix $\varepsilon > 0$. By (4.10),

$$\begin{aligned}\big|P_N(k) - G_N(k)\big| \le{}& \mathbb{P}(A_N = k,\ M_N < \eta L) + |\mathrm{Cen}_g(k) - \kappa(j)T_+(\hat{L})| \\ &+ |\mathrm{Int}_g(k) - \gamma(j)f_+(\hat{L})| + \mathbb{E}\big[(\mathcal{G}_g - 1)^+ \mathbf{1}_{\{A_N = k\}}\big].\end{aligned}$$

In the zone $k \ge pN + K\lambda_N B_N$ we have $L \ge k - pN \ge K\lambda_N B_N \ge KB_N$. By Lemma 4.6 the second and third terms are at most $\varepsilon[\kappa(j)T_+(\hat{L}) + \gamma(j)f_+(\hat{L})] = \varepsilon G_N(k)$ for $K \ge K_1(\varepsilon)$ and $N \ge N_1(K)$. It remains to compare the first and last terms with $G_N(k)$. We use (2.4) for the plus sign.

Small j. Let $j < J_0$. For large N we have $\eta L \le \eta N \le (N-j)/(j+1)$, so Lemma 2.7 gives $\mathbb{P}(A_N = k, M_N < \eta L) = 0$. If $G_N(k) = 0$, then $\kappa(j) = \gamma(j) = 0$, so $\mathrm{Cen}_g = \mathrm{Int}_g = 0$

and $\mathcal{G}_g = 0$ almost surely on $\{A_N = k\}$, and by (4.10) $P_N(k) = 0$. If $G_N(k) > 0$, then $G_N(k) \geq \min(\kappa_* T_+(\hat{L}), \gamma_* f_+(\hat{L})) \geq cT_+(L)/L$ by (LR). Also $L \geq N/2$ for large N. By Lemma 4.4 the last term is at most $CJ_0^2 T_+(L)^2$, and its ratio to $G_N(k)$ is at most $CJ_0^2 L T_+(L)$, which tends to zero since $\alpha_+ > 1$.

Large j. Let $j \geq J_0$. By (LR), $f_+(\hat{L}) \geq \alpha_+ T_+(\hat{L})/(2L)$ for large L. Since $js/\mu_-^2 = (N-L)/\mu_-$ by (4.1),

$$G_N(k) \geq \frac{T_+(L)}{2\mu_-}\Big(\frac{\alpha_+(N-L)}{2L} + 1\Big) \geq c_6 \frac{N}{L} T_+(L).$$

By Lemma 4.4, with $j+1 \leq N$, the last term divided by $G_N(k)$ is at most $C(j+1)T_+(L) \leq CNT_+(L) \leq CK^{-(\alpha_++1)/2}$ by (2.4), since $L/b_{+,N} \geq L/B_N \geq K$. By Lemma 4.3, for $K \geq C_0$, the first term divided by $G_N(k)$ is at most C times

$$\big(NT_+(L)\big)^{r-1} \frac{L}{b_{+,N}} + \frac{L}{b_{-,N}} e^{-cL/b_{-,N}} \frac{1}{NT_+(L)} + \frac{e^{-N^c}}{T_+(N)}.$$

The last part tends to zero. The first part is at most $C(L/b_{+,N})^{(1-\alpha_+)/2}$ by (2.4) and $r \geq 2$, and this is at most $CK^{(1-\alpha_+)/2}$. For the middle part put $x = L/b_{-,N}$ and $\varrho = b_{-,N}/b_{+,N}$, so $L/b_{+,N} = x\varrho$. By (2.4) the middle part is at most $Cx^{\alpha_++2}\varrho^{\alpha_++1}e^{-cx}$. If $\varrho \leq 1$, then $x \geq L/B_N \geq K$ and the bound is at most $C \sup_{x \geq K} x^{\alpha_++2}e^{-cx}$. If $\varrho > 1$, then $B_N = b_{-,N}$, $\lambda_N = 1 + \log \varrho$ and $x \geq K(1 + \log \varrho)$. For $cK \geq 2(\alpha_+ + 1)$ this gives $\varrho^{\alpha_++1}e^{-cx/2} \leq e^{(\alpha_++1)\log\varrho - cK(1+\log\varrho)/2} \leq e^{-cK/2}$, and the bound is at most $Ce^{-cK/2} \sup_x x^{\alpha_++2}e^{-cx/2}$. In every case the first term is at most $\varepsilon G_N(k)$ for K large and then N large. This proves the ratio form of (b).

For the additive form note that $\gamma(j) \leq (1+\mu_+)j \leq (1+\mu_+)N$ by (4.2), $(N-L)/\mu_- \leq N/\mu_-$, $f_+(\hat{L}) \leq 2C_{\mathrm{LR}} T_+(\hat{L})/L$ by (LR), and $\kappa \leq 1 \leq N/L$. Hence

$$G_N(k) \leq C_{\mathrm{G}} \widehat{G}_N(k), \qquad \overline{G}_N(k) \leq C_{\mathrm{G}} \widehat{G}_N(k), \tag{4.14}$$

with C_{G} depending only on the laws, and the same holds at every horizon n and for both starting signs whenever $L_k^{(n)} \geq 1$. The ratio form gives the additive form. Replacing $\gamma(j)$ by $js/\mu_-^2 = (N-L)/\mu_-$ changes $G_N(k)$ by $|\gamma(j) - js/\mu_-^2| f_+(\hat{L}) \leq 2C_{\mathrm{LR}}|\gamma(j) - js/\mu_-^2|\, T_+(\hat{L})/L$. By (4.4) the factor $|\gamma(j) - js/\mu_-^2|$ is at most εN for $j \geq J_\varepsilon$ and at most a constant C_ε for $j < J_\varepsilon$, and in the second case $L \geq N/2$. In both cases the change is at most $(2C_{\mathrm{LR}}\varepsilon + 4C_{\mathrm{LR}}C_\varepsilon/N)\widehat{G}_N(k)$. This proves (b).

For (c), let $\delta_N \to 0$ and $0 \leq j \leq \delta_N N$. Then $k - pN \geq (1-p)N/2$ for large N, and $\lambda_N B_N = o(N)$ by Lemma 2.1(v). So for every K these bins lie in the zone of (b) once N is large, and (b) gives $\sup_{j \leq \delta_N N} |P_N(N-j)/G_N(N-j) - 1| \to 0$. Here $L = N - js/\mu_- \geq N(1 - \delta_N s/\mu_-)$, so $T_+(\hat{L})/T_+(N) \to 1$ uniformly by Lemma 2.1(i). By (LR) and $\gamma(j) \leq (1+\mu_+)j$,

$$\frac{\gamma(j) f_+(\hat{L})}{\kappa(j) T_+(N)} \leq \frac{C(j+1)}{\kappa_* N} \leq \frac{C(\delta_N + 1/N)}{\kappa_*} \to 0$$

for $\kappa(j) > 0$. Hence $G_N(N-j)/(\kappa(j)T_+(N)) \to 1$ uniformly over these j, which proves the second statement of (c). For the first, the same bound gives $\gamma(j) f_+(\hat{L}) \leq C(\delta_N + 1/N)T_+(N)$ for all $j \leq \delta_N N$, so $|G_N(N-j) - \kappa(j)T_+(N)| = o(T_+(N))$ uniformly. Also $\widehat{G}_N(N-j) \leq 2T_+(\hat{L}) \leq 4T_+(N)$ there, and the additive form of (b) gives $|P_N(N-j) - G_N(N-j)| = o(T_+(N))$ uniformly. This proves (c). □

Corollary 4.7 (the lower bound)**.** *Under the hypotheses of Theorem B(b),*

$$\lim_{K \to \infty} \liminf_{N \to \infty} \inf_{pN + KB_N \leq k \leq N,\ G_N(k) > 0} \frac{P_N(k)}{G_N(k)} \geq 1.$$

That is, for every $\varepsilon > 0$ there is K such that $P_N(k) \geq (1-\varepsilon)G_N(k)$ for all $pN + KB_N \leq k \leq N$ and all large N.

Proof. The first term of (4.10) is nonnegative, and it is the only term whose bound in the proof of (b) used λ_N. So $P_N(k) \geq \mathrm{Cen}_g(k) + \mathrm{Int}_g(k) - \mathbb{E}[(\mathcal{G}_g - 1)^+ \mathbf{1}_{\{A_N = k\}}]$, with J_0, η and g as in that proof. For $pN + KB_N \leq k \leq N$ we have $L_k \geq KB_N$. Lemma 4.6 needs only this, and it gives $\mathrm{Cen}_g(k) + \mathrm{Int}_g(k) \geq (1-\varepsilon)G_N(k)$ for K large and then N large. The bounds for the last term in the proof of (b) also use only $L_k \geq KB_N$, and they make it at most $\varepsilon G_N(k)$. □

4.4 Consequences and an example

Corollary 4.8 (the infinite density)**.** *Assume (RV) and (SPAN) for both signs and (LR) for the plus sign. Fix $v \in (p, 1)$ and put $\ell = (v-p)/(1-p)$. For a fresh start of either sign,*

$$\frac{P_N(\lfloor vN \rfloor)}{T_+(N)} \longrightarrow I_+(v) = \frac{1}{\mu_-}\Big[\alpha_+(1-\ell)\ell^{-\alpha_+-1} + \ell^{-\alpha_+}\Big].$$

The function I_+ is not integrable at $v \downarrow p$.

Proof. Let $k = \lfloor vN \rfloor$. Then $k - pN \sim (v-p)N$, which exceeds $K\lambda_N B_N$ for every K once N is large, by Lemma 2.1(v). So Theorem B(b) gives $P_N(k) \sim G_N(k)$. Here $j = N - k \sim (1-v)N = (1-p)(1-\ell)N$ and $L_k \sim \ell N$. By (4.4), $\gamma_+^{(\sigma_0)}(j) \sim (1-\ell)N/\mu_-$. By (LR) and (RV), $f_+(\lfloor L_k \rfloor) \sim \alpha_+ \ell^{-\alpha_+-1} T_+(N)/N$ and $T_+(\lfloor L_k \rfloor) \sim \ell^{-\alpha_+} T_+(N)$. By (2.10), $\kappa_+^{(\sigma_0)}(j) \to 1/\mu_-$. Adding the two terms of (4.3) gives $I_+(v)$. Near $v = p$ the first term is of order $(v-p)^{-\alpha_+-1}$, which is not integrable. □

Rebenshtok, Denisov, Hänggi and Barkai [35, Eqs. (8) and (10)] (see also [36, Eqs. (38) and (52)]) obtained for the two-state velocity model an infinite density proportional to $|v|^{-1-\alpha}[\alpha - (\alpha-1)|v|]$. They derived it formally, in continuous time, from the Montroll and Weiss equation through the moments. Since $\alpha(1-\ell)\ell^{-\alpha-1} + \ell^{-\alpha} = \ell^{-\alpha-1}[\alpha - (\alpha-1)\ell]$, the function I_+ has this shape, and the term $-(\alpha-1)\ell$ comes from the censored final run. So their formula already contains the censored-run term. Vezzani, Barkai and Burioni [38] derive the same tail from a single big jump. What is new here is the proof on the lattice at fixed time, uniform up to the end with the exact kernel at bounded j, together with the boundary layer and, in Section 5, the preparation. When $b_{+,N} = B_N$, Theorem A(c) on $|k - pN| \leq KB_N$ and Theorem B(b) on $k - pN \geq KB_N$ together give every bin on the plus side with relative error $o(1)$ in the double limit. The matching between the stable bulk and the infinite density, which is formal in the physics literature, is this statement.

Remark 4.9 (the double limit)**.** The bulk profile and the big-jump kernel are different expressions on their common ground. Neither the bulk profile nor the big-jump kernel nor their sum has relative error $o(1)$ uniformly over the plus side. Assume $b_{+,N} = B_N$ and (LR) for the plus sign, and let $k = \lceil pN + KB_N \rceil$ with K fixed. By Theorem A(c), $B_N P_N(k)/h_N(K) \to 1$, and $h_N(K)$ is bounded below by Lemma 2.2(d). On the other hand, $j = N - k \sim (1-p)N$ and $L_k \sim KB_N/(1-p)$, so by (4.4) and (LR)

$$B_N G_N(k) = \frac{\alpha_+}{s}\,\frac{NT_+(L_k)}{K}\,(1 + o(1)) = \alpha_+(1-p)^{\alpha_+} K^{-\alpha_+-1}(1 + o(1)),$$

where the second step uses $NT_+(b_{+,N}) \to s$ and Lemma 2.1(i), and the censored term is negligible because $B_N T_+(L_k) \to 0$. The right side is the leading term of the tail of h_N, but it is not $h_N(K)$. So $P_N(k)/G_N(k)$ does not tend to one at fixed K, and $P_N(k)/(B_N^{-1}h_N(K) + G_N(k))$ tends to a limit below one. Section 8 reports the check. The ratio P_N/G_N tends to one only when $K \to \infty$ after $N \to \infty$, which is the form of Theorem B(b).

Remark 4.10 (the start of the zone)**.** The zone of Theorem B(b) cannot start at $pN+KB_N$ with K fixed when $\theta_{+,N} \to 0$. Some factor in front of B_N that tends to infinity is needed. Assume (LR) for the plus sign and let $\theta_{+,N} \to 0$ along a subsequence. Fix K and let $k = \lceil pN + KB_N \rceil$. By Theorem A(c) and Lemma 2.2(d), $B_N P_N(k)$ is bounded below. By (LR) and the bounds $\gamma(j) \le (1+\mu_+)N$ and $\kappa \le 1$,

$$B_N G_N(k) \le C\, NT_+(L_k)\, \frac{B_N}{L_k} + B_N T_+(L_k) \le \frac{C}{K}\, NT_+(L_k) + B_N T_+(B_N).$$

Since $L_k \ge KB_N = Kb_{+,N}/\theta_{+,N}$, the bound (2.4) gives $NT_+(L_k) \le 4s\,(K/\theta_{+,N})^{-(\alpha_+ +1)/2}$, which tends to zero, and $B_N T_+(B_N) \to 0$. So $P_N(k)/G_N(k) \to \infty$ along the subsequence. The bins there are carried by short minus runs, the second term of Lemma 4.3, and not by a giant plus run. We do not show that the factor must be as large as λ_N. A left-tail bound of order $\exp(-cy^\beta)$ in place of Lemma 3.1(iv) would replace λ_N by $\lambda_N^{1/\beta}$, and we do not pursue this. The lower bound needs no factor at all (Corollary 4.7).

Example 4.11 ((RV) without (LR))**.** Fix $1 < \alpha < 2$ and let $n_i = 2^{2^i}$ for $i \ge 1$ and $n_0 = 0$. Put $h(n) = 1/\log n_i$ for $n_{i-1} < n \le n_i$, and

$$T(n) = \frac{n^{-\alpha}\big(1+h(n)\big)}{1+h(1)}, \qquad f(n) = T(n) - T(n+1) \qquad (n \ge 1).$$

Let both run laws be this law, so $p = 1/2$ and $L_k = 2k - N$. Then (RV) and (SPAN) hold and (LR) fails. Let $N_i = 2n_i$ and $k_i = 3n_i/2$. Then $k_i - pN_i = n_i/2$, so k_i lies in the zone of Theorem B(b) for every K once i is large, and

$$\frac{P_{N_i}(k_i)}{G_{N_i}(k_i)} \to 0.$$

So the conclusion of Theorem B(b), whose kernel reads f_+ at a single point, fails under (RV) and (SPAN) alone, while (a) holds.

Verification. Write $h_i = 1/\log n_i = 1/(2^i \log 2)$, so $h_{i+1} = h_i/2$ and h is nonincreasing and tends to zero. Put $c = 1/(1+h(1))$. Then $T(1) = 1$, and T is decreasing and tends to zero, so f is a probability mass function with $f(n) > 0$ for all n. This gives (SPAN), and $T(n) \sim cn^{-\alpha}$ gives (RV). For $n_{i-1} < n < n_i$ the points $n, n+1$ lie in the same block, so $f(n) = c(1+h_i)(n^{-\alpha} - (n+1)^{-\alpha}) \le 2c\alpha n^{-\alpha-1}$. At the block ends,

$$f(n_i) = c\Big[\big(n_i^{-\alpha} - (n_i+1)^{-\alpha}\big)(1+h_{i+1}) + n_i^{-\alpha}\,\frac{h_i}{2}\Big],$$

so $\frac{c}{2} h_i n_i^{-\alpha} \le f(n_i) \le C h_i n_i^{-\alpha}$. Since $n_i h_i \to \infty$, $n_i f(n_i)/T(n_i) \to \infty$ and (LR) fails.

Let $N = N_i$, $k = k_i$, $j = n_i/2$, so $L_k = n_i$. By (4.4), $\gamma(j) \ge n_i/(2\mu)$ for large i, where μ is the common mean. Hence

$$G_N(k) \ge \gamma(j) f(n_i) \ge \frac{c}{4\mu}\, h_i n_i^{1-\alpha}.$$

For the upper bound take $\eta = 1/32$ and $g = \lceil n_i/32 \rceil$ in Lemma 4.1, and drop the last term of (4.10). Lemma 4.3 uses only (RV) and (SPAN). With $L_k = n_i$, $N = 2n_i$ and $r = 2$ it bounds the first term by $Cn_i^{2-2\alpha}$ plus terms that decay faster than every power of n_i, and $n_i^{2-2\alpha} = o(h_i n_i^{1-\alpha})$. By (4.11), $\mathrm{Cen}_g(k) \le T(g) \le Cn_i^{-\alpha}$. For Int_g note that $n_{i-1} = \sqrt{n_i} < n_i/32$ and $n_{i+1} = n_i^2 > 2n_i$ for large i. So n_i is the only block end in $[g, N]$, and $f(n) \le Cn_i^{-\alpha-1}$ for the other n there. With the measure χ_j of the proof of Lemma 4.6,

$$\mathrm{Int}_g(k) \le Cn_i^{-\alpha-1}\gamma(j) + f(n_i) \sup_z \chi_j(z) \le Cn_i^{-\alpha} + Ch_i n_i^{-\alpha} \sup_z \chi_j(z).$$

It remains to show $\sup_z \chi_j(z) = o(j)$. Since $\chi_j = \sum_{i'<j} \rho_{i'} * \beta_{j-i'}$ and the two factors have masses at most 1 and $1+\mu$, $\sup_z(\rho_{i'} * \beta_{j-i'})(z) \le \min\{(1+\mu)\sup_x \rho_{i'}(x),\ \sup_x \beta_{j-i'}(x)\}$. One of i' and $j-i'$ is at least $j/2$. So it suffices that $\sup_x \rho_n(x) \to 0$ and $\sup_x \beta_n(x) \to 0$ as $n \to \infty$. Let $m_1 = \lfloor n/(2\mu) \rfloor$. By (2.13) and Lemma 3.1(ii), $\rho_n(x) \le \mathbb{P}(S_{m_1} \ge n) + \bar{u}/a(m_1 - 1)$, because a renewal at n with at most m_1 runs forces $S_{m_1} \ge n$. Both terms tend to zero, by the law of large numbers and since $a(m) \to \infty$. By (4.8) and the independence of X_J^+, $\beta_n(x) \le \sum_{r \ge 0} T(r)\mathbb{P}(S_{J-1} = x - r)$ with $T(0) = 1$, so $\beta_n(x) \le (1+\mu)[\mathbb{P}(J \le m_1) + \bar{u}/a(m_1 - 1)]$, and $\mathbb{P}(J \le m_1) \le \mathbb{P}(S_{m_1} \ge n)$, which tends to zero. Hence $P_N(k) = o(h_i n_i^{1-\alpha})$, and the ratio tends to zero. □

The spike at n_i enters $G_N(k_i)$ at full weight but enters $P_N(k_i)$ only through a window of width of order B_N around L_k, as in the local big-jump asymptotics of Denisov, Dieker and Shneer [11, Theorem 9.1], which assume a local hypothesis.

Remark 4.12 (the hypothesis (LR))**.** The proof of Theorem B(b) uses (LR) to replace $f_+(n)$ by $f_+(\lfloor L_k \rfloor)$ for $|n - L_k| \le ZB_N$, to bound $f_+^*(\eta L_k)$ by $CT_+(L_k)/L_k$, and to bound G_N below. The sharp condition is presumably a regularity of f_+ over windows of width of order B_N, and we leave it open. At the junction $L_k \asymp KB_N$ such windows have relative width of order $1/K$, and window regularity at relative width $o(1)$ together with (RV) implies (LR), since $T_+(n) - T_+((1+\delta)n)$ is then both $\delta n f_+(n)(1 + o_\delta(1))$ and $T_+(n)(\alpha_+\delta + O(\delta^2))$.

5 Preparation

The first run is the only part of the walk that the two fresh laws do not describe, and Proposition 2.4 says exactly how it enters. This section turns that identity into a criterion for each zone of the row.

5.1 Statements

For a sign σ and $b \in \mathbb{Z}$ define the directional defect

$$\mathcal{D}_N^\sigma(b) = B_N \sum_{1 \le m < N/2} \nu_\sigma(N-m)\, \Pi_m^{\bar{\sigma}}(b), \qquad \Delta_{\sigma,N} = \sup_{b \ge 1} \mathcal{D}_N^\sigma(b), \tag{5.1}$$

with $\Pi_m^{\bar{\sigma}}(b) = 0$ for $b \notin \{0, \dots, m\}$. Note that $\Pi_m^{\bar{\sigma}}(0) = 0$ for $m \ge 1$, since a fresh walk has at least one slot of its own sign. So $\mathcal{D}_N^\sigma(b) = 0$ for $b \le 0$ and for $b \ge N/2$. Take a first run of sign σ that ends within the last $N/2$ slots. Times B_N, its contribution to bin k is $\mathcal{D}_N^+(N-k)$ for $\sigma = +$ and $\mathcal{D}_N^-(k)$ for $\sigma = -$. So the argument b is the number of slots of the opposite sign. We say that the preparation satisfies the *bulk criterion* if $B_n \mathbb{P}(D = n) \to 0$.

For the big-jump zone put $\pi_\sigma = \mathbb{P}(\Sigma = \sigma)$, let L_k be as in (4.1) and $\widehat{G}_N$ as in (4.6), and define for $pN < k \le N$ and $j \ge 0$

$$\kappa_+(j) = \pi_+ u_-(j)\mathbf{1}_{\{j \ge 1\}} + \sum_{d=1}^{j} \nu_-(d)\, u_-(j-d), \tag{5.2}$$

$$G_N^\nu(k) = \frac{N - L_k}{\mu_-}\, f_+(\lfloor L_k \rfloor) + T_+(\lfloor L_k \rfloor)\, \kappa_+(N-k), \tag{5.3}$$

$$\Psi_N(k) = \sum_{L_k/2 \le d < N} \nu_+(d)\, F_{N-d}^-(k-d). \tag{5.4}$$

The kernel $\kappa_+(j)$ is the probability that a plus run other than the first run starts immediately after the j-th minus slot. For a fresh start of sign σ_0 it equals $\kappa_+^{(\sigma_0)}(j)$ for $j \ge 1$, by (2.11). The kernel G_N^ν uses the limit coefficient $(N - L_k)/\mu_-$. The term $\Psi_N(k)$ is the

exact contribution of a first plus run that supplies at least half of the big jump. We also use two conditions on ν_+. The first is the regularity condition

$$\begin{aligned} &\nu_+(n) > 0 \ \text{ and } \ \sup_{d \ge n/2} \nu_+(d) \le C_\nu \, \nu_+(n) \ \text{ for all large } n, \\ &\lim_{\delta \downarrow 0} \ \limsup_{n \to \infty} \ \sup_{|d-n| \le \delta n} \Big| \frac{\nu_+(d)}{\nu_+(n)} - 1 \Big| = 0. \end{aligned} \tag{Rν}$$

Every regularly varying ν_+ with a negative index satisfies (Rν), by Lemma 2.1(i) and (ii). The second is a condition on the window averages

$$W_n = \frac{1}{B_n} \sum_{|d-n| \le B_n} \nu_+(d).$$

Theorem C (preparation)**.** *Fix a preparation (Σ, D).*

(a) (The exact defect.) Assume (RV) and (SPAN) for both signs. Uniformly over $0 \le k \le N$,

$$\begin{aligned} B_N P_N(k) = h_N(x_{N,k}) + B_N Q_+(N) \mathbf{1}_{\{k=N\}} + B_N Q_-(N) \mathbf{1}_{\{k=0\}} \\ + \mathcal{D}_N^+(N-k) + \mathcal{D}_N^-(k) + o(1). \end{aligned}$$

(b) (The bulk criterion.) Under the hypotheses of (a),

$$\begin{aligned} &\sup_{1 \le k < N} \big| B_N P_N(k) - h_N(x_{N,k}) \big| = \max\big(\Delta_{+,N}, \Delta_{-,N}\big) + o(1), \\ &\sup_{0 \le k \le N} \big| B_N P_N(k) - h_N(x_{N,k}) \big| = \max\big(\Delta_{+,N}, \Delta_{-,N}, B_N Q_+(N), B_N Q_-(N)\big) + o(1). \end{aligned}$$

The first error tends to zero if and only if $B_n \mathbb{P}(D = n) \to 0$, and the second if and only if $B_n \mathbb{P}(D \ge n) \to 0$. For every fixed $\varepsilon > 0$ the approximation holds on $\varepsilon N \le k \le (1-\varepsilon)N$ with no condition on the preparation.

(c) (The big-jump zone.) Assume (RV) and (SPAN) for both signs and (LR) for the plus sign, and let λ_N be as in (4.5)*. Then*

$$\lim_{K \to \infty} \ \limsup_{N \to \infty} \ \sup_{pN + K\lambda_N B_N \le k \le N} \frac{\big| P_N(k) - Q_+(N) \mathbf{1}_{\{k=N\}} - G_N^\nu(k) - \Psi_N(k) \big|}{\widehat{G}_N(k)} = 0.$$

The same holds with Ψ_N removed if and only if $\nu_+(n) = o(T_+(n))$. If ν_+ satisfies (Rν), it holds with $\Psi_N(k)$ replaced by $\nu_+(\lfloor L_k \rfloor)(1 + \mu_+ u_-(N-k)) \mathbf{1}_{\{k<N\}}$ and $\widehat{G}_N(k)$ replaced by $\widehat{G}_N(k) + \nu_+(\lfloor L_k \rfloor)$. Now restrict to the interior zones $\varepsilon N \le L_k \le (1-\varepsilon)N$, and ask that $\Psi_N = o(\widehat{G}_N)$ there for every $\varepsilon > 0$. The condition $\nu_+(n) = o(T_+(n))$ is sufficient for this. The window condition $W_n = o(T_+(n))$ is necessary, and under (LR) for both signs it is also sufficient.

(d) (The boundary layer.) Assume (RV) for the plus sign. For every fixed $j \ge 0$,

$$P_N(N-j) = Q_+(N) \mathbf{1}_{\{j=0\}} + T_+(N) \, \kappa_+(j) + B_N^{-1} \mathcal{D}_N^+(j) + o(T_+(N)),$$

where $B_N^{-1} \mathcal{D}_N^+(j) = \sum_{1 \le m < N/2} \nu_+(N-m) \Pi_m^-(j)$ does not involve the scales. Hence $P_N(N-j)/T_+(N) \to \kappa_+(j)$ for one $j \ge 1$, equivalently for every $j \ge 1$, if and only if $\nu_+(n) = o(T_+(n))$. If ν_+ satisfies (Rν), then $B_N^{-1} \mathcal{D}_N^+(j) = \nu_+(N)\big(1 + \mu_+ u_-(j)\big)(1 + o(1))$ for every $j \ge 1$.

At the plus end the criteria are nested. Since $\nu_+ \le Q_+$, the atom condition $Q_+(n) = o(T_+(n))$ implies the big-jump condition $\nu_+(n) = o(T_+(n))$. Since $B_n T_+(n) \to 0$, the big-jump condition implies $B_n \nu_+(n) \to 0$, which is the plus half of the bulk criterion.

The zone in (c) starts where the zone of Theorem B(b) starts, and by Remark 4.10 the factor λ_N cannot be dropped there, already for a fresh start. At $k = N$ we have $\Psi_N(N) = 0$, $G^{\nu}_N(N) = 0$ and $\mathcal{D}^+_N(0) = 0$, so (c) and (d) both reduce to the identity $P_N(N) = Q_+(N)$ of Proposition 2.4.

Remark 5.1 (the prepared boundary kernel)**.** By (2.10) and dominated convergence, $\kappa_+(j) \to 1/\mu_-$, the fresh value. With generating functions F, $U = 1/(1-F)$ and $\widehat{\nu}_-$ of f_-, u_- and ν_-, the identity $\kappa_+ = u_-$ on $j \ge 1$ reads $\widehat{\nu}_- U = \pi_-(U-1)$, that is, $\nu_- = \pi_- f_-$. So the prepared kernel is the fresh one exactly when the first run, given that it is a minus run, has the law of X_-.

5.2 Tools

For a horizon n and a bin $k > pn$ we write $L^{(n)}_k = (k-pn)/(1-p)$ for the quantity (4.1) at the horizon n, and $\widehat{G}_n(k)$, $G^{(\sigma_0)}_n(k)$ and λ_n for (4.6), (4.3) and (4.5) at the horizon n. Recall from Lemma 2.1(v) that B_n is nondecreasing and regularly varying with index $1/\alpha_*$.

Fix $c \in (0,1)$. We have $B_n/b_{+,n} = \max(1, b_{-,n}/b_{+,n})$, and the ratio $b_{-,n}/b_{+,n}$ is regularly varying. By Lemma 2.1(i) its logarithm changes by $o(1)$, uniformly in $n \in [cN, N]$, when n is replaced by N. The map $x \mapsto \log\max(1,x)$ changes by at most $|\log(x/y)|$ when x is replaced by y. Hence $\lambda_n = \lambda_N + o(1)$ uniformly in $n \in [cN, N]$, and since $\lambda_n \ge 1$,

$$\lambda_n \le 2\lambda_N \qquad (cN \le n \le N) \tag{5.5}$$

once N is large. By Lemma 2.1(v), $\lambda_N B_N = o(N)$. So for each $c > 0$ the bins with $k - pN \ge cN$ lie in the zone of Theorem C(c) for every K once N is large.

Lemma 5.2 (tightness of the count)**.** *Assume (RV) and (SPAN). For a fresh start of either sign,* $\sup_{n\ge1} \mathbb{P}(|A_n - pn| > RB_n) \to 0$ *as* $R \to \infty$.

Proof. Fix the starting sign and put $\varepsilon_n = \sup_k |B_n F_n(k) - h_n(x_{n,k})|$, which tends to zero by Theorem A(a). Let $R \ge 1$ and let k run over the integers with $|k - pn| \le RB_n$. The points $x_{n,k}$ are spaced $1/B_n$ apart, and the intervals $[x_{n,k}, x_{n,k} + 1/B_n)$ cover $[-R + 1/B_n, R]$. On each of them h_n exceeds $h_n(x_{n,k})$ by at most C_2/B_n, by Lemma 2.2(a). There are at most $2RB_n + 1$ such k. Hence, for $B_n \ge 1$,

$$\begin{aligned}\mathbb{P}(|A_n - pn| \le RB_n) &\ge \sum_k \frac{h_n(x_{n,k}) - \varepsilon_n}{B_n} \ge \int_{-R+1/B_n}^{R} h_n - 3R\Big(\varepsilon_n + \frac{C_2}{B_n}\Big) \\ &\ge 1 - \varepsilon'_h(R-1) - 3R\Big(\varepsilon_n + \frac{C_2}{B_n}\Big).\end{aligned}$$

So $\limsup_n \mathbb{P}(|A_n - pn| > RB_n) \le \varepsilon'_h(R-1)$, which tends to zero by Lemma 2.2(b). For each of the finitely many remaining n the probability is zero once $RB_n > n$. □

The measure $m \mapsto \Pi^-_m(j)$ of the Green identity sits within $O(B_N)$ of $j/(1-p)$, the time at which a typical walk collects j minus slots.

Lemma 5.3 (the Green measure is concentrated)**.** *Assume (RV) and (SPAN). For every $\varepsilon > 0$ there are R and N_0 such that for all $N \ge N_0$ and all $1 \le j \le N$,*

$$\sum_{m\ge1:\ |m - j/(1-p)| > RB_N} \Pi^-_m(j) \le \varepsilon.$$

Proof. Take a fresh minus walk continued forever, as in the proof of Lemma 2.5. Its minus runs $X_1^-, X_2^-, \dots$ and its plus runs $X_1^+, X_2^+, \dots$ are independent, and the i-th plus run comes right after the i-th minus run. Let C_m be the number of minus slots among the first m, let t_j be the slot of the j-th minus slot, and let r be the index of the minus run that contains it. Put $\xi_j = X_r^+$ if $S_r^- = j$ and $\xi_j = 0$ otherwise. As in the proof of Lemma 2.5, $C_m = j$ exactly for $t_j \le m \le t_j + \xi_j$. The index r, the event $\{S_r^- = j\}$ and $t_j = j + X_1^+ + \cdots + X_{r-1}^+$ are functions of the minus runs and of $X_1^+, \dots, X_{r-1}^+$. So X_r^+ is independent of them and has the law of X_+.

Put $c = j/(1-p)$, $\mathcal{S} = \{m : |m - c| > RB_N\}$ and $\mathcal{V} = \{m : |m - c| \le RB_N/2\}$. If $t_j \in \mathcal{V}$ and $\xi_j < RB_N/2$, then the slots $t_j, \dots, t_j + \xi_j$ all lie outside $\mathcal{S}$. Hence, by Tonelli's theorem,

$$\sum_{m \in \mathcal{S}} \Pi_m^-(j) = \mathbb{E}\,\#\{m \in \mathcal{S} : C_m = j\} \le \mathbb{E}\big[(1 + X_r^+)\mathbf{1}_{\{t_j \notin \mathcal{V}\}}\big] + \mathbb{E}\big[(1 + X_r^+)\mathbf{1}_{\{X_r^+ \ge RB_N/2\}}\big].$$

By the independence just noted, the first term is $(1 + \mu_+)\mathbb{P}(t_j \notin \mathcal{V})$ and the second is $\mathbb{E}[(1 + X_+)\mathbf{1}_{\{X_+ \ge RB_N/2\}}]$, which tends to zero as $N \to \infty$ because $\mu_+ < \infty$.

It remains to bound $\mathbb{P}(t_j \notin \mathcal{V})$. Let $m_1 = \lfloor c + RB_N/2 \rfloor$ and $m_2 = \lceil c - RB_N/2 \rceil$. If $t_j \notin \mathcal{V}$, then $t_j > m_1$ or $t_j \le m_2$. Now $\{t_j > m_1\} = \{C_{m_1} < j\} = \{A_{m_1} > m_1 - j\}$, and $m_1 - j - pm_1 = (1-p)m_1 - j \ge (1-p)(RB_N/2 - 1) \ge (1-p)RB_N/4$ for large N. In the same way $\{t_j \le m_2\} = \{A_{m_2} \le m_2 - j\}$ and $(1-p)m_2 - j \le -(1-p)RB_N/4$, and this event is empty if $m_2 < 1$. For $N \ge N_0$ we have $RB_N \le N$, so $m_1, m_2 \le c_0 N$ with $c_0 = 2 + 1/(1-p)$, and $B_{m_i} \le B_{c_0 N} \le c_1 B_N$ for some c_1 by Lemma 2.1(ii). Hence

$$\mathbb{P}(t_j \notin \mathcal{V}) \le 2 \sup_{n \ge 1} \mathbb{P}\Big(|A_n - pn| > \frac{(1-p)R}{4c_1}\, B_n\Big),$$

which is small for large R by Lemma 5.2. □

Finally we record the fresh big-jump asymptotics at a general horizon. For a horizon n and a bin $k \le n$ with $L = L_k^{(n)} \ge 1$ let $\overline{G}_n^{(\sigma_0)}(k) = (n-L)f_+(\lfloor L \rfloor)/\mu_- + T_+(\lfloor L \rfloor)\kappa_+^{(\sigma_0)}(n-k)$ be the kernel (4.3) at the horizon n with the limit coefficient. Assume (RV) and (SPAN) for both signs and (LR) for the plus sign. By the last clause of Theorem B(b), applied at the horizon n, for every $\varepsilon > 0$ there are K_0 and n_0 such that

$$\big|F_n^{\sigma_0}(k) - \overline{G}_n^{(\sigma_0)}(k)\big| \le \varepsilon\, \widehat{G}_n(k) \qquad (n \ge n_0,\ pn + K_0 \lambda_n B_n \le k \le n) \tag{5.6}$$

for both starting signs, and by (4.14) both $\overline{G}_n^{(\sigma_0)}(k)$ and $G_n^{(\sigma_0)}(k)$ are at most $C_{\mathrm{G}} \widehat{G}_n(k)$ whenever $L_k^{(n)} \ge 1$.

5.3 The bulk, parts (a) and (b)

Proof of Theorem C(a). By the definition of Π_m^σ we have $F_{N-d}^-(k-d) = \Pi_{N-d}^-(N-k)$ and $F_{N-d}^+(k) = \Pi_{N-d}^+(k)$. Fix $J \ge 1$. By Proposition 2.4, for $N > 2J$,

$$B_N P_N(k) = B_N Q_+(N)\mathbf{1}_{\{k=N\}} + B_N Q_-(N)\mathbf{1}_{\{k=0\}} + E_1(k) + E_2(k) + E_3(k),$$

where E_1, E_2 and E_3 collect the first-run lengths $d \le J$, $J < d \le N/2$ and $N/2 < d < N$, each multiplied by B_N. In E_3 put $m = N - d$, so that $1 \le m < N/2$. By (5.1) this gives exactly $E_3(k) = \mathcal{D}_N^+(N-k) + \mathcal{D}_N^-(k)$.

For E_2 the horizon is $N - d \ge N/2$. By Theorem A(b) there is C_0 with $B_n F_n^{\sigma_0}(k) \le C_0$ for all n, k and σ_0. By Lemma 2.1(v), $B_N/B_{N-d} \le B_N/B_{\lceil N/2 \rceil} \le C$. Hence $0 \le E_2(k) \le CC_0\mathbb{P}(D > J)$ for all k.

For E_1 we claim that for each fixed $d \geq 1$

$$\sup_{k\in\mathbb{Z}}\big|B_N F^-_{N-d}(k-d) - h_N(x_{N,k})\big| \to 0, \qquad \sup_{k\in\mathbb{Z}}\big|B_N F^+_{N-d}(k) - h_N(x_{N,k})\big| \to 0. \tag{5.7}$$

We prove the first, and the second is the same with pd in place of $-(1-p)d$ below. Put $n = N - d$. By Theorem A(a), $B_n F^-_n(k-d) = h_n(y_k) + o(1)$ uniformly in k, with $y_k = (k - d - pn)/B_n = (k - pN - (1-p)d)/B_n$. Also $B_N/B_n \to 1$ and h_n is bounded. By Lemma 2.2(c), $\|h_n - h_N\|_\infty \to 0$. Fix R. If $|x_{N,k}| \leq R$, then $|y_k - x_{N,k}| \leq R|B_N/B_n - 1| + (1-p)d/B_n \to 0$, and the Lipschitz bound of Lemma 2.2(a) gives $|h_N(y_k) - h_N(x_{N,k})| \to 0$. If $|x_{N,k}| > R$, then $|y_k| \geq R/2$ for large N, and both values are at most $\varepsilon_h(R/2)$. Let $N \to \infty$ and then $R \to \infty$. This proves (5.7). Summing it over the finitely many $d \leq J$ gives $\sup_k |E_1(k) - \mathbb{P}(D \leq J) h_N(x_{N,k})| \to 0$.

Let $e_N(k)$ be the difference between $B_N P_N(k)$ and the right side of (a) without the term $o(1)$. Collecting the three parts, $\limsup_{N\to\infty} \sup_{0\leq k\leq N} |e_N(k)| \leq (C_1 + CC_0)\mathbb{P}(D > J)$, with C_1 the bound of Lemma 2.2(a). This tends to zero as $J \to \infty$, since D is finite almost surely. □

Proof of Theorem C(b). The two defects are nonnegative. If $\mathcal{D}^+_N(N-k) \neq 0$, then some $m < N/2$ has $\Pi^-_m(N-k) > 0$, so $N - k \leq m < N/2$ and $k > N/2$. In the same way $\mathcal{D}^-_N(k) \neq 0$ forces $k < N/2$. So the two defects have disjoint supports in $1 \leq k < N$, and as k runs over $1 \leq k < N$ the arguments $N - k$ and k run over all b at which $\mathcal{D}^\sigma_N(b)$ can be positive. Hence $\sup_{1\leq k<N}\big(\mathcal{D}^+_N(N-k) + \mathcal{D}^-_N(k)\big) = \max(\Delta_{+,N}, \Delta_{-,N})$. Part (a) and the inequality $|\sup|f+g| - \sup|f|| \leq \sup|g|$ then give

$$\sup_{1\leq k<N}\big|B_N P_N(k) - h_N(x_{N,k})\big| = \max(\Delta_{+,N}, \Delta_{-,N}) + o(1).$$

At $k = N$ part (a) gives $B_N P_N(N) = B_N Q_+(N) + h_N(x_{N,N}) + o(1)$, since $\mathcal{D}^+_N(0) = 0$ and $\mathcal{D}^-_N(N) = 0$. Here $x_{N,N} = (1-p)N/B_N \to \infty$, so $h_N(x_{N,N}) \to 0$ by Lemma 2.2(b). The same holds at $k = 0$ with Q_-. This proves the formula for the whole-row error.

Now we compare the defects with the atoms. The term $m = 1$ of (5.1) at $b = 1$ is $B_N \nu_+(N-1)\Pi^-_1(1) = B_N \nu_+(N-1)$, since a fresh minus walk of one slot has one minus slot. So $\Delta_{+,N} \geq B_N \nu_+(N-1)$. In the other direction, $\nu_+(N-m) \leq \sup_{d>N/2} \nu_+(d)$ for $m < N/2$, and the Green identity, Lemma 2.5, together with $u_- \leq 1$ gives

$$\Delta_{+,N} \leq (1+\mu_+)\, B_N \sup_{d>N/2} \nu_+(d) \leq (1+\mu_+)\, C \sup_{d>N/2} B_d\, \nu_+(d), \tag{5.8}$$

with C a bound for B_N/B_d on $d > N/2$. The same bounds hold for the minus sign. Since $B_N/B_{N-1} \to 1$ and $\mathbb{P}(D = n) = \nu_+(n) + \nu_-(n)$, the two bounds show that $\max(\Delta_{+,N}, \Delta_{-,N}) \to 0$ if and only if $B_n \mathbb{P}(D = n) \to 0$. This is the interior criterion. For the whole row we need in addition $B_N Q_\sigma(N) \to 0$ for both signs, and since $\nu_\sigma \leq Q_\sigma$ this condition implies the interior one. This is the criterion $B_n \mathbb{P}(D \geq n) \to 0$.

Finally let $\varepsilon N \leq k \leq (1-\varepsilon)N$. A nonzero term of $\mathcal{D}^+_N(N-k)$ has $m \geq N - k \geq \varepsilon N$, and Theorem A(b) gives $\Pi^-_m(N-k) \leq C_0/B_m \leq C_0/B_{\lceil \varepsilon N\rceil}$. Hence $\mathcal{D}^+_N(N-k) \leq C_0(B_N/B_{\lceil\varepsilon N\rceil})\, Q_+(\lceil N/2\rceil) \to 0$, since D is finite almost surely. The minus defect is the same, and part (a) gives the approximation on this range. □

5.4 The boundary layer, part (d)

Proof of Theorem C(d). The proof uses only (RV) for the plus sign, through Theorem B(a) and the bound $T_+(\lceil N/2\rceil) \leq CT_+(N)$ of Lemma 2.1(ii). Fix $j \geq 0$ and let $N > 2j$. Proposition 2.4 at $k = N - j$ gives

$$P_N(N-j) = Q_+(N)\mathbf{1}_{\{j=0\}} + \sum_{d<N} \nu_-(d)\, F^+_{N-d}(N-j) + \sum_{d<N} \nu_+(d)\, F^-_{N-d}(N-d-j).$$

In the first sum the fresh plus walk on $N-d$ slots has $N-j$ plus slots, so $d \le j$, and it then has $j-d$ minus slots. For each such d, Theorem B(a) at the horizon $N-d$ and (2.11) give $F^+_{N-d}(N-j)/T_+(N-d) \to u_-(j-d)$, and $T_+(N-d)/T_+(N) \to 1$ by (RV). The sum has at most j terms. So the first sum equals $T_+(N)\big(\sum_{d=1}^{j} \nu_-(d)u_-(j-d) + o(1)\big)$.

Split the second sum at J and at $N/2$, with J fixed. For $d \le J$, Theorem B(a) for a fresh minus start and (2.11) give $F^-_{N-d}(N-d-j)/T_+(N-d) \to u_-(j)\mathbf{1}_{\{j\ge 1\}}$, so this part equals $T_+(N)\big(\mathbb{P}(\Sigma=+, D\le J)u_-(j)\mathbf{1}_{\{j\ge1\}} + o(1)\big)$. For $J < d \le N/2$ the horizon is $n = N-d \ge N/2$. The convergent sequence $F^-_n(n-j)/T_+(n)$ is bounded by some C_j for $n \ge N/2$ and large N, and $T_+(n) \le T_+(\lceil N/2\rceil) \le CT_+(N)$. So this part is at most $CC_jQ_+(J+1)T_+(N)$. For $N/2 < d < N$ put $m = N-d$. Then $F^-_{N-d}(N-d-j) = \Pi^-_m(j)$, and by (5.1) this part equals $B_N^{-1}\mathcal{D}^+_N(j)$ exactly. Collecting terms and using (5.2),

$$\limsup_{N\to\infty} \frac{\big|P_N(N-j) - Q_+(N)\mathbf{1}_{\{j=0\}} - T_+(N)\kappa_+(j) - B_N^{-1}\mathcal{D}^+_N(j)\big|}{T_+(N)} \le (1+CC_j)\,Q_+(J+1),$$

and the right side tends to zero as $J\to\infty$. This is the expansion.

Let $j \ge 1$. If $\nu_+(n) = o(T_+(n))$, then the Green identity, Lemma 2.5, and $u_- \le 1$ give $B_N^{-1}\mathcal{D}^+_N(j) \le (1+\mu_+)\sup_{d>N/2}\nu_+(d) = o(T_+(\lceil N/2\rceil)) = o(T_+(N))$, so $P_N(N-j)/T_+(N) \to \kappa_+(j)$. Conversely, suppose this limit holds for one $j \ge 1$. The term $m = j$ of (5.1) gives $B_N^{-1}\mathcal{D}^+_N(j) \ge \nu_+(N-j)\Pi^-_j(j) = \nu_+(N-j)T_-(j)$, because a fresh minus walk has j minus slots among its first j exactly when its first run has length at least j. Since $T_-(j) > 0$ and $T_+(N)/T_+(N-j) \to 1$, we get $\nu_+(n) = o(T_+(n))$.

Finally let ν_+ satisfy (Rν) and $j \ge 1$. For each fixed m, $\nu_+(N-m)/\nu_+(N) \to 1$, and $\nu_+(N-m) \le C_\nu\nu_+(N)$ for $m < N/2$. By Lemma 2.5 the weights $\Pi^-_m(j)$ have the finite sum $1+\mu_+u_-(j)$. Dominated convergence gives $B_N^{-1}\mathcal{D}^+_N(j)/\nu_+(N) = \sum_{m<N/2}(\nu_+(N-m)/\nu_+(N))\Pi^-_m(j) \to 1+\mu_+u_-(j)$. □

5.5 The big-jump zone, part (c)

Proof of the decomposition in Theorem C(c). Fix $\varepsilon \in (0,1]$ and take K_0 and n_0 from (5.6). Let $K \ge 4K_0$ and $J \ge 1$. Let $pN + K\lambda_N B_N \le k \le N$, and write $L = L_k$ and $j = N-k$. Then $L \ge K\lambda_N B_N/(1-p) \ge KB_N/(1-p)$, since $\lambda_N \ge 1$, and $N - L = j/(1-p)$. Take N so large that $\min(p, \frac12)N \ge n_0$, that $J < L/2$, and that (5.5) holds with $c = \min(p,\frac12)$. By Proposition 2.4,

$$P_N(k) = Q_+(N)\mathbf{1}_{\{k=N\}} + E_1 + E_2 + E_3 + E_4 + \Psi_N(k),$$

where E_1 and E_2 collect $\nu_-(d)F^+_{N-d}(k)$ over $d \le \min(J,j)$ and over $J < d \le j$, and E_3 and E_4 collect $\nu_+(d)F^-_{N-d}(k-d)$ over $d \le J$ and over $J < d < L/2$. The terms $\nu_-(d)F^+_{N-d}(k)$ with $d > j$ vanish, since then $k > N-d$.

Step 1 (geometry). Let $d \le j$ and $n = N-d$. Then $n \ge k > pN$, so $n \ge n_0$, and $L^{(n)}_k = L + pd/(1-p) \ge L$. Also $k - pn \ge k - pN \ge K\lambda_N B_N \ge (K/2)\lambda_n B_n$, by (5.5) and since B is nondecreasing. So k lies in the zone of (5.6) at the horizon n, and $\widehat G_n(k) \le (N/L)T_+(\lfloor L\rfloor) = \widehat G_N(k)$ because T_+ is nonincreasing. Let $d < L/2$ and $n = N-d \ge N/2$. Then $L^{(n)}_{k-d} = L - d \ge L/2$, and $(k-d) - pn = (1-p)(L-d) \ge K\lambda_N B_N/2 \ge (K/4)\lambda_n B_n$, again by (5.5), so $k-d$ lies in that zone at the horizon n. By Lemma 2.1(ii), $\widehat G_n(k-d) \le (2N/L)T_+(\lfloor L/2\rfloor) \le C_*\widehat G_N(k)$ for large N.

Step 2 (intermediate first-run lengths). By Step 1 and (5.6), every fresh probability in E_2 and E_4 is at most $(C_{\mathrm G}+1)$ times the corresponding $\widehat G_n$, since $\varepsilon \le 1$. Hence $E_2 + E_4 \le (C_{\mathrm G}+1)(1+C_*)\mathbb{P}(D>J)\,\widehat G_N(k)$.

Step 3 (short first-run lengths). Let $d \le J$. By Step 1 and (5.6),

$$|F^+_{N-d}(k) - \overline G^{(+)}_{N-d}(k)| \le \varepsilon\widehat G_N(k), \qquad |F^-_{N-d}(k-d) - \overline G^{(-)}_{N-d}(k-d)| \le \varepsilon C_*\widehat G_N(k).$$

With $L' = L + pd/(1-p)$ and $(N-d) - L' = (j-d)/(1-p)$, and with (2.11), the kernels are

$$\overline{G}^{(+)}_{N-d}(k) = \frac{j-d}{(1-p)\mu_-} f_+(\lfloor L' \rfloor) + T_+(\lfloor L' \rfloor)\, u_-(j-d),$$
$$\overline{G}^{(-)}_{N-d}(k-d) = \frac{N-L}{\mu_-} f_+(\lfloor L-d \rfloor) + T_+(\lfloor L-d \rfloor)\, u_-(j)\mathbf{1}_{\{j\geq 1\}}.$$

By (LR) and (RV), $f_+(m+i)/f_+(m) \to 1$ and $T_+(m+i)/T_+(m) \to 1$ as $m \to \infty$, uniformly in $|i| \leq 2J/(1-p)$. Let r_N be the largest deviation from one of these ratios over $m \geq KB_N/(2(1-p))$. Then $r_N \to 0$. Note that $(j-d)/(1-p) = (N-L) - d/(1-p)$ and that $f_+(\lfloor L' \rfloor) \leq 2CT_+(\lfloor L \rfloor)/L$ for large N, by (LR). Hence

$$\Big|\overline{G}^{(+)}_{N-d}(k) - \frac{N-L}{\mu_-} f_+(\lfloor L \rfloor) - T_+(\lfloor L \rfloor) u_-(j-d)\Big| \leq C\Big(r_N + \frac{J}{L}\Big)\widehat{G}_N(k),$$

and $\overline{G}^{(-)}_{N-d}(k-d)$ differs from $(N-L)f_+(\lfloor L \rfloor)/\mu_- + T_+(\lfloor L \rfloor)u_-(j)\mathbf{1}_{\{j\geq 1\}}$ by at most $Cr_N\widehat{G}_N(k)$. Here we used $(N-L)f_+(\lfloor L \rfloor) \leq C\widehat{G}_N(k)$ and $T_+(\lfloor L \rfloor) \leq \widehat{G}_N(k)$.

Now we sum over d. In E_3 the weights sum to $\mathbb{P}(\Sigma = +, D \leq J)$, which differs from π_+ by at most $\mathbb{P}(D > J)$. In E_1 the weights of the interior term sum to $\mathbb{P}(\Sigma = -, D \leq \min(J, j))$. If $j \geq J$ this differs from π_- by at most $\mathbb{P}(D > J)$. If $j < J$, then $N - L < J/(1-p)$ and the whole interior term is at most $CJ\, T_+(\lfloor L \rfloor)/L \leq C(J/L)\widehat{G}_N(k)$. The boundary terms in E_1 sum to $T_+(\lfloor L \rfloor)\sum_{d\leq \min(J,j)} \nu_-(d)u_-(j-d)$, which differs from $T_+(\lfloor L \rfloor)\sum_{d\leq j}\nu_-(d)u_-(j-d)$ by at most $\mathbb{P}(D > J)\widehat{G}_N(k)$, since $u_- \leq 1$. By (5.2) and (5.3),

$$|E_1 + E_3 - G^\nu_N(k)| \leq C\Big(\varepsilon + r_N + \frac{J}{L} + \mathbb{P}(D > J)\Big)\widehat{G}_N(k).$$

Step 4 (conclusion). Steps 2 and 3 and $J/L \leq (1-p)J/(KB_N)$ give, for $K \geq 4K_0$ and large N,

$$\sup_{pN + K\lambda_N B_N \leq k \leq N} \frac{|P_N(k) - Q_+(N)\mathbf{1}_{\{k=N\}} - G^\nu_N(k) - \Psi_N(k)|}{\widehat{G}_N(k)} \leq C\Big(\varepsilon + r_N + \frac{(1-p)J}{KB_N} + \mathbb{P}(D > J)\Big).$$

Let $N \to \infty$, then $J \to \infty$. The bound becomes $C\varepsilon$ for every $K \geq 4K_0(\varepsilon)$, and ε is arbitrary. □

Proof of the criteria in Theorem C(c). The whole zone. By the decomposition, the statement without Ψ_N holds if and only if $\lim_K \limsup_N \sup_k \Psi_N(k)/\widehat{G}_N(k) = 0$, the supremum being over the zone. Suppose $\nu_+(n) = o(T_+(n))$. Every d in (5.4) has $d \geq L_k/2$, and Lemma 2.5 with $u_- \leq 1$ gives, with $j = N - k$,

$$\Psi_N(k) \leq \sup_{d\geq L_k/2} \nu_+(d) \sum_{m\geq 1} \Pi^-_m(j) \leq (1+\mu_+) \sup_{d\geq L_k/2} \nu_+(d), \tag{5.9}$$

because $F^-_{N-d}(k-d) = \Pi^-_{N-d}(j)$ and distinct d give distinct $m = N - d$. The right side is $o(T_+(\lceil L_k/2 \rceil)) = o(T_+(\lfloor L_k \rfloor))$ by Lemma 2.1(ii), and $T_+(\lfloor L_k \rfloor) \leq \widehat{G}_N(k)$. Conversely, for large N the bin $k = N-1$ lies in the zone for every K, because $\lambda_N B_N = o(N)$, and it has $L = N - 1/(1-p) \geq N/2$. The term $d = N-1$ of (5.4) gives $\Psi_N(N-1) \geq \nu_+(N-1)F^-_1(0) = \nu_+(N-1)$, and $\widehat{G}_N(N-1) \leq 2T_+(\lfloor N/2 \rfloor) \leq CT_+(N)$. So the criterion forces $\nu_+(N-1) = o(T_+(N))$, that is, $\nu_+(n) = o(T_+(n))$.

The regular case. Let ν_+ satisfy (Rν). Fix $\varepsilon > 0$. Take R and N_0 from Lemma 5.3, and $\delta > 0$ with $|\nu_+(d)/\nu_+(n) - 1| \leq \varepsilon$ for $|d-n| \leq \delta n$ and large n. Let $K \geq R(1-p)/\delta + 1$

and let $k < N$ lie in the zone, so $j = N - k \geq 1$. Put $\mathcal{V} = \{d : |d - L_k| \leq RB_N\}$. Since $m = N - d$ satisfies $|m - j/(1-p)| = |d - L_k|$, Lemma 5.3 gives $\sum_{d \notin \mathcal{V}} \Pi^-_{N-d}(j) \leq \varepsilon$, the sum running over all d with $N - d \geq 1$. Every $d \in \mathcal{V}$ satisfies $|d - L_k| \leq RB_N \leq \delta L_k$, because $L_k \geq KB_N/(1-p)$, so $d \geq L_k/2$ and $\nu_+(d) = \nu_+(\lfloor L_k \rfloor)(1 + O(\varepsilon))$. Hence

$$\Psi_N(k) = \nu_+(\lfloor L_k \rfloor)\big(1 + O(\varepsilon)\big) \sum_{d \in \mathcal{V}} \Pi^-_{N-d}(j) + O\big(\varepsilon C_\nu \nu_+(\lfloor L_k \rfloor)\big).$$

By Lemma 2.5 and the choice of R, the last sum lies between $1 + \mu_+ u_-(j) - \varepsilon$ and $1 + \mu_+ u_-(j)$. So $|\Psi_N(k) - \nu_+(\lfloor L_k \rfloor)(1 + \mu_+ u_-(j))| \leq C\varepsilon\, \nu_+(\lfloor L_k \rfloor)$, and the decomposition gives the stated form. At $k = N$ both Ψ_N and the replacement vanish.

Interior zones. Fix $\varepsilon \in (0, \frac{1}{2})$ and let $Z^\varepsilon_N = \{k : \varepsilon N \leq L_k \leq (1-\varepsilon)N\}$. For large N it lies in the zone for every K, because $\lambda_N B_N = o(N)$, and $T_+(N) \leq \widehat{G}_N(k) \leq \varepsilon^{-1} T_+(\lfloor \varepsilon N \rfloor) \leq C_\varepsilon T_+(N)$ on it. So $\Psi_N = o(\widehat{G}_N)$ on Z^ε_N means $\sup_{k \in Z^\varepsilon_N} \Psi_N(k) = o(T_+(N))$. Sufficiency of $\nu_+(n) = o(T_+(n))$ follows from (5.9).

For the necessity of the window condition, let n be large, put $N = 2n$ and choose k with $|L_k - n| \leq 1/(1-p)$, which is possible because L_k moves in steps of $1/(1-p)$. Then $k \in Z^{1/4}_N$. For $|d - n| \leq B_n$ put $m = N - d$. Then $m \in [N/4, 3N/4]$, and $(m - j) - pm = (1-p)(L_k - d)$ has absolute value at most $2(1-p)B_n \leq c_2 B_m$, with c_2 depending only on the laws, by Lemma 2.1(ii). Theorem A(c) and Lemma 2.2(d) give $B_m \Pi^-_m(j) = B_m F^-_m(m - j) \geq c_3 > 0$ for large n. Also $d \geq L_k/2$ and $B_m \leq CB_n$. Hence $\Psi_N(k) \geq (c_3/C) W_n$. If $\Psi_N = o(\widehat{G}_N)$ on $Z^{1/4}_N$, then $W_n = o(T_+(2n)) = o(T_+(n))$.

Sufficiency of the window condition under (LR) for both signs. Let $W_n = o(T_+(n))$ and $k \in Z^\varepsilon_N$, with $j = N - k \geq (1-p)\varepsilon N$. A nonzero term of (5.4) has $m = N - d \geq j$, so $m \in [(1-p)\varepsilon N, N]$ and $d \in [\varepsilon N/2, N]$. Hence $B_m \geq c_\varepsilon B_N$, and Theorem A(b) gives $\Pi^-_m(j) \leq C_\varepsilon / B_N$. Put $\overline{W}_N = \sup_{\varepsilon N/4 \leq n \leq N} W_n$, which is $o(T_+(N))$ by (RV). Split the d axis into the blocks $\mathcal{B}_i = \{d : iB_N \leq d - L_k < (i+1)B_N\}$, $i \in \mathbb{Z}$. Each block meets $[\varepsilon N/2, N]$ in a set covered by at most C_ε windows $[n - B_n, n + B_n]$ with $\varepsilon N/4 \leq n \leq N$, so $\nu_+(\mathcal{B}_i \cap [\varepsilon N/2, N]) \leq C_\varepsilon B_N \overline{W}_N$. The blocks with $|i| \leq R$ contribute at most $(2R+2) C^2_\varepsilon \overline{W}_N$ to $\Psi_N(k)$.

For the blocks with $|i| > R$ we do not use (5.6), since its zone, and its analogue at the minus end, carry logarithmic factors that are unbounded when $\alpha_+ \neq \alpha_-$. We use a bound that needs only an excess of order B_m. Let m be a large horizon and $k' > pm$ a bin, and put $L = L^{(m)}_{k'}$ and $g = \lceil L/32 \rceil$. We apply (4.10) at the horizon m for a fresh minus start and drop its last term. We bound its first term by Lemma 4.3 with $\eta = 1/32$ and $r = 2$, and Cen_g and Int_g by (4.11), with $\kappa^{(-)}_+ \leq 1$ and $\gamma^{(-)}_+(m - k') \leq (1 + \mu_+)m$. By (LR) for the plus sign, $f^*_+(g) \leq C_{\mathrm{LR}} T_+(g)/g$, and $T_+(g) \leq C T_+(L)$ by Lemma 2.1(ii). Since $L \leq m$, this gives

$$F^-_m(k') \leq C\, \frac{(mT_+(L))^2}{b_{+,m}} + \frac{C}{b_{-,m}}\, e^{-cL/b_{-,m}} + Ce^{-m^c} + C\, \frac{m}{L}\, T_+(L) \qquad (L \geq C_0 B_m), \tag{5.10}$$

with C, c and C_0 depending only on the laws. The minus-end forms of Lemmas 4.1 and 4.3, with (LR) for the minus sign, give the same bound for $k' < pm$ with the signs exchanged, with the same constants. There L is replaced by $L^- = (pm - k')/p$, T_+ by T_-, and $b_{+,m}$ and $b_{-,m}$ by $b_{-,m}$ and $b_{+,m}$.

Now let $|i| > R$ and $d \in \mathcal{B}_i$, so that $m = N - d \in [(1-p)\varepsilon N, N]$ and $\Pi^-_m(j) = F^-_m(k')$ with $k' = m - j$. Then $k' - pm = (1-p)(L_k - d)$. If $d < L_k$, then $k' > pm$ and $L = L_k - d > (|i| - 1)B_N$. If $d > L_k$, then $k' < pm$ and $L^- = (1-p)(d - L_k)/p \geq (1-p)|i| B_N/p$. Put $c_* = \min(1, (1-p)/p)$, so the excess is at least $y_i = c_*(|i| - 1)B_N$ in both cases. Choose R with $c_* R \geq \max(C_0, 1)$. Then $y_i \geq C_0 B_N \geq C_0 B_m$, so (5.10) or its minus-end form applies, and $y_i \geq B_N \geq b_{\sigma,N}$ for both signs. Each term of (5.10) decreases in L. For these m we

have $mT_\sigma \le NT_\sigma$ and $c_\varepsilon b_{\sigma,N} \le b_{\sigma,m} \le b_{\sigma,N}$. We treat the blocks with $i<0$, and those with $i>0$ are the same with the signs exchanged. We multiply the four terms of (5.10) at $L=y_i$ by the block mass $C_\varepsilon B_N \overline{W}_N$, sum over $i<-R$, and leave out the common factor $C_\varepsilon \overline{W}_N$ below. Put $t=|i|-1\ge 1$.

The first term. By (2.4), $(NT_+(y_i))^2 \le 16 s^2 (y_i/b_{+,N})^{-\alpha_+-1}$, so the term is at most $C(B_N/b_{+,N})^{-\alpha_+} t^{-\alpha_+-1} \le Ct^{-\alpha_+-1}$. These sum to a constant.

The second term. Put $x = B_N/b_{-,N} \ge 1$. The term is at most $Cxe^{-cc_* tx}$, and $\sum_{t\ge 1} xe^{-cc_* tx} = x/(e^{cc_* x}-1) \le 1/(cc_*)$.

The third term. At most N/B_N+2 blocks meet $[0,N]$, so these terms add up to at most $C(N+2B_N)e^{-((1-p)\varepsilon N)^c}$, which tends to zero.

The fourth term. Fix $\delta' \in (0,\alpha_*)$. Since $c_* t \ge 1$, Lemma 2.1(ii) gives $T_+(y_i) \le Ct^{-\alpha_+-\delta'} T_+(B_N)$. So the term is at most $CNT_+(y_i)/t \le CNT_+(B_N) t^{-1-\alpha_+-\delta'}$, and $NT_+(B_N) \le NT_+(b_{+,N}) \to s$ by Lemma 2.1(v). These sum to a constant.

For $i>R$ the same four steps use (2.4) for the minus sign, $x = B_N/b_{+,N}$, and $NT_-(B_N) \le NT_-(b_{-,N}) \to s$. So the blocks with $|i|>R$ contribute at most $C_\varepsilon \overline{W}_N$ to $\Psi_N(k)$. Hence $\sup_{k\in Z_N^\varepsilon} \Psi_N(k) \le C_{\varepsilon,R}\overline{W}_N = o(T_+(N))$. □

The gap between the two interior conditions is real. If $\nu_+(n_i) = T_+(n_i)$ on a sparse sequence and $\nu_+ = 0$ near it, then $W_n = O(T_+(n)/B_n)$, so under (LR) for both signs the preparation is invisible on every interior zone, by the window criterion (the sufficiency just proved). It is visible at the bin $N-1$ for $N = n_i+1$, where $P_N(N-1) \ge T_+(N-1)$, and $\nu_+(n) \ne o(T_+(n))$. So the interior of the big-jump zone sees only window averages of the preparation, while the edge of the zone sees single atoms.

5.6 The stationary preparation

Corollary 5.4 (stationary preparation)**.** *Assume (RV) and (SPAN), and let the preparation be stationary, $\nu_\sigma = T_\sigma/s$. Then the bulk criterion holds, and*

(a) the atoms are $Q_\sigma(N) = s^{-1}\sum_{d\ge N} T_\sigma(d) \sim NT_\sigma(N)/(s(\alpha_\sigma-1))$.

(b) if (LR) holds for the plus sign, then, with λ_N as in (4.5)*,*

$$\lim_{K\to\infty} \limsup_{N\to\infty} \sup_{pN+K\lambda_N B_N \le k < N} \Big| \frac{P_N(k)}{G_N^{\mathrm{st}}(k)} - 1 \Big| = 0,$$

where $G_N^{\mathrm{st}}(k) = (N-L_k) f_+(\lfloor L_k \rfloor)/\mu_- + T_+(\lfloor L_k \rfloor)\, \kappa_+^{\mathrm{st}}(N-k)$, and for fixed $v\in(p,1)$, with $\ell = (v-p)/(1-p)$, $P_N(\lfloor vN \rfloor)/T_+(N) \to I_+(v) + \ell^{-\alpha_+}/\mu_-$. So in the interior of the big-jump zone the kernel gains the term $T_+(\lfloor L_k \rfloor)/\mu_-$, which is the complete initial plus run.

(c) the boundary kernel at the plus end is

$$\frac{P_N(N-j)}{T_+(N)} \to \kappa_+^{\mathrm{st}}(j) = \frac{2}{s}\big(1+\mu_+ u_-(j)\big) \qquad (j\ge 1),$$

while $P_N(N)/T_+(N) \to \infty$. Moreover $\kappa_+^{\mathrm{st}}(j) \to 2/\mu_-$ as $j\to\infty$, which is the value of $I_+(v) + \ell^{-\alpha_+}/\mu_-$ at $v=1$.

Proof. The bulk criterion holds because $B_n T_\sigma(n) \to 0$ by Lemma 2.1(v). Part (a) is Lemma 2.1(iv).

We first compute κ_+. Here $\pi_+ = \sum_d T_+(d)/s = \mu_+/s = p$, and $\sum_{d=1}^{j} T_-(d) u_-(j-d) = 1$ for $j\ge 1$ by (2.9). So (5.2) gives $\kappa_+(j) = pu_-(j) + 1/s$ for $j\ge 1$.

(c) The function $\nu_+ = T_+/s$ is regularly varying with index $-\alpha_+ < 0$, so it satisfies (Rν). Theorem C(d) gives, for $j \geq 1$,

$$\frac{P_N(N-j)}{T_+(N)} \to pu_-(j) + \frac{1}{s} + \frac{1}{s}\big(1+\mu_+ u_-(j)\big) = \frac{2}{s} + \frac{2\mu_+}{s}u_-(j),$$

since $p = \mu_+/s$. Also $P_N(N) = Q_+(N)$, and $Q_+(N)/T_+(N) \to \infty$ by (a). By (2.10), $\kappa_+^{\mathrm{st}}(j) \to (2/s)(1+\mu_+/\mu_-) = 2/\mu_-$.

(b) By the regular case of Theorem C(c), uniformly in the zone with $k < N$ and in the double limit,

$$P_N(k) = G_N^{\nu}(k) + \frac{T_+(\lfloor L_k \rfloor)}{s}\big(1+\mu_+ u_-(N-k)\big) + o\big(\widehat{G}_N(k)\big),$$

where we used $T_+(\lfloor L_k \rfloor) \leq \widehat{G}_N(k)$. By the computation of κ_+ the right side is $G_N^{\mathrm{st}}(k) + o(\widehat{G}_N(k))$. For the ratio form we need $G_N^{\mathrm{st}}(k) \geq c\widehat{G}_N(k)$. Since $\kappa_+^{\mathrm{st}} \geq 2/s$, this holds when $L_k > N/2$. When $L_k \leq N/2$, (LR) gives $(N-L_k)f_+(\lfloor L_k \rfloor)/\mu_- \geq c(N/L_k)T_+(\lfloor L_k \rfloor)$ for large N. Now fix $v \in (p,1)$ and let $k = \lfloor vN \rfloor$. Then k lies in the zone for every K when N is large, because $\lambda_N B_N = o(N)$. Also $L_k = \ell N + O(1)$ and $N - k \to \infty$. By (LR) and (RV), $(N-L_k)f_+(\lfloor L_k \rfloor)/T_+(N) \to (1-\ell)\alpha_+ \ell^{-\alpha_+ - 1}$ and $T_+(\lfloor L_k \rfloor)/T_+(N) \to \ell^{-\alpha_+}$, and $\kappa_+^{\mathrm{st}}(N-k) \to 2/\mu_-$. This gives the limit $I_+(v) + \ell^{-\alpha_+}/\mu_-$, with I_+ as in Corollary 4.8. At $v = 1$ it equals $2/\mu_-$. □

In $\kappa_+^{\mathrm{st}}(j)$ the term $pu_-(j)$ comes from a short first plus run, the term $1/s$ from a short first minus run, and the term $(1+\mu_+u_-(j))/s$ from a first plus run that ends within $O(1)$ slots of the horizon. Section 8 reports a check of part (c) against the exact law.

The same index comparisons place other preparations. Suppose $Q_\sigma \in \mathrm{RV}_{-\rho}$ with $\rho > 0$ and $\nu_\sigma(n) \sim \rho Q_\sigma(n)/n$. Then $\nu_\sigma \in \mathrm{RV}_{-\rho-1}$, the bulk criterion holds, and the whole-row criterion holds for $\rho > 1/\alpha_*$ and fails for $\rho < 1/\alpha_*$. At the plus end $\nu_+/T_+ \in \mathrm{RV}_{\alpha_+ - 1 - \rho}$, so by Theorem C(d), and by (c) under (LR) for the plus sign, the preparation is invisible in the boundary layer and the big-jump zone for $\rho > \alpha_+ - 1$ and visible for $\rho < \alpha_+ - 1$, with the extra term $\nu_+(\lfloor L_k \rfloor)(1+\mu_+u_-(N-k))$. The atom $Q_+(N)$ is negligible against $T_+(N)$ for $\rho > \alpha_+$ and dominates it for $\rho < \alpha_+$. The stationary preparation has $\rho = \alpha_\sigma - 1$ and $\nu_\sigma/T_\sigma = 1/s$, so it sits on the line of the second comparison and is visible there. The fresh start of sign σ has $\rho = \alpha_\sigma$ and $Q_\sigma/T_\sigma = 1$, so it sits on the line of the third, where the atom and the layer are of the same order.

5.7 No universal trim

Part (b) of Theorem C holds on every macroscopic interior with no condition. The next corollary shows that no sublinear trim of the row does the same.

Corollary 5.5 (no universal trim)**.** *Assume (RV) and (SPAN). For every sequence of nonnegative integers $r_N = o(N)$ there is a preparation with a fixed deterministic sign for which the bulk approximation fails on $r_N \leq k \leq N - r_N$ along a subsequence.*

Proof. Put $R(n) = \max_{1 \leq i \leq 2n} r_i$ and

$$m(n) = \Big\lceil \max\Big\{\sqrt{n},\ \frac{4(R(n)+1)}{1-p}\Big\}\Big\rceil.$$

Since $r_i = o(i)$, for every $\eta > 0$ we have $r_i \leq \eta i$ for $i \geq i_\eta$, so $R(n) \leq \max_{i < i_\eta} r_i + 2\eta n$ and $R(n) = o(n)$. Hence $m(n) \to \infty$ and $m(n) = o(n)$. Since B is regularly varying with the positive index $1/\alpha_*$, Lemma 2.1(ii) gives $B_{m(n)}/B_n \to 0$. Choose $n_1 < n_2 < \cdots$ so that $n_{i+1} > 2n_i$, $m(n_i) \leq n_i$ and $\sum_i B_{m(n_i)}/B_{n_i} \leq \frac{1}{2}$. Put $m_i = m(n_i)$, $N_i = n_i + m_i \leq 2n_i$ and

$v_i = B_{m_i}/B_{n_i}$. The preparation has $\Sigma = +$ almost surely, $\nu_+(n_i) = v_i$ for every i, and the remaining mass at $d = 1$. Then D is finite almost surely.

Fix i and let $k_i' = \lfloor pm_i \rfloor$. By Theorem A(c) and Lemma 2.2(d) there is $c > 0$ with $B_{m_i} F^-_{m_i}(k_i') \geq c$ for large i. Let $k_i = n_i + k_i'$. Proposition 2.4, keeping only the term $d = n_i$, gives

$$B_{N_i} P_{N_i}(k_i) \geq B_{N_i}\, v_i\, F^-_{m_i}(k_i') \geq c\, \frac{B_{N_i}}{B_{n_i}} \geq c,$$

because B is nondecreasing. On the other hand $k_i - pN_i = (1-p)n_i + (k_i' - pm_i) \geq (1-p)n_i - 1$, so $x_{N_i,k_i} \geq ((1-p)n_i - 1)/B_{2n_i} \to \infty$, and $h_{N_i}(x_{N_i,k_i}) \to 0$ by Lemma 2.2(b). So $|B_{N_i} P_{N_i}(k_i) - h_{N_i}(x_{N_i,k_i})| \geq c/2$ for large i.

It remains to check that k_i is retained. Since $N_i \leq 2n_i$, we have $r_{N_i} \leq R(n_i) = o(n_i)$, so $k_i \geq n_i \geq r_{N_i}$ for large i. Also $N_i - k_i = m_i - k_i' \geq (1-p)m_i \geq 4(R(n_i)+1) > r_{N_i}$. Hence $r_{N_i} \leq k_i \leq N_i - r_{N_i}$, and the approximation fails along N_i. □

The preparation is fixed once, it violates the bulk criterion since $B_{n_i}\nu_+(n_i) = B_{m_i} \to \infty$, and its atoms may be as sparse as we like.

6 The shape of the law

This section reads two features of the shape off Theorems A to C. The first is the global mode, where the golden ratio enters, and the second is the interior minimum between the bulk and the plus end.

6.1 The global mode

Let $P_N^{\max} = \max_{1 \leq k < N} P_N(k)$ be the largest interior bin and $H_N = \sup_x h_N(x)$ the height of the profile. By Proposition 2.4 the two endpoint bins are $P_N(N) = Q_+(N)$ and $P_N(0) = Q_-(N)$.

Theorem D (the global mode)**.** *Assume (RV) and (SPAN), and let the preparation satisfy the bulk criterion $B_n \mathbb{P}(D = n) \to 0$.*

(a) $P_N^{\max} = (H_N + o(1))/B_N$, every interior maximizer lies in $pN + O(B_N)$, and along any subsequence on which h_N converges uniformly the scaled maximizers $(k - pN)/B_N$ converge to the set of maximizers of the limit profile.

(b) If $\liminf_N B_N Q_\sigma(N)/H_N > 1$ for some sign σ, the endpoint bin of sign σ exceeds every interior bin for all large N, so the global mode of P_N is an endpoint. If $\limsup_N B_N Q_\sigma(N)/H_N < 1$ for both signs, the global mode is interior for all large N.

(c) If $Q_\sigma \in \mathrm{RV}_{-\rho_\sigma}$, the endpoint of sign σ beats every interior bin for all large N when $\rho_\sigma < 1/\alpha_*$, and loses to the largest interior bin when $\rho_\sigma > 1/\alpha_*$.

(d) (The golden ratio.) For the stationary preparation, $\rho_\sigma = \alpha_\sigma - 1$. With equal indices $\alpha_+ = \alpha_- = \alpha$, the global mode is an endpoint for all large N when $\alpha < \varphi = (1+\sqrt{5})/2$ and is interior for all large N when $\alpha > \varphi$. For two equal laws $T_+(n) = T_-(n) = n^{-\alpha}$ and the stationary preparation, $B_N Q_\sigma(N)/H_N \sim \mathcal{C}(\alpha) N^{1-\alpha+1/\alpha}$ for each sign, where

$$\mathcal{C}(\alpha) = \frac{\pi\big(K_\alpha |\cos\tfrac{\pi\alpha}{2}|\big)^{1/\alpha}}{4(\alpha-1)\,\Gamma(1+\tfrac{1}{\alpha})\,\zeta(\alpha)^{1+1/\alpha}}, \tag{6.1}$$

and at $\alpha = \varphi$ the ratio tends to $\mathcal{C}(\varphi) = 0.7761317\ldots < 1$, so the global mode is interior. For a fresh start of sign σ_0, $Q_{\sigma_0} = T_{\sigma_0}$, $Q_{\bar\sigma_0} = 0$ and $\rho_{\sigma_0} = \alpha_{\sigma_0} > 1 > 1/\alpha_*$, so the global mode is interior for all large N and every pair of indices.

Proof. (a) Put $h_* = \inf_N H_N$, which is positive by Lemma 2.2(d). By Theorem C(b) the bulk criterion gives

$$\varepsilon_N = \sup_{1\le k<N} \bigl|B_N P_N(k) - h_N(x_{N,k})\bigr| \to 0. \tag{6.2}$$

Hence $B_N P_N^{\max} = \max_{1\le k<N} h_N(x_{N,k}) + O(\varepsilon_N)$, and $\max_k h_N(x_{N,k}) \le H_N$. For the lower bound, Lemma 2.2(b) gives K with $\sup_{|x|>K} h_N(x) < h_*/2$ for all N, so H_N is the supremum of h_N over $[-K,K]$. For large N the points $x_{N,k}$ with $1 \le k < N$ cover $[-K,K]$ with mesh $1/B_N$, since $pN/B_N \to \infty$ and $(1-p)N/B_N \to \infty$. The common Lipschitz constant of Lemma 2.2(a) gives $\max_k h_N(x_{N,k}) \ge H_N - C_2/B_N$. Hence $B_N P_N^{\max} = H_N + o(1)$.

Let k_N be an interior maximizer. Then $h_N(x_{N,k_N}) \ge B_N P_N^{\max} - \varepsilon_N = H_N - o(1) > h_*/2$ for large N, so $|x_{N,k_N}| \le K$, which is the statement $k_N \in pN + O(B_N)$. Along a subsequence with $h_N \to h$ uniformly we have $H_N \to \sup h$ and $h(x_{N,k_N}) \ge H_N - o(1)$. Every limit point x of the bounded sequence x_{N,k_N} satisfies $h(x) = \sup h$ by continuity. So the distance from x_{N,k_N} to the set of maximizers of h tends to zero.

(b) If $B_N Q_\sigma(N) \ge (1+\eta) H_N$ for all large N, then by (a) $B_N Q_\sigma(N) - B_N P_N^{\max} \ge \eta H_N - o(1) \ge \eta h_* - o(1) > 0$, so the endpoint bin of sign σ exceeds every interior bin. If $B_N Q_\sigma(N) \le (1-\eta) H_N$ for both signs and all large N, then (a) gives $B_N P_N^{\max} - B_N Q_\sigma(N) \ge \eta h_* - o(1) > 0$ for both signs.

(c) By Lemma 2.1(v), $B_N \in \mathrm{RV}_{1/\alpha_*}$. Hence $B_N Q_\sigma(N) \in \mathrm{RV}_{1/\alpha_* - \rho_\sigma}$. A regularly varying function with a positive index tends to infinity and one with a negative index tends to zero [5, Proposition 1.5.1]. Since H_N is bounded above and below by Lemma 2.2(a) and (d), part (b) applies.

(d) By Corollary 5.4(a) the stationary preparation satisfies the bulk criterion and $Q_\sigma(N) \sim N T_\sigma(N)/(s(\alpha_\sigma - 1))$, so $\rho_\sigma = \alpha_\sigma - 1$. With equal indices $\alpha_* = \alpha$, and $\alpha - 1 < 1/\alpha$ if and only if $\alpha^2 - \alpha - 1 < 0$, that is, $\alpha < \varphi$. Part (c) then shows that both endpoints beat the interior when $\alpha < \varphi$ and that both lose when $\alpha > \varphi$. For unequal indices part (c) shows that the endpoint of sign σ beats the interior if and only if $\alpha_\sigma - 1 < 1/\alpha_*$, away from equality.

Now let $T_+(n) = T_-(n) = n^{-\alpha}$. Then $\mu_\sigma = \sum_{n\ge1} n^{-\alpha} = \zeta(\alpha)$, $s = 2\zeta(\alpha)$, $p = 1/2$, $\theta_{+,N} = \theta_{-,N} = 1$, and we may take $a_\sigma(t) = t^{1/\alpha}$, so $B_N = (N/s)^{1/\alpha}$. The profile does not depend on N. It is the density h of $\frac12(Z_+ - Z_-)$ with Z_+ and Z_- independent copies of Z_α. Its characteristic function is

$$\exp\bigl(K_\alpha(-it/2)^\alpha + K_\alpha(it/2)^\alpha\bigr) = \exp\bigl(-c_\alpha |t|^\alpha\bigr), \qquad c_\alpha = 2^{1-\alpha} K_\alpha \bigl|\cos\tfrac{\pi\alpha}{2}\bigr|,$$

because $(-iy)^\alpha + (iy)^\alpha = 2y^\alpha \cos\frac{\pi\alpha}{2}$ for $y > 0$ and $\cos\frac{\pi\alpha}{2} < 0$. This function is positive and integrable, so by Fourier inversion $h(x) \le h(0)$ for every x, and

$$H_N = h(0) = \frac{1}{\pi}\int_0^\infty e^{-c_\alpha t^\alpha}\,dt = \frac{\Gamma(1+\frac1\alpha)}{\pi\, c_\alpha^{1/\alpha}}.$$

The stationary atom is $Q_\sigma(N) = s^{-1}\sum_{d\ge N} d^{-\alpha} \sim N^{1-\alpha}/(2(\alpha-1)\zeta(\alpha))$. Multiplying,

$$\frac{B_N Q_\sigma(N)}{H_N} \sim \frac{N^{1/\alpha+1-\alpha}}{(2\zeta(\alpha))^{1/\alpha}\cdot 2(\alpha-1)\zeta(\alpha)} \cdot \frac{\pi\, 2^{(1-\alpha)/\alpha}\bigl(K_\alpha|\cos\frac{\pi\alpha}{2}|\bigr)^{1/\alpha}}{\Gamma(1+\frac1\alpha)} = \mathcal{C}(\alpha)\, N^{1-\alpha+1/\alpha},$$

since $2^{(1-\alpha)/\alpha} 2^{-1/\alpha} 2^{-1} = 2^{-2}$. At $\alpha = \varphi$ the exponent vanishes, and $\alpha - 1 = 1/\varphi$, $1 + 1/\alpha = \varphi$ and $2 - \alpha = \varphi^{-2}$, so $K_\varphi = \varphi\,\Gamma(\varphi^{-2})$ and

$$\mathcal{C}(\varphi) = \frac{\pi\varphi\bigl(\varphi\,\Gamma(\varphi^{-2})|\cos\frac{\pi\varphi}{2}|\bigr)^{1/\varphi}}{4\,\Gamma(\varphi)\,\zeta(\varphi)^\varphi} = 0.7761317205\ldots.$$

That $\mathcal{C}(\varphi) < 1$ needs only four elementary bounds. Since $x^{-\varphi}$ is convex, the trapezoid rule gives $\zeta(\varphi) \geq \frac{1}{2} + 1/(\varphi - 1) = \varphi + \frac{1}{2}$. The minimum of Γ on $(0, \infty)$ exceeds 0.8856 [32, Section 5.4(iii)], so $\Gamma(\varphi) \geq 0.8856$. Since $\Gamma \leq 1$ on $[1, 2]$, $\Gamma(\varphi^{-2}) = \varphi^2 \Gamma(1 + \varphi^{-2}) \leq \varphi^2$. Finally $\frac{\pi}{2} < \frac{\pi\varphi}{2} < \frac{5\pi}{6}$, so $|\cos \frac{\pi\varphi}{2}| \leq \frac{\sqrt{3}}{2}$. Together these give $\mathcal{C}(\varphi) \leq \pi\varphi(\sqrt{3}\varphi^3/2)^{1/\varphi}/(4 \cdot 0.8856\,(\varphi + \frac{1}{2})^{\varphi}) = 0.9513\ldots < 1$, and (b) shows that the global mode is interior.

For a fresh start of sign σ_0 we have $\mathbb{P}(D = n) = f_{\sigma_0}(n) \leq T_{\sigma_0}(n)$, and $B_n T_{\sigma_0}(n) \to 0$ by Lemma 2.1(v), so the bulk criterion holds. Moreover $Q_{\sigma_0} = T_{\sigma_0} \in \mathrm{RV}_{-\alpha_{\sigma_0}}$ with $\alpha_{\sigma_0} > 1 > 1/\alpha_*$, and the other atom vanishes. Part (c) concludes. □

For two equal laws $H_N = h(0)$, so $\mathcal{C}(\alpha)$ also compares the endpoint with the center bin. On the line $\rho_\sigma = 1/\alpha_*$ the ratio $B_N Q_\sigma(N)/H_N$ is slowly varying, the comparison depends on the slowly varying factors, and we leave it open.

6.2 The boundary layer at second order

By Theorem B the law follows the renewal mass function u_- in the boundary layer, and $u_-(j) \to 1/\mu_-$ by (2.10). If u_- stays above its limit, the layer lies above the level $T_+(N)/\mu_-$ at which it meets the big-jump zone, while the big-jump kernel increases as k decreases from N. To locate the interior minimum between the two we need the boundary layer at second order, with an exact coefficient.

Theorem 6.1 (second order in the boundary layer)**.** *Assume (RV) and (LR) for the plus sign, and let the start be fresh with sign $+$. Put*

$$\begin{aligned} C(j) &= \alpha_+ \Big[j\, u_-(j) + 2\mu_+ v_-(j) + \sum_{i=0}^{j-1} u_-(i) \Big] \\ &= \alpha_+ \Big[j\, u_-(j) + \mu_+ v_-(j) + \sum_{i=0}^{j-1} u_-(i) \big(1 + \mu_+ u_-(j-i)\big) \Big]. \end{aligned} \tag{6.3}$$

(a) For every fixed $j \geq 0$,

$$\frac{P_N(N-j)}{T_+(N)} = u_-(j) + \frac{C(j)}{N} + o\Big(\frac{1}{N}\Big).$$

(b) For every $\eta \in (0, 1)$,

$$\sup_{1 \leq j \leq N^{1-\eta}} \frac{N}{j} \Big| \frac{P_N(N-j)}{T_+(N)} - u_-(j) - \frac{C(j)}{N} \Big| \to 0.$$

If in addition (RV) and (SPAN) hold for the minus sign, then $C(j)/j \to 2\alpha_+ s/\mu_-^2$ as $j \to \infty$.

In the second form of $C(j)$ the terms $j\, u_-(j) + \mu_+ v_-(j)$ come from a censored giant run that starts after a slots, through $T_+(N-a)/T_+(N) = 1 + \alpha_+ a/N + o(a/N)$, and the sum comes from a complete giant run followed by a fresh minus walk, through Lemma 2.5. We prove the first form. The convergence of $C(j)/j$ is slow, of the order of $jT_-(j)$ by (6.7) below, and Section 8 reports the check.

The proof uses the final-run identity (2.14) and two renewal identities besides (2.9). Write $u = u_-$ and $v = v_-$. For $j \geq 1$,

$$\sum_{m=1}^{j} T_-(m)\, v(j-m) = \sum_{i=1}^{j-1} u(i), \tag{6.4}$$

$$\sum_{i=0}^{j-1} u(i)\, u(j-i) = v(j). \tag{6.5}$$

For (6.4), $v(j-m)T_-(m) = \sum_r r\mathbb{P}(S_r^- = j-m,\, X_{r+1}^- \geq m)$, so the sum is the expectation of the index r of the run that covers level j, minus one. That index minus one equals the number of $r \geq 1$ with $S_r^- \leq j-1$, whose expectation is $\sum_{i=1}^{j-1} u(i)$. For (6.5), $\sum_{i=0}^{j} u(i)u(j-i) = \sum_{a,b\geq 0} u_{a+b}^-(j) = \sum_n (n+1)u_n^-(j) = v(j)+u(j)$, and the term $i=j$ equals $u(j)$. By (6.5) the two forms of (6.3) agree.

We need two facts about sums of plus runs. Write $f = f_+$, $T = T_+$, $\alpha = \alpha_+$, $\mu = \mu_+$ and $S_r = S_r^+$ in Lemmas 6.2 and 6.3. Under (LR) the sequence f is regularly varying with index $-\alpha-1$, and Lemma 2.1(i) gives, for every $c>0$ and large n,

$$\sup_{m\geq cn} \frac{f(m)}{f(n)} \leq C_c, \qquad \sup_{0\leq w\leq w_n} \Big|\frac{f(n-w)}{f(n)} - 1\Big| \to 0 \quad \text{whenever } w_n = o(n). \tag{6.6}$$

By Lemma 2.1(iv), $\mathbb{E}[X_+^2; X_+ \leq y] \leq \sum_{n\leq y}(2n-1)T(n) \leq Cy^2T(y)$ and $\mathbb{E}[X_+; X_+ > y] \to 0$. Finally $nT(n)\to 0$, since $nT(n) \in \mathrm{RV}_{1-\alpha}$.

Lemma 6.2 (uniform local big jump)**.** *Assume (RV) and (LR) for the plus sign, and fix $\eta \in (0,1)$. Then*

$$\sup_{1\leq r\leq n^{1-\eta}} \Big|\frac{f^{*r}(n)}{r\,f(n)} - 1\Big| \to 0 \qquad (n\to\infty).$$

Proof. Put $\delta = \eta/2$. By Lemma 2.1(ii), $rT(n) \leq n^{1-\eta}T(n) \leq n^{-\delta}$ for large n. Fix $\varepsilon \in (0,1/4)$ with $\varepsilon < \delta/(16(1+\delta))$ and put $y = \varepsilon n$. Let $1 \leq r \leq n^{1-\eta}$ and split the event $\{S_r = n\}$ according to the number of the r runs that exceed y.

Exactly one run exceeds y. By symmetry this part equals

$$E_1 = r \sum_{w<n-y} \mathbb{P}\big(S_{r-1} = w,\, \max_{i<r} X_i^+ \leq y\big)\, f(n-w).$$

Put $w_n = n^{1-\eta/2}$. For $w \leq w_n$, (6.6) gives $f(n-w) = f(n)(1+o(1))$ uniformly. For $w_n < w < n-y$ we have $n-w > y$ and $f(n-w) \leq C_\varepsilon f(n)$ by (6.6), while $\mathbb{P}(S_{r-1} > w_n) \leq r\mu/w_n \leq \mu n^{-\eta/2}$ by Markov's inequality. Hence $E_1 \leq rf(n)(1+o(1))$. Keeping only $w \leq w_n$,

$$E_1 \geq rf(n)(1-o(1))\big(1 - \mathbb{P}(S_{r-1} > w_n) - rT(y)\big) = rf(n)(1-o(1)),$$

since $rT(y) \leq C_\varepsilon n^{1-\eta}T(n) \to 0$ by Lemma 2.1(ii).

At least two runs exceed y. For two given runs, $\mathbb{P}(S_r = n,\, X_1^+ > y,\, X_2^+ > y) \leq T(y)\sup_{x>y} f(x)$, by conditioning on all runs but the second. So this part is at most $r^2T(y)\sup_{x>y} f(x) \leq C_\varepsilon r^2 T(n) f(n)$, which is $o(rf(n))$ uniformly, since $rT(n)\to 0$.

No run exceeds y. For $r=1$ this part vanishes. For $r \geq 2$ write $S_r = S' + S''$, the sums of the first $\lceil r/2\rceil$ and of the remaining runs. On $\{S_r = n\}$ one of the two halves is at least $n/2$. Bounding the probability that the other half takes a given value by one,

$$E_3 = \mathbb{P}\big(S_r = n,\, \max_i X_i^+ \leq y\big) \leq 2 \max_{1\leq q\leq r} \mathbb{P}\big(S_q \geq n/2,\, \max_{i\leq q} X_i^+ \leq y\big).$$

We apply Lemma 4.2 with $m = q$, $w = \lfloor \varepsilon n\rfloor + 1$ and $z = n/2 - q\mu$. For large n we have $q\mu \leq n^{1-\eta}\mu \leq n/4$, so $z \geq n/4$, and $w \leq 2\varepsilon n$, so $z/(2w) \geq 1/(16\varepsilon)$. Also $qT(w) \leq rT(\varepsilon n) \leq C_\varepsilon rT(n) < 1$ for large n, by Lemma 2.1(ii). Since a^e decreases in e for $0<a<1$, the lemma gives

$$E_3 \leq 2e^{C_{\mathrm{K}}}\big(C_\varepsilon rT(n)\big)^{1/(16\varepsilon)}.$$

By (LR), $f(n) \geq cT(n)/n$. Hence

$$\frac{E_3}{rf(n)} \leq C'_\varepsilon\, n\,(rT(n))^{1/(16\varepsilon)-1} \leq C'_\varepsilon\, n^{1-\delta(1/(16\varepsilon)-1)},$$

which tends to zero because $\varepsilon < \delta/(16(1+\delta))$ gives $\delta(1/(16\varepsilon)-1) > 1$. The three parts together prove the lemma. □

Lemma 6.3 (sums of plus runs)**.** *Assume (RV) for the plus sign. There is a constant C such that, for all $r \ge 1$ and real $x > 0$,*

$$\mathbb{E}[S_r; S_r > x] \le r\Big(\mathbb{E}[X_+; X_+ > x/2] + \frac{2r\mu^2}{x}\Big),$$

and, if in addition $x \ge 4r\mu$,

$$\mathbb{P}(S_r \ge x) \le C\, rT(x).$$

Proof. If $S_r \ge x$, either some run is at least $x/2$, or the sum of the runs truncated at $x/2$ is at least x. The truncated sum has mean at most $r\mu \le x/4$ and variance at most $r\,\mathbb{E}[X_+^2; X_+ < x/2] \le Crx^2T(x)$. By Chebyshev's inequality the second event has probability at most $(4/(3x))^2 Crx^2T(x)$, and the first at most $rT(x/2) \le CrT(x)$ by Lemma 2.1(ii), since $x \ge 4r\mu \ge 4$. For the other bound, which uses no condition on x, on $\{S_r > x\}$ either $X_i^+ > x/2$ or $S_r - X_i^+ > x/2$. Hence $\mathbb{E}[X_i^+; S_r > x] \le \mathbb{E}[X_+; X_+ > x/2] + \mu\mathbb{P}(S_r - X_i^+ > x/2)$, and the last probability is at most $2r\mu/x$ by Markov's inequality. Sum over i. □

Proof of Theorem 6.1. We write $u = u_-$, $v = v_-$ and keep the notation of the two lemmas. By (LR), $Nf(N)/T(N) \to \alpha$. We use $v(j) \le j\,u(j) \le j$, which holds because $u_r^-(j) = 0$ for $r > j$.

(b) Fix $\eta \in (0,1)$, put $J_N = N^{1-\eta}$ and $x_N = N^{1-\eta/2}$, and let N be so large that $2J_N \le x_N$, $4J_N\mu \le N/2$ and $J_N \le (N/2)^{1-\eta/2}$. Every $o(\cdot)$ below is uniform in $1 \le r \le j \le J_N$. In (2.14) the first sum runs over $1 \le r \le j$, since $u_0^-(j) = 0$ for $j \ge 1$, and the second over $1 \le m \le j$ and $0 \le r \le j - m$.

The censored final run. Split the expectation in the first sum of (2.14) at $S_r = N/2$. For integers $0 \le x \le 3N/4$ put

$$\mathcal{T}_N(x) = \frac{N}{T(N)}\big(T(N-x) - T(N)\big) = \frac{N}{T(N)} \sum_{n=N-x}^{N-1} f(n).$$

By (6.6) and $Nf(N)/T(N) \to \alpha$, $\mathcal{T}_N(x) = \alpha x(1 + o(1))$ uniformly in $x \le x_N$, and $\mathcal{T}_N(x) \le Cx$ for $x \le 3N/4$, since $f(n) \le C_{1/4} f(N)$ for $n \ge N/4$. Since $j \le N/4$,

$$\mathbb{E}\big[T(N - j - S_r); S_r \le N/2\big] = T(N)\Big(\mathbb{P}(S_r \le N/2) + \frac{1}{N}\mathbb{E}\big[\mathcal{T}_N(j + S_r); S_r \le N/2\big]\Big).$$

On $\{S_r \le x_N/2\}$ we have $j + S_r \le x_N$. With $\mathbb{E}[j + S_r] = j + r\mu$ this gives

$$\big|\mathbb{E}[\mathcal{T}_N(j + S_r); S_r \le N/2] - \alpha(j + r\mu)\big| \le o(j + r) + C\,\mathbb{E}[j + S_r; S_r > x_N/2].$$

By Markov's inequality and Lemma 6.3, $\mathbb{E}[j + S_r; S_r > x_N/2] \le 2jr\mu/x_N + r\big(\mathbb{E}[X_+; X_+ > x_N/4] + 4r\mu^2/x_N\big) = o(j)$, since $r \le j \le J_N = o(x_N)$. By Lemma 6.3 and Lemma 2.1(ii), $\mathbb{P}(S_r > N/2) \le CrT(N)$. On the other part,

$$\mathbb{E}\big[T(N - j - S_r); N/2 < S_r \le N - j - 1\big] = \sum_{1 \le l < N/2 - j} f^{*r}(N - j - l)\, T(l).$$

Here $N - j - l \ge N/2$ and $r \le J_N \le (N - j - l)^{1-\eta/2}$. So Lemma 6.2 with $\eta/2$ in place of η and (6.6) give $f^{*r}(N - j - l) = rf(N)(1 + o(1))$ for $l \le x_N$, since then $j + l = o(N)$, and $f^{*r}(N - j - l) \le CrT(N)/N$ for all l in the range. Since $\sum_{l \le x_N} T(l) \to \mu$ and $\sum_{l > x_N} T(l) \to 0$, the sum equals $r\alpha\mu T(N)N^{-1}(1 + o(1))$. Together,

$$\mathbb{E}\big[T(N - j - S_r); S_r \le N - j - 1\big] = T(N)\Big(1 - O\big(rT(N)\big) + \frac{\alpha(j + 2r\mu)}{N} + o\Big(\frac{j}{N}\Big)\Big).$$

Multiply by $u_r^-(j)$ and sum over r. Since $\sum_r u_r^-(j) = u(j) \le 1$ and $\sum_r r u_r^-(j) = v(j)$, the first sum of (2.14) divided by $T(N)$ equals

$$u(j) - O\big(v(j)T(N)\big) + \frac{\alpha}{N}\big(j\,u(j) + 2\mu v(j)\big) + o\Big(\frac{j}{N}\Big).$$

Since $v(j)T(N) \le jT(N) = (j/N)\,NT(N)$ and $NT(N) \to 0$, the second term is $o(j/N)$.

The final minus run. Since $r+1 \le j \le J_N \le (N-J_N)^{1-\eta/2}$, Lemma 6.2 with $\eta/2$ in place of η and (6.6) give $u_{r+1}^+(N-j) = f^{*(r+1)}(N-j) = (r+1)\alpha T(N)N^{-1}(1+o(1))$. So the second sum of (2.14) equals

$$\begin{aligned}&\frac{\alpha T(N)}{N}(1+o(1))\sum_{m=1}^{j} T_-(m)\sum_r (r+1)u_r^-(j-m)\\ &\quad= \frac{\alpha T(N)}{N}(1+o(1))\sum_{m=1}^{j} T_-(m)\big(u(j-m)+v(j-m)\big),\end{aligned}$$

with $v(0)=0$. By (2.9) and (6.4) the last sum equals $1+\sum_{i=1}^{j-1}u(i) = \sum_{i=0}^{j-1}u(i)$, which is at most j. Adding the two contributions gives $u(j) + C(j)/N + o(j/N)$ with the first form of (6.3), uniformly in $1 \le j \le J_N$. This is (b).

(a) For $j=0$, (2.14) gives $P_N(N) = T(N)$ exactly, and $C(0)=0$. For fixed $j \ge 1$, part (b) gives an error $o(j/N) = o(1/N)$.

The limit. By (2.10) for the minus sign, $u(j) \to 1/\mu_-$. By (6.5) and Cesàro's theorem, $v(j)/j = j^{-1}\sum_{i<j}u(i)u(j-i) \to \mu_-^{-2}$, and $j^{-1}\sum_{i<j}u(i) \to 1/\mu_-$. Hence $C(j)/j \to \alpha_+(2/\mu_- + 2\mu_+/\mu_-^2) = 2\alpha_+ s/\mu_-^2$. □

6.3 The antimode

Two classical facts about the renewal mass function enter. The first is the second-order renewal theorem. Under (RV) and (SPAN) for the minus sign,

$$u_-(j) - \frac{1}{\mu_-} \sim \frac{j\,T_-(j)}{\mu_-^2(\alpha_- - 1)} \qquad (j \to \infty). \tag{6.7}$$

This is Lemma 4 of Frenk [19], stated there for an aperiodic recurrent event whose tail $\mathbb{P}(X > n) \sim T(n)$ is regularly varying with index $-\alpha$, $\alpha > 1$ (see also [37] and [15, Theorem 1]). Section 8 reports the check. The second fact gives a simple condition for the hypothesis of the next theorem.

Lemma 6.4 (log-convex run laws)**.** *Assume (RV) and (SPAN) for the minus sign. If $f_-(n) > 0$ for all $n \ge 1$ and $f_-(n-1)f_-(n+1) > f_-(n)^2$ for all $n \ge 2$, then u_- is strictly decreasing and $u_-(j) > 1/\mu_-$ for every $j \ge 0$.*

Proof. By Theorem 1 of de Bruijn and Erdős [10], the renewal sequence of a strictly log-convex law is strictly log-convex, that is, $u_-(m-1)u_-(m+1) > u_-(m)^2$ for $m \ge 1$. Hence the ratios $u_-(m+1)/u_-(m)$ increase strictly in m. If one of them were at least 1, all later ones would exceed some $q > 1$, and u_- would grow geometrically, which is impossible since $u_- \le 1$. So every ratio is smaller than 1, u_- is strictly decreasing, and it decreases to its limit $1/\mu_-$ given by (2.10). □

Theorem E (the antimode)**.** *Assume (RV) and (SPAN) for both signs and (LR) for the plus sign, and let the start be fresh with sign $+$. Let j_N^* be a minimizer of $j \mapsto P_N(N-j)$ over $0 \le j \le (1-p)N$.*

(a) If $u_-(j) > 1/\mu_-$ for every $j \ge 1$, then $j_N^* \to \infty$, $j_N^* = o(N)$, and

$$T_-(j_N^*) \sim \frac{2\alpha_+ s}{N}.$$

Moreover $j_N^*/i_N \to 1$ for every choice of minimizers i_N of $j \mapsto u_-(j) + C(j)/N$ over $j \ge 0$. So j_N^* is of the order of $b_{-,N}$, the stable scale of the minus runs, and $j_N^* \asymp N^{1/\alpha_-}$ up to a slowly varying factor.

(b) If $\inf_{j\ge1} u_-(j) < 1/\mu_-$, then j_N^* is bounded and every value that j_N^* takes infinitely often is a minimizer of u_-.

The borderline case, in which $u_- \ge 1/\mu_-$ with equality at some $j \ge 1$, is not covered. By Lemma 6.4 part (a) applies to every strictly log-convex law of the minus runs, for instance the zeta law $f(n) = n^{-\alpha-1}/\zeta(\alpha+1)$.

Proof. Write $R_N(j) = P_N(N-j)/T_+(N)$, $u = u_-$ and $c_{\mathrm{a}} = 2\alpha_+ s/\mu_-^2$. In this proof μ_- and μ_+ are always written with their signs. We first prove two lower bounds on R_N and then treat (a) and (b).

Lower bound in the layer. Let $1 \le j \le N/(8\mu_+)$. In the first sum of (2.14), $T_+(N - j - S_r^+) \ge T_+(N-j)$, and $\mathbb{P}(S_r^+ \le N-j-1) \ge 1 - \mathbb{P}(S_r^+ \ge N/2) \ge 1 - CrT_+(N)$ by Lemma 6.3, since $4r\mu_+ \le N/2$. Dropping the second sum,

$$R_N(j) \ge \frac{T_+(N-j)}{T_+(N)}\big(u(j) - C\,v(j)\,T_+(N)\big) \ge \frac{T_+(N-j)}{T_+(N)}\big(u(j) - C\,j\,T_+(N)\big). \tag{6.8}$$

By (LR), $f_+(n) \ge \frac34\alpha_+ T_+(n)/n \ge \frac34\alpha_+ T_+(N)/N$ for $N/2 \le n \le N$ and large N, because $T_+(n)/n$ is nonincreasing. Hence $T_+(N-j)/T_+(N) \ge 1 + \frac34\alpha_+ j/N$ for $j \le N/2$. We use (6.8) below only for $j \le \delta N$ with $\delta \le 1/(8\mu_+)$ and on ranges where $u(j)$ is bounded below by a positive constant. There $CjT_+(N) \le C\delta N T_+(N) \to 0$, so the second factor is positive for large N. Since the first factor is at least 1, we may then drop it or expand the product.

Lower bound far from the layer. Fix $\delta \in (0, 1-p)$. We claim that

$$\liminf_N \min_{\delta N \le j \le (1-p)N} R_N(j) \ge \frac{(1-\delta)^{-\alpha_+}}{\mu_-}. \tag{6.9}$$

Let $\varepsilon > 0$. By Corollary 4.7 there is K such that $P_N(k) \ge (1-\varepsilon)G_N^{(+)}(k)$ for $pN + KB_N \le k \le N$ and large N. For $k = N - j$ with $j \ge \delta N$ we have $L_k = N - j/(1-p) \le (1-\delta)N$, so $G_N^{(+)}(k) \ge T_+(\lfloor L_k \rfloor)u(j) \ge T_+(\lfloor (1-\delta)N \rfloor)u(j)$, and $T_+(\lfloor(1-\delta)N\rfloor)/T_+(N) \to (1-\delta)^{-\alpha_+}$ by (RV). For $pN \le k < pN + KB_N$, Theorem A(c) and Lemma 2.2(d) give $B_N P_N(k) \ge \frac12 \inf_N \inf_{[-K,K]} h_N > 0$ for large N, so $R_N(N-k) \ge c_K/(B_N T_+(N)) \to \infty$, since $B_N T_+(N) \to 0$ by Lemma 2.1(v). As ε is arbitrary and $u(j) \to 1/\mu_-$, the claim follows.

(b) Let $m_* = \min_j u(j) < 1/\mu_-$, which is attained on a finite set because $u(j) \to 1/\mu_-$, and let $\varepsilon_* = (1/\mu_- - m_*)/3$. Choose J with $u(j) \ge 1/\mu_- - \varepsilon_* > 0$ for $j \ge J$, and let $\delta = \min\{1/(8\mu_+), (1-p)/2\}$. For $J \le j \le \delta N$, (6.8) gives $R_N(j) \ge u(j) - CjT_+(N) \ge 1/\mu_- - \varepsilon_* - C\delta N T_+(N)$, which exceeds $m_* + \varepsilon_*$ for large N, since $NT_+(N) \to 0$. For $j \ge \delta N$, (6.9) gives $R_N(j) \ge m_* + \varepsilon_*$ for large N. For $j < J$, Theorem 6.1(a) gives $R_N(j) \to u(j)$. So for large N every minimizer is smaller than J, and a value j' taken infinitely often with $u(j') > m_*$ would give $R_N(j') > R_N(j_0)$ for large N, where $u(j_0) = m_*$.

(a) Let $\Lambda_N = a_-(N)$, so that $NT_-(\Lambda_N) \to 1$ and $\Lambda_N \in \mathrm{RV}_{1/\alpha_-}$. Put $w(j) = u(j) - 1/\mu_-$ and $c_{\mathrm{r}} = 1/(\mu_-^2(\alpha_- - 1))$, so $w(j) \sim c_{\mathrm{r}} j T_-(j)$ by (6.7), and $w(j) > 0$ for $j \ge 1$. Let

$$\Upsilon(x) = c_{\mathrm{r}} x^{1-\alpha_-} + c_{\mathrm{a}} x, \qquad x > 0.$$

The function Υ is strictly convex with a unique minimizer x_*, given by $c_{\mathrm{r}}(\alpha_- - 1)x_*^{-\alpha_-} = c_{\mathrm{a}}$, that is, $x_*^{-\alpha_-} = c_{\mathrm{a}}\mu_-^2 = 2\alpha_+ s$.

We fix the constants first. They depend only on c_{r}, c_{a}, $\alpha_\pm$ and $\mu_\pm$. Choose $\varepsilon \in (0, x_*)$ so small that $\frac{c_{\mathrm{r}}}{5}\varepsilon^{-(\alpha_- - 1)/2} > \Upsilon(x_*) + 2$, and choose $K > x_*$ so large that $\alpha_+ K/(2\mu_-) > \Upsilon(x_*) + 1$. So $\varepsilon < x_* < K$.

Step 1 (the window). By (6.7) and Lemma 2.1(i), $(N/\Lambda_N)\, w(j) = c_{\mathrm{r}}(j/\Lambda_N)^{1-\alpha_-}(1 + o(1))$ uniformly in $\varepsilon\Lambda_N \le j \le K\Lambda_N$, because $NT_-(j) = NT_-(\Lambda_N) \cdot T_-(j)/T_-(\Lambda_N)$. By Theorem 6.1, $C(j) = c_{\mathrm{a}} j(1 + o(1))$ as $j \to \infty$. Choose $\eta > 0$ with $1 - \eta > 1/\alpha_-$, so that $K\Lambda_N \le N^{1-\eta}$ for large N. Theorem 6.1(b) then gives

$$\frac{N}{\Lambda_N}\Big(R_N(j) - \frac{1}{\mu_-}\Big) = \Upsilon\Big(\frac{j}{\Lambda_N}\Big) + o(1) \quad \text{and} \quad \frac{N}{\Lambda_N}\Big(u(j) + \frac{C(j)}{N} - \frac{1}{\mu_-}\Big) = \Upsilon\Big(\frac{j}{\Lambda_N}\Big) + o(1), \tag{6.10}$$

uniformly in $\varepsilon\Lambda_N \le j \le K\Lambda_N$. Since $\varepsilon < x_* < K$, in particular the minimum of R_N is at most $1/\mu_- + (\Lambda_N/N)(\Upsilon(x_*) + o(1))$.

Step 2 (small j). We have $R_N(0) = 1$, and $u(0) + C(0)/N = 1$ for the comparison function. Since $\mu_- > 1$ and $\Lambda_N/N \to 0$, both exceed $1/\mu_- + (\Lambda_N/N)(\Upsilon(x_*) + 1)$ for large N. Let $1 \le j \le \varepsilon\Lambda_N$. Then $u(j) > 1/\mu_-$, and (6.8) gives $R_N(j) \ge u(j) - C\varepsilon\Lambda_N T_+(N) = 1/\mu_- + w(j) - o(\Lambda_N/N)$. Choose j_0 with $w(j) \ge \frac{c_{\mathrm{r}}}{2} jT_-(j)$ for $j \ge j_0$. By Lemma 2.1(ii) applied to $jT_-(j) \in \mathrm{RV}_{1-\alpha_-}$, for $j_0 \le j \le \varepsilon\Lambda_N$ and j_0 large,

$$\frac{jT_-(j)}{\Lambda_N T_-(\Lambda_N)} \ge \frac{1}{2}\Big(\frac{j}{\Lambda_N}\Big)^{-(\alpha_- - 1)/2} \ge \frac{1}{2}\varepsilon^{-(\alpha_- - 1)/2}.$$

Since $\Lambda_N T_-(\Lambda_N) \sim \Lambda_N/N$, we get $w(j) \ge \frac{c_{\mathrm{r}}}{5}\varepsilon^{-(\alpha_- - 1)/2}\Lambda_N/N$ for large N. For $1 \le j < j_0$, $w(j) \ge \min_{i<j_0} w(i) > 0$, which is much larger than Λ_N/N. By the choice of ε, $R_N(j) > 1/\mu_- + (\Lambda_N/N)(\Upsilon(x_*) + 1)$ for $1 \le j \le \varepsilon\Lambda_N$ and large N. The same bound holds for $u(j) + C(j)/N \ge 1/\mu_- + w(j)$.

Step 3 (intermediate j). Let $\delta = \min\{1/(8\mu_+), (1-p)/2\}$ and $K\Lambda_N \le j \le \delta N$. By (6.8) and $u(j) > 1/\mu_-$,

$$R_N(j) \ge \Big(1 + \frac{3\alpha_+ j}{4N}\Big)\Big(\frac{1}{\mu_-} - CjT_+(N)\Big) \ge \frac{1}{\mu_-} + \frac{\alpha_+ j}{2\mu_- N} \ge \frac{1}{\mu_-} + \frac{\alpha_+ K}{2\mu_-} \cdot \frac{\Lambda_N}{N}$$

for large N, since $NT_+(N) \to 0$. By the choice of K this exceeds $1/\mu_- + (\Lambda_N/N)(\Upsilon(x_*) + 1)$. For the comparison function, $C(j) \ge \alpha_+ j\, u(j) \ge \alpha_+ j/\mu_-$, so $u(j) + C(j)/N \ge 1/\mu_- + \alpha_+ j/(\mu_- N)$ for every j, and the same bound holds for all $j \ge K\Lambda_N$.

Step 4 (large j). By (6.9), $R_N(j) \ge (1 + \varepsilon_\delta)/\mu_-$ for $\delta N \le j \le (1-p)N$ and large N, with $\varepsilon_\delta = ((1-\delta)^{-\alpha_+} - 1)/2 > 0$, while $\Lambda_N/N \to 0$.

Conclusion. By Steps 2 to 4 every minimizer of R_N, and every minimizer of $u(j) + C(j)/N$, lies in $[\varepsilon\Lambda_N, K\Lambda_N]$ for large N. Fix $\eta' > 0$. Since Υ is continuous with a unique minimizer, $\Upsilon \ge \Upsilon(x_*) + \varepsilon'$ on $[\varepsilon, K] \setminus (x_* - \eta', x_* + \eta')$ for some $\varepsilon' > 0$, and (6.10) shows that both minimizers lie in $((x_* - \eta')\Lambda_N, (x_* + \eta')\Lambda_N)$ for large N. Hence $j_N^*/\Lambda_N \to x_*$ and $i_N/\Lambda_N \to x_*$, so $j_N^*/i_N \to 1$. By (RV), $NT_-(j_N^*) = NT_-(\Lambda_N) \cdot T_-(j_N^*)/T_-(\Lambda_N) \to x_*^{-\alpha_-} = 2\alpha_+ s$. Finally $\Lambda_N = a_-(N)$ and $b_{-,N} = a_-(N/s)$ differ by a bounded factor, and $a_- \in \mathrm{RV}_{1/\alpha_-}$. □

The leading law $T_-(j_N^*) \sim 2\alpha_+ s/N$ is approached slowly, because $C(j)/j$ converges slowly, and the minimizer of $u_-(j) + C(j)/N$ is the better prediction at moderate N. Section 8 reports the check.

7 Reading the run laws off the ends

The boundary layer is a renewal mass function, and a renewal mass function determines its interarrival law. So the ends of the row identify the run laws, provided the end

where the start puts no atom is normalized with care.

Theorem F (identification from the boundary layer)**.** *Assume (RV) for both signs.*

(a) Let the start be fresh with sign $+$. For every fixed $j \geq 1$,

$$\frac{P_N(N-j)}{P_N(N)} \to u_-(j), \qquad \frac{P_N(j)}{T_-(N)} \to u_+(j).$$

The first limits determine f_- through

$$f_-(j) = u_-(j) - \sum_{i=1}^{j-1} f_-(i)\, u_-(j-i), \qquad j \geq 1, \tag{7.1}$$

hence T_-, and then the second limits determine f_+ by the same recursion with the signs exchanged. In particular the family $(P_N)_{N\geq 1}$ determines both run laws.

(b) Let the preparation satisfy $Q_+(N)/T_+(N) \to q \in (0,\infty)$ and $\nu_+(n) = o(T_+(n))$, and put $\pi_+ = \mathbb{P}(\Sigma = +)$. Then $\pi_+ > 0$, and for every $j \geq 1$

$$\frac{P_N(N-j)}{T_+(N)} \to \kappa_+(j) = \pi_+ u_-(j) + \sum_{d=1}^{j} \nu_-(d)\, u_-(j-d).$$

Hence $P_N(N-j)/P_N(N) \to \kappa_+(j)/q$, and when the preparation and q are known these limits determine u_-, hence f_-.

The theorem uses only the boundary layer at fixed j, so (LR) is not needed. The minus end in (a) is normalized by $T_-(N)$, which is known only after the plus end has been read, and Remark 7.1 shows that this normalization cannot be replaced by $\sum_n f_+(n) = 1$.

Proof of Theorem F(a). For a fresh plus start $P_N(N) = F_N^+(N) = T_+(N)$ by Proposition 2.3, and $T_+(N) > 0$. Theorem B(a) with $\sigma_0 = +$ and (2.11) give $P_N(N-j)/T_+(N) \to \kappa_+^{(+)}(j) = u_-(j)$, which is the first limit. At the minus end, Theorem B(a) with the signs exchanged gives $P_N(j)/T_-(N) \to \kappa_-^{(+)}(j)$, and (2.11) with the signs exchanged gives $\kappa_-^{(+)}(j) = u_+(j)$ for $j \geq 1$. This is the second limit.

For the recursion, let $j \geq 1$ and split the sum defining $u_-(j)$ according to the length i of the first run. The term $m = 1$ is $f_-(j)$. For $m \geq 2$ the first run has length $1 \leq i \leq j-1$ and the other $m-1$ runs sum to $j - i \geq 1$. Hence

$$u_-(j) = f_-(j) + \sum_{i=1}^{j-1} f_-(i) \sum_{m\geq 1} f_-^{*m}(j-i) = f_-(j) + \sum_{i=1}^{j-1} f_-(i)\, u_-(j-i),$$

which is (7.1). It gives $f_-(1) = u_-(1)$, and for $j \geq 2$ it expresses $f_-(j)$ through $u_-(1), \ldots, u_-(j)$ and $f_-(1), \ldots, f_-(j-1)$. By induction on j the sequence u_- determines f_-, and $T_-(n) = 1 - \sum_{i<n} f_-(i)$. So the first limits determine $T_-(N)$ for every N, and the second limits then determine $u_+(j)$ for every $j \geq 1$. Since $u_+(0) = 1$, the same recursion with the signs exchanged determines f_+. □

Remark 7.1 (what the minus end alone determines)**.** Let the start be fresh with sign $+$, and suppose only the ratios at the two ends are read. At the minus end the second limit of Theorem F(a) gives $P_N(j)/P_N(i) \to u_+(j)/u_+(i)$, which determines $(u_+(j))_{j\geq 1}$ only up to a positive factor. The normalization $\sum_n f_+(n) = 1$ does not fix this factor. To see this, let $0 < c \leq 1$, let G be independent of the runs with $\mathbb{P}(G = n) = c(1-c)^{n-1}$ for $n \geq 1$, and let X_+^c be the sum of G independent copies of X_+. With generating

functions $F(z) = \sum_n f_+(n) z^n$ and $U = 1/(1-F)$, the law of X_+^c has generating function $F_c = cF/(1-(1-c)F)$, which equals one at $z = 1$. Then

$$\frac{1}{1-F_c} = \frac{1-(1-c)F}{1-F} = 1 + c\,(U-1),$$

so $u_+^c(j) = c\,u_+(j)$ for every $j \geq 1$. In words, the renewals of X_+^c are those of X_+ kept independently with probability c. The support of X_+^c contains that of X_+, so (SPAN) is inherited, and $\mathbb{P}(X_+^c \geq n) \sim T_+(n)/c$ by the tail asymptotics of random sums of subexponential variables [16], so (RV) holds with the same index. Replacing X_+ by X_+^c leaves u_- and every ratio $u_+(j)/u_+(i)$ unchanged. Hence the ratios at both ends do not distinguish f_+ from the one-parameter family f_+^c, each member of which sums to one. The normalization by $T_-(N)$ in Theorem F(a) separates them, since $P_N(j)/T_-(N)$ changes by the factor c. The bulk also separates them, since the mean of X_+^c is μ_+/c and p changes.

Proof of Theorem F(b). By Theorem C(d), which needs only (RV) for the plus sign, and $\nu_+(n) = o(T_+(n))$, we have $P_N(N-j)/T_+(N) \to \kappa_+(j)$ for every $j \geq 1$, with κ_+ as in (5.2). Since $Q_+(N) \leq \pi_+$ and $Q_+(N) > 0$ for large N, we have $\pi_+ > 0$.

Proposition 2.4 gives $P_N(N) = Q_+(N)$, so $P_N(N-j)/P_N(N) \to \kappa_+(j)/q$. Suppose the preparation and q are known. Then $\kappa_+(j)$ is known for every $j \geq 1$, and since $\nu_-(0) = 0$ and $u_-(0) = 1$,

$$\pi_+ u_-(j) = \kappa_+(j) - \nu_-(j) - \sum_{d=1}^{j-1} \nu_-(d)\,u_-(j-d),$$

which determines $u_-(j)$ by induction on j, because $\pi_+ > 0$. Then (7.1) determines f_-. □

The hypotheses of (b) hold, for example, when $\nu_+ = \pi_+ f_+$ with $\pi_+ > 0$ and ν_- arbitrary, since then $Q_+ = \pi_+ T_+$, so $q = \pi_+$, and $\nu_+ = o(T_+)$ by (4.12). The stationary preparation fails the first hypothesis, since its atom is of order $NT_+(N)$ by Corollary 5.4(a).

8 Verification

No proof uses a computed number. The computations check the exact identities, produce Figure 1 and Table 1, and test the limit theorems at moderate N. Except for the identities, all of them use the zeta law $f_\sigma(n) = n^{-5/2}/\zeta(5/2)$ for both signs, so $\alpha_\pm = 3/2$, $p = 1/2$ and $s = 3.894745$, with a fresh plus start unless we say otherwise.

The exact law. For a fresh plus start let $a_t(k)$ and $b_t(k)$ be the probabilities that a plus run, respectively a minus run, starts at slot $t+1$ and the first t slots contain exactly k plus slots. So $a_t(k) = \rho_{t-k}^{(+)}(k)$, $a_0(0) = 1$ and $b_0(0) = 0$. Since the runs alternate, with $a_t(k) = b_t(k) = 0$ for $k < 0$,

$$b_t(k) = \sum_{\ell=1}^{t} a_{t-\ell}(k-\ell)\,f_+(\ell), \qquad a_t(k) = \sum_{\ell=1}^{t} b_{t-\ell}(k)\,f_-(\ell) \qquad (t \geq 1). \tag{8.1}$$

By (2.13) the identity (2.14) of Proposition 2.6 then reads

$$P_N(N-j) = \sum_{t=j}^{N-1} a_t(t-j)\,T_+(N-t) + \sum_{m=1}^{j} b_{N-m}(N-j)\,T_-(m). \tag{8.2}$$

The recursion gives the $O(N^2)$ numbers $a_t(k)$ and $b_t(k)$ with $t \leq N-1$ in $O(N^3)$ operations, and (8.2) then gives the whole row in $O(N^2)$ more. The script certify_atlas.py runs

it in double precision. Every entry is one dot product of earlier entries with nonnegative terms. So every computed bin has relative error at most $\varepsilon_N = (1+\gamma_{N+2})^{N+2} - 1$, with $\gamma_n = nu/(1-nu)$ and $u = 2^{-53}$, whatever the order of summation. This bound is $4.0 \cdot 10^{-11}$ at $N = 600$ and $2.8 \cdot 10^{-9}$ at $N = 5000$. At $N = 600$ we also run the recursion in 128-bit ball arithmetic. The double-precision row lies in the balls widened by ε_{600}, with a largest relative deviation of $5.9 \cdot 10^{-15}$.

The figure. The script figures/make_atlas_figure.py draws Figure 1 from the law at $N = 5000$, with h_N obtained by inverting its characteristic function. The horizontal axis of panel (a) is linear on $[250, 4750]$ and stretched on the last 250 bins at each end, which makes the two kinks. The shaded bands have width $B_N = 74$ and mark the junctions between the pieces. The largest relative error of the bulk on $|k - pN| \le 3B_N$ is 0.071. That of the big-jump kernel where $|k - pN| \ge 4B_N$ and $\min(k, N-k) \ge 2B_N$ is 0.38, attained at the junction with the bulk, and that of the boundary kernel on $1 \le \min(k, N-k) \le B_N$ is 0.094. The atom is exact.

The antimode. Table 1 compares the leading law and the refined prediction of Theorem E with the exact law. The script certify_antimode.py finds the minimizer of P_N on $[pN, N]$. It is certified when every other bin there exceeds the minimum by a relative margin above $2\varepsilon_N/(1-\varepsilon_N)$, which is at most $1.5 \cdot 10^{-8}$. The smallest margins are $2.6 \cdot 10^{-4}$, $2.3 \cdot 10^{-4}$, $1.7 \cdot 10^{-5}$, $3.1 \cdot 10^{-5}$ and $1.1 \cdot 10^{-5}$ for the five rows.

N	observed j_N^*	refined i_N	leading	$NT_-(j_N^*)/(2\alpha_+ s)$
600	7	7	10	1.534
1200	11	12	15	1.498
2400	19	20	23	1.282
4000	27	28	32	1.247
8000	44	45	50	1.186

Table 1: The antimode for the zeta law with $\alpha_+ = \alpha_- = 3/2$ and a fresh plus start. The observed j_N^* minimizes $P_N(N-j)$ over $0 \le j \le N/2$, which is the range $0 \le j \le (1-p)N$ of Theorem E. The refined i_N minimizes $u_-(j) + C(j)/N$, and the leading prediction is the smallest j with $T_-(j) < 2\alpha_+ s/N$.

The coefficient. The script verify_coefficient.py checks that the two forms of (6.3) agree for $j \le 6000$ and every entry of Table 1. The limit $2\alpha_+ s/\mu_-^2$ of $C(j)/j$ in Theorem 6.1 is 3.0811, while $C(j)/j$ equals 4.968, 4.430, 4.041, 3.762, 3.563, 3.422 and 3.323 at $j = 5, 10, 20, 40, 80, 160, 320$. At $j = 5$, 10, 20 and 40 the script compares $C(j)/j$ with $[P_N(N-j)/T_+(N) - u_-(j)]N/j$ for $N = 1200$, 2400 and 4800. The distance halves each time N doubles, from 0.090 to 0.022 at $j = 5$ and from 0.592 to 0.135 at $j = 40$. The ratio of the two sides of (6.7) is 0.978, 0.995 and 0.999 at $j = 40$, 160 and 640. The script also checks the constant $\mathcal{C}(\varphi)$ and the bound $0.9513\ldots$ of Theorem D(d).

The preparation and the big-jump zone. The script verify_preparation.py computes the law for a general preparation by the recursion with the first run inserted, and checks it against Proposition 2.4. It takes the fresh plus start, the stationary start and a start with $Q_+(n) = n^{-0.3}/2$, for which $\rho = 0.3 < \alpha_+ - 1$. For $j \in \{1, 2, 3, 5, 10, 20\}$ the largest relative deviation of $P_N(N-j)$ from the exact defect form of Theorem C(d) falls from 0.12 at $N = 1200$ to 0.059 at $N = 2400$ for the fresh start, from 0.057 to 0.028 for the stationary start and from 0.013 to 0.006 for the third start. For the stationary start the ratio of $P_N(N-j)/T_+(N)$ to $\kappa_+^{\mathrm{st}}(j)$ in Corollary 5.4(c) is 1.0007, 1.0031 and 1.0098 at $j = 1$, 2 and 5 for $N = 2400$, and each deviation is about half of its value at $N = 1200$. The ratios in the big-jump zone are printed without a test, since Theorem C(c) is a double limit. The script verify_bigjump.py checks Remark 4.9. The limit of $P_N(k)/G_N(k)$ at

$k = \lceil pN + KB_N \rceil$ is 0.31, 1.68 and 1.09 at $K = 1$, 4 and 16. The exact ratio is 0.354, 1.482 and 1.036 at $N = 1200$, and 0.337, 1.538 and 1.059 at $N = 3000$. These move toward the limits, and at $N = 3000$ they are still about 8 percent off at $K = 1$ and $K = 4$.

The identities. The script verify_identities.py checks Propositions 2.3, 2.4 and 2.6, the latter in both forms (2.14) and (8.2) with the recursion (8.1), Lemmas 2.5 and 2.7, (2.8), (2.11), (2.9) to (6.5) and the inversion of Theorem F in exact rational arithmetic. It uses two pairs of run laws with finite support and a preparation with atoms of both signs, some beyond the horizon. The law is computed as a sum over all 2^N sign strings for $N \leq 12$, and independently by a chain that moves slot by slot with the hazard $f_\sigma(a)/T_\sigma(a)$ of the current run. At $N = 1200$ and $j \leq 40$ the script verify_coefficient.py matches the two forms of Proposition 2.6 in double precision to $2.2 \cdot 10^{-15}$.

The Lean library. The library PaperF in kagey-problems/problem131/lean has 7 modules, in which a walk of N slots is a word of signs. The Lean kernel checks the fresh identities (`fresh_plus`, `fresh_minus`), the kernel sums (`kernel_sum_u`, `kernel_sum_q`, `kernel_sum_q_succ`), Proposition 2.4 (`first_run`), Proposition 2.6 in the form (8.2) with the recursion (8.1) (`final_run`, `A_succ`, `B_succ`), Lemma 2.5 as a convergent series (`green_plus`, `green_minus`), the first part of Lemma 2.7 (`counting`) and the inversion of Theorem F(a) (`renewal_inverse`, `renewal_injective`). No module uses `sorry` or `native_decide`, and `#print axioms` reports only `propext`, `Classical.choice` and `Quot.sound`. The bound of Lemma 2.7, (2.11), the limits in Theorem F and every asymptotic statement are not formalized. The file lean/PaperF/MAP.md maps the labels to the Lean names.

Software. The scripts, their outputs with the hash of each script and the date, and the Lean files are in the folder kagey-problems/problem131 of https://github.com/apemm/Kagey-Problems. In paperF, `python -I verify_all.py` runs the four `verify` scripts in about half a minute, and the option `--full` reruns the two `certify` scripts first, in about two minutes. In lean, `lake exe cache get` and `lake build PaperF` build the library. We used Python 3.14.3, numpy 2.4.2, scipy 1.17.1, mpmath 1.3.0, python-flint 0.9.0 (FLINT 3.6.0, with Arb), matplotlib 3.10.8, Lean 4.33.1 and Mathlib v4.33.1 (commit `0df444a360ea`).

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